Object-attribute data:

A Journey from Philosophy to Visualization, Venturing into Physics, Mathematics, Logic, and Programming

Speaker: Alexey Neznanov, PhD
Senior Data Scientist, STISS
Associate Professor, HSE
2026-10-20

V. 1.00.04

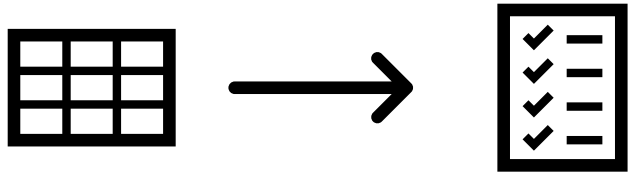


About the speaker

- Alexey Neznanov
  - PhD, IEEE member
  - Senior Data Scientist at STISS
    - Associate professor at NRU HSE (http://www.hse.ru/staff/aneznanov/)
  - Author of textbooks, instructional guidelines, 11 courses, and more than 70 research papers
    - «Small Guide to Big Data» consultant at PostScience portal (http://postnauka.ru/author/a_neznanov)



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INTRODUCTION INTO OBJECT-ATTRIBUTE DATA

- Topics
- Some sources
- Illustration of object-attribute data
- Word “attribute” as term

FANTASTIC ENTITIES AND WHERE TO FIND THEM

- Semiotics and sign systems
- Entities and concepts
- Hermeneutic circle, natural language as a language of philosophy
- Ontology, epistemology, gnoseology
- Grounding
- Measurements and scales

THE CONCEPT OF “MEASUREMENT”

- Representational measurement theory
- Context of value in object-attribute data representation
- Scales
- ISO standards
- Ontologies of units of Measure

FORMALIZATION OF CONCEPTS

- Partially Ordered Sets (posets)
- Formal Concept Analysis
  - Objects and attributes
  - Formal context and formal concept
  - Extension → extent, intension → intent
  - Formal concept lattice and line diagrams
- Implications in context and association rules

ENTITIES AND DATA MODELS

- Formal Concept Analysis
  - Objects and attributes
  - Formal context and formal concept
  - Extension → extent, intension → intent
  - Formal concept lattice and line diagrams
- Implications in context and association rules

OBJECT-ATTRIBUTE DATA IN HCI AND UI

- HCI and UI
- Tables and spreadsheets
  - Pivot tables
- Rows and columns in UI
  - Implicit and explicit ordering

OBJECT-ATTRIBUTE DATA IN PROGRAMMING

- Dataframes and tables
- Dataframes interoperability
- Rich table construction and visualization

OBJECT-ATTRIBUTE DATA AND AI

- Data and decision making
- AI and AGI
- Machine leaning : train, test, and validate

VISUALIZATION OF “TABLES”

- Object-attribute data representation
  - Objects, attributes, values, tuples, and tables
- Statistics
- Visual design
  - Tables and charts
  - Charts classification
- Methodologies and tools

COMMON WORDS ABOUT DATAVIZ (PREWORDS)

- Common problems
- Subtopics
- Some sources and references

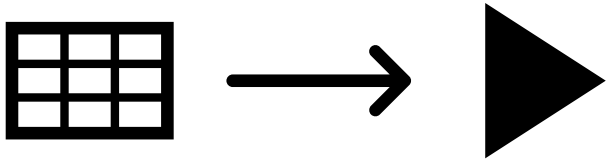
DASHBOARDING

- Concepts
- Classification of dashboards
- BI and classical/modern methodology

TEMPORAL DATA VISUALIZATION

- Essentials
- Standartization
  - ISO 8601
- Classification
  - TimeVis Browser
- Tools

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INTRODUCTION INTO OBJECT-ATTRIBUTE DATA

- ✓ Topics
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Some discussed topics by terms

- Entity and concept in Philosophy (epistemology and ontology)
- Object and class in Object-oriented analysis and ontology engineering
- Attribute and attribute value in object-attribute data representation
- Measure, scale, and unit of measure in attributes grounding
- Formal context and formal concept in Formal Concept Analysis
- Tuple, relation and relationship in Relational theory
- Record, table and recordset in RDBMS
- Type and format in Computer science
- Interface and protocol in IT
- Table (another one ☺), pivot table, and dataframe in modern Data Science
- Dataset and feature in Machine Learning
- Chart in DataViz

‼ Information, data and metadata → in **other** presentation

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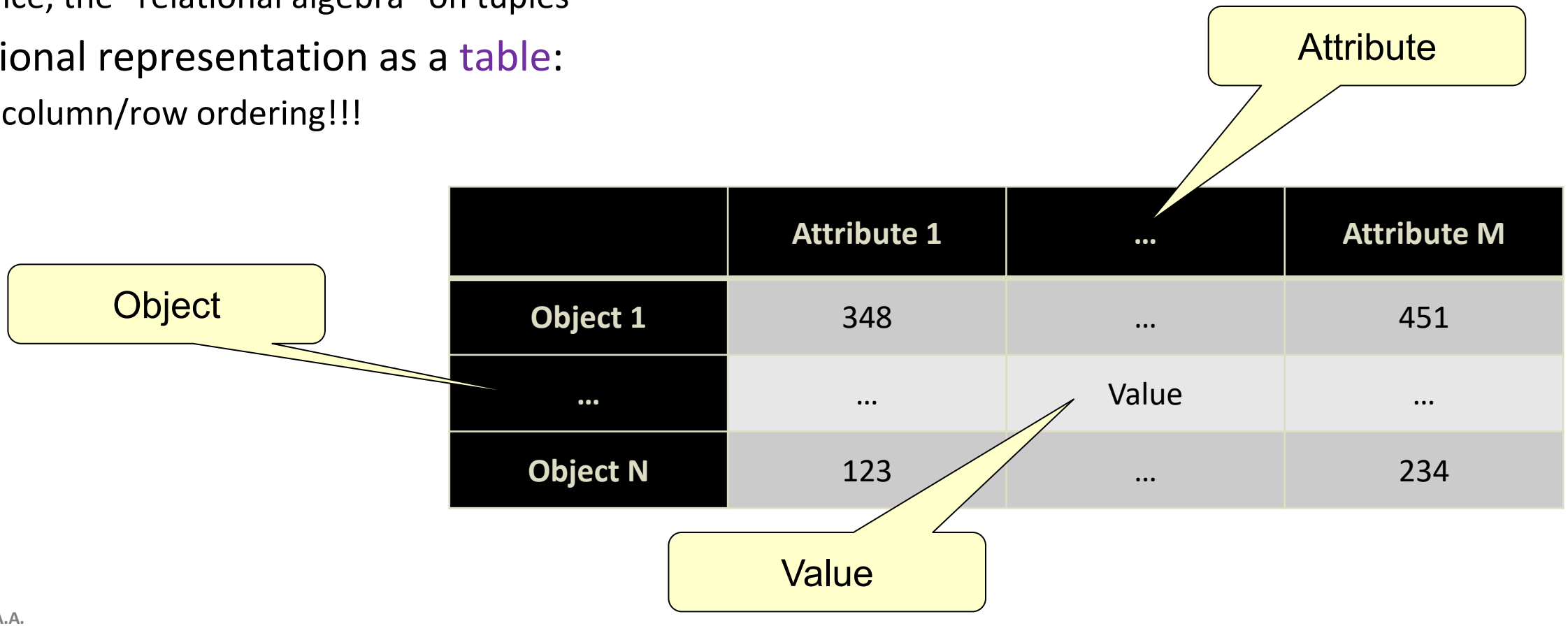
Basic sources

- Logic and set theory
  - Full Open-Source, Collaborative Logic Text, aimed at a non-mathematical audience (http://openlogicproject.org)
- Formal Concept Analysis (FCA)
  - Formal Concept Analysis Homepage by Uta Priss (http://upriss.github.io/fca/fca.html)
  - Introduction to Formal Concept Analysis by Radim Belohlavek (http://phoenix.inf.upol.cz/esf/ucebni/formal.pdf)
- Relational model
  - Codd, E. A Relational Model of Data for Large Shared Data Banks. Communications of the ACM, 13(6), 1970, pp. 377-387. (http://www.seas.upenn.edu/~zives/03f/cis550/codd.pdf)
  - Codd E.F. The relational model for database management: version 2. Addison-Wesley, 1990. 538 p. (http://dl.acm.org/doi/pdf/10.5555/77708)
  - Date C.J. Database Design and Relational Theory. Normal Forms and All That Jazz. O'Reilly, 2012. 260 p.
  - Mancas C. Conceptual Data Modeling and Database Design: a Fully Algorithmic Approach. Volume 1, The Shortest Advisable Path. Apple Academic Press, 2016. 634 p.

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Object-attribute data (table?)

- **Object** – an instance of some entity
- **Attribute** – a description of an attribute of some entity
- **Attribute value** (assembled into a tuple)
  - Hence, the “relational algebra” on tuples
- Traditional representation as a table:
  - No column/row ordering!!!



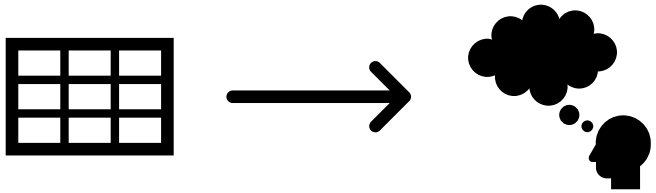
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Attribute synonyms

- Ability (capability)
- Accent
- Aspect
- **Attribute**
- Capacity
- Characteristic
- Condition
- Connotation
- Degree
- Differentia
- Distinctor (distinction)
- Facet
- **Feature**
- Genus
- Indicator (indication)
- Inherent
- Kind
- Mark (marker)
- Peculiarity
- Predicate
- **Property**
- Quality
- Quirk
- Score
- Sign
- Specific
- Stamp
- Status
- Symptom
- Token
- Trait
- Type
- Variety
- …

See also: Merriam-Webster Thesaurus – attribute (http://www.merriam-webster.com/thesaurus/attribute)

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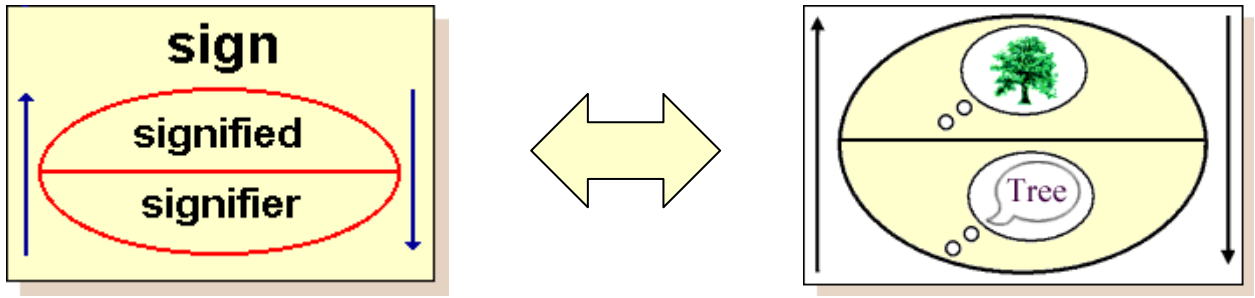
FANTASTIC ENTITIES AND WHERE TO FIND THEM

- ✓ Semiotics and sign systems
- ✓ Entities and concepts
- ✓ Hermeneutic circle, natural language as a language of philosophy
- ✓ Ontology, epistemology, gnoseology
- ✓ Grounding
- ✓ Measurements and scales

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Semiotics and sign system

- **Sign system** – any system of signs and relations between signs
  - Sign system – algebraic system intended to capture the systematic structure of complex signs, since in all but the simplest cases, most signs are composed from other signs
    - Goguen J.A. Semiotic Morphisms. 2004
  - Semiotics for Beginners (http://www.cs.princeton.edu/~chazelle/courses/BIB/semio2.htm)

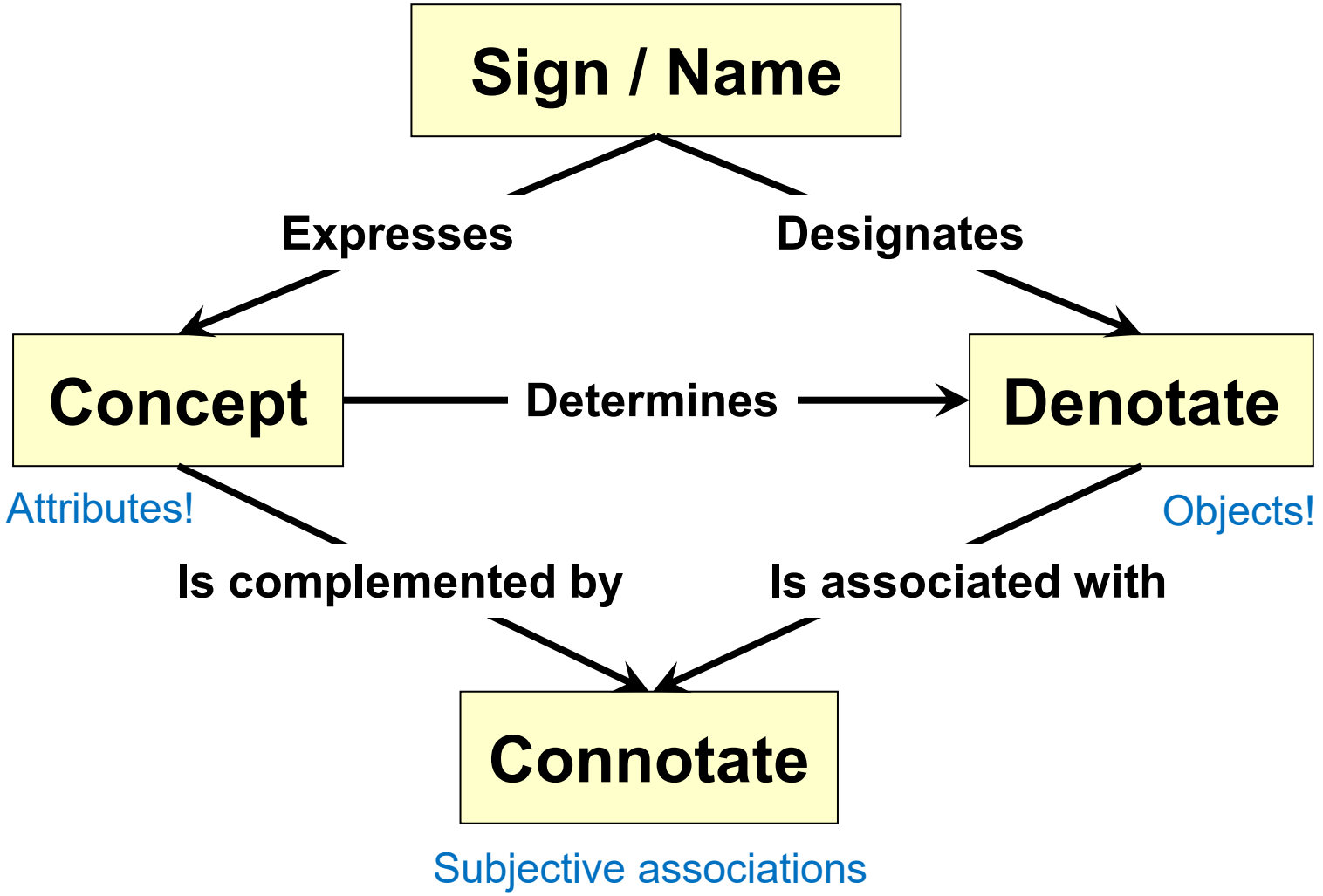


- We interpret things as signs largely unconsciously by relating them to familiar systems of conventions!
  - **Codes** for messages transmission
    - Encoding is the process of creating a message for transmission by an addresser (sender) to an addressee (receiver)
    - Decoding is the process of interpreting a message sent by an addresser (sender) to an addressee (receiver)

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Frege triangle – most popular model of “sign”

- F.L.G. Frege, 1848-1925
- Basic triangle + connotations:



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Semiotics deconstruction: semantics, syntactics, and pragmatics

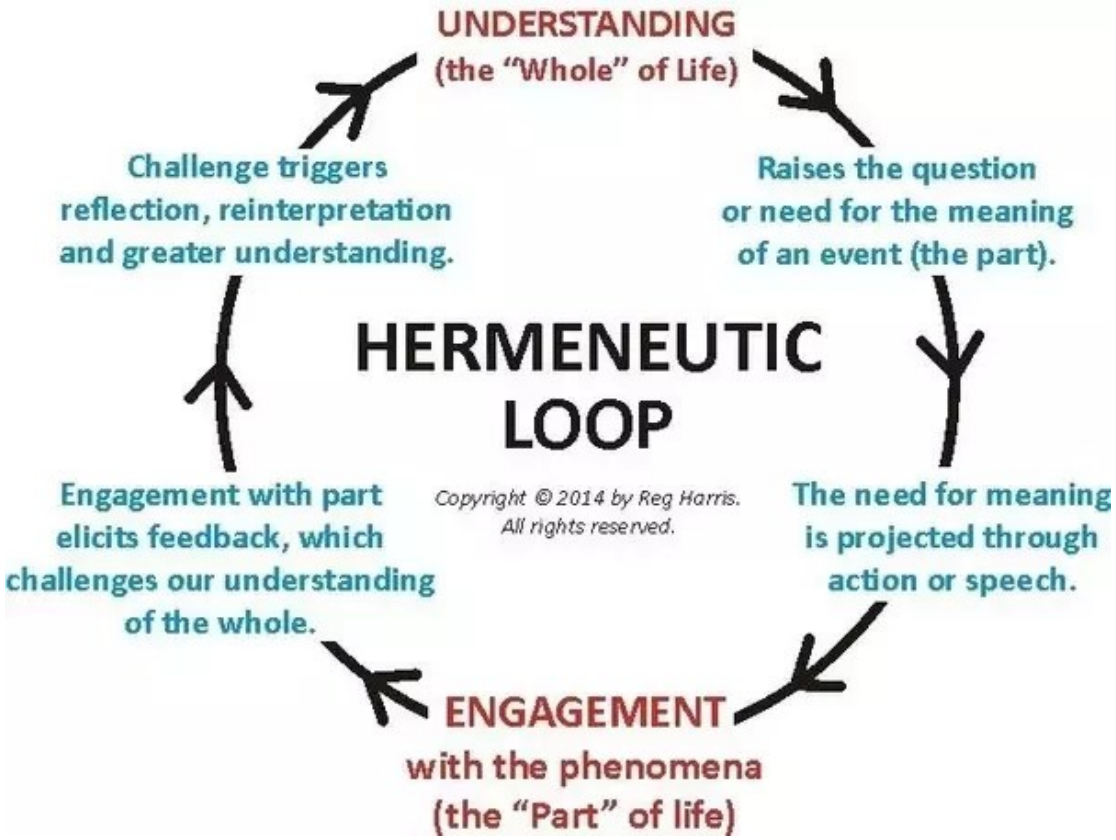
- Three points of view (the branches of semiotics):
  - **Semantics** – this is the “how” of semiotics, and is concerned with this relationship between a signified and signifier – the sign and what it stands in for
  - **Syntactics** – this refers to structural relations in the set of signs
    - One structural relation in language is **grammar**, but syntactics in semiotics refers to the formal relationship between signs that lets them build into sign systems
  - **Pragmatics** – according to Morris, is the relationship of sign to the person reading or understanding that sign
    - Morris C. Foundations of the Theory of Signs (in International Encyclopaedia of Unified Science, V.1). University of Chicago Press, 1938. 59 p.

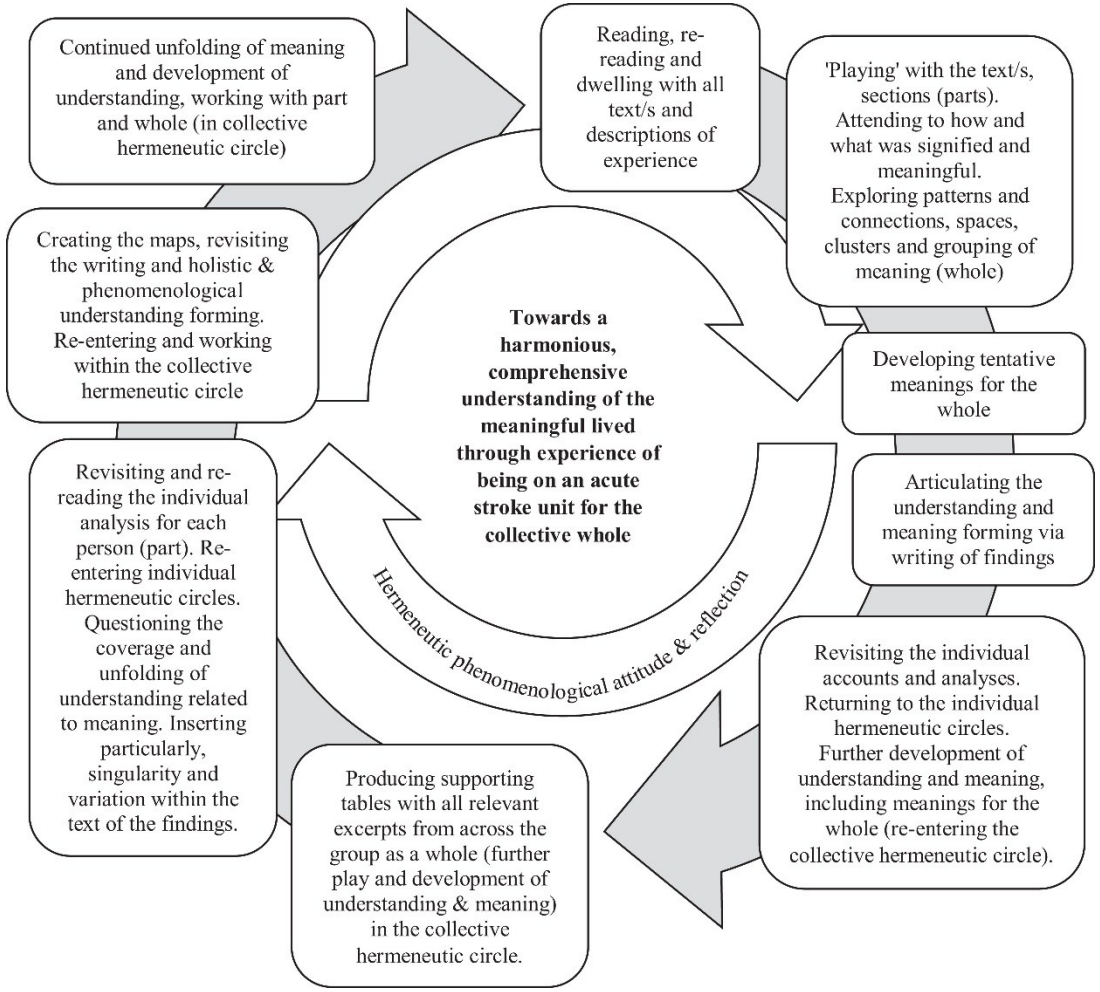
- “The syntactic and semantic structure of natural languages evidently offers many mysteries, both of fact and of principle, and any attempt to delimit these domains must certainly be quite tentative”
  - Chomsky N. Aspects of the Theory of Syntax. MIT Press, 1965. 261 p.

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Hermeneutics

- Hermeneutic loop/circle
  - Martin Heidegger – Being and Time, 1927



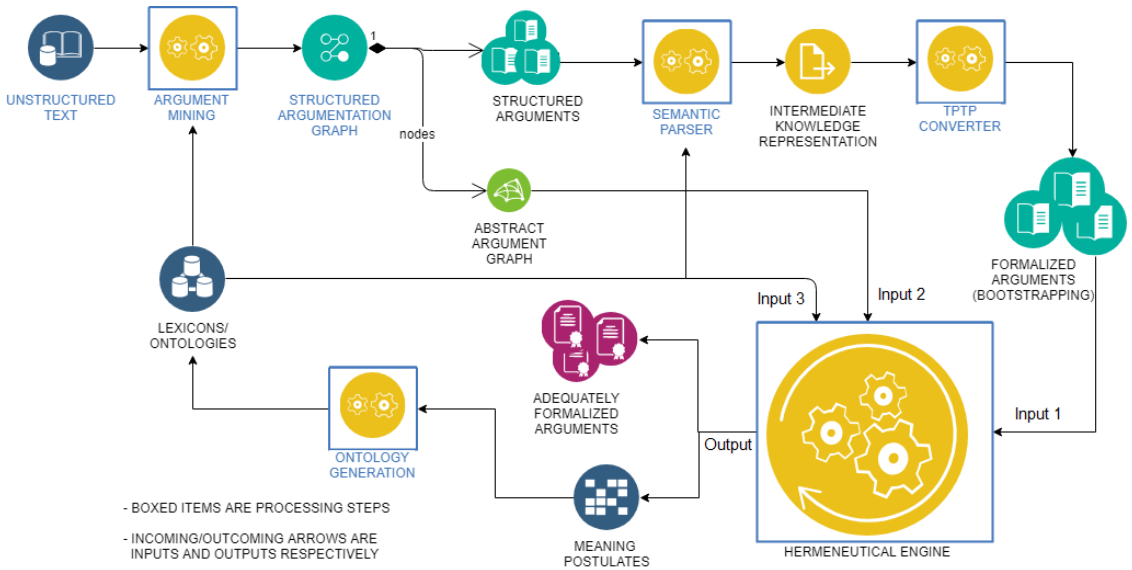


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Hermeneutic circle in practice

- From “Basic Formal Ontology 2.0 – Specification and User’s Guide” (http://github.com/BFO-ontology/BFO/raw/master/docs/bfo2-reference/BFO2-Reference.pdf):
  - “… we distinguish between primitive and defined. ‘Entity’ is an example of one such primitive term.
  - Primitive terms in a highest-level ontology such as BFO are terms that are so basic to our BFO 2.0 Specification and User’s Guide understanding of reality that there is **no way** of defining them in a **non-circular fashion**.
  - For these, therefore, we can provide only elucidations, supplemented by examples and by axioms”

- Fuenmayor D., Benzmüller C.
  A Computational-Hermeneutic Approach for Conceptual Explicitation.
  arXiv:1906.06582, 2019
  (http://arxiv.org/abs/1906.06582)

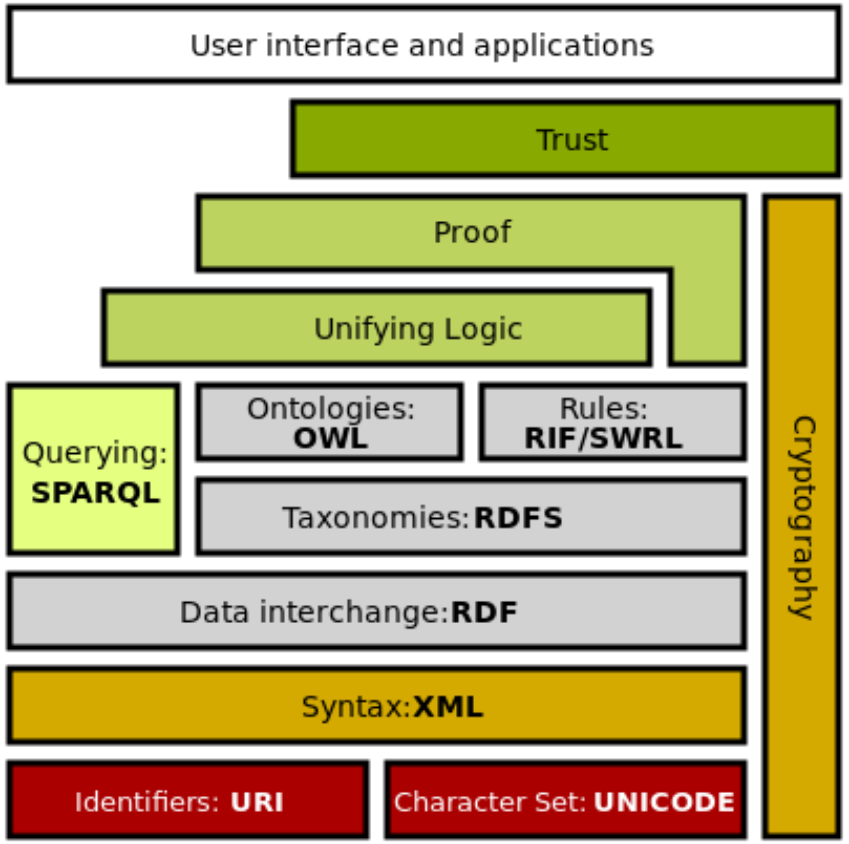


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Ontology (+ontics), epistemology, gnoseology

- Technical ontology – formalized computer-readable domain knowledge
- T. Gruber, 1993:
  "**Explicit specification of a conceptualization**"
  - The most cited definition!
  - + "**Shared**"

- **Semantic Web** (http://www.w3.org/standards/semanticweb/)
  - XML (http://www.w3.org/XML/) from 1996
  - RDF (http://www.w3.org/RDF/) from 1999
  - OWL (http://www.w3.org/OWL/) from **2004**
  - SPARQL (http://www.w3.org/TR/sparql11-overview/) from 2008
  - SKOS (http://www.w3.org/TR/skos-reference/) from 2005
  - +++

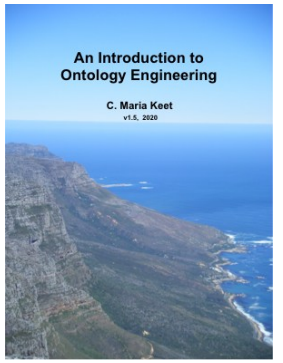


From wikipedia, license: CC BY-SA

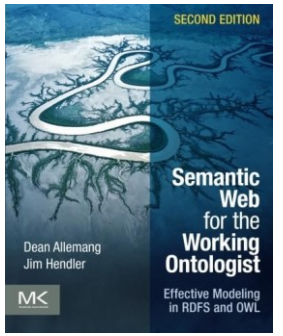
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An introduction to ontology engineering

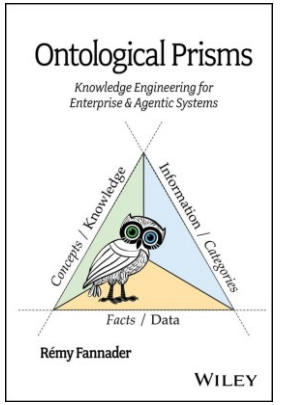
- Keet M. An introduction to ontology engineering. 2020 (http://people.cs.uct.ac.za/~mkeet/OEbook/)
  - The best introduction
  + Keet M. The What and How of Modelling Information and Knowledge. From Mind Maps to Ontologies. Springer, 2023. 184 p.



- Allemang D., Hendler J. Semantic Web for the Working Ontologist: Effective Modeling in RDFS and OWL. 2nd Ed. Morgan Kaufmann, 2011. 384 p.
  - A practical perspective on ontological modeling in the Semantic Web



- Rémy Fannader. Ontological Prisms – Knowledge Engineering for Enterprise & Agentic Systems. Wiley, 2026. 348 p.
  - A game-changing paradigm that unfolds the geometry of knowledge into a triptych of observed worlds (facts), meanings (concepts), and shared representations (categories)



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The problem of definitions

- Different domains → different languages → different conceptualizations
  - **Vocabulary**: the body of words used in a particular language
  - **Glossary**: link terms to concepts, described informally by glosses
  - **Thesaurus**: add structural (semantic) relationships among terms (and concepts), i.e. generalization, part, dependence, causation, …
- Michie S., West R., Hastings J. Creating ontological definitions for use in science. 2022 (http://www.qeios.com/read/YGIF9B)
  - Vocabulary definitions are often ambiguous or circular
  - We try to use constructive distinctive non-circular definitions!
- Bjørner D. Domain Science and Engineering. Springer, 2021. 401 p.

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Ontologies in various scientific fields

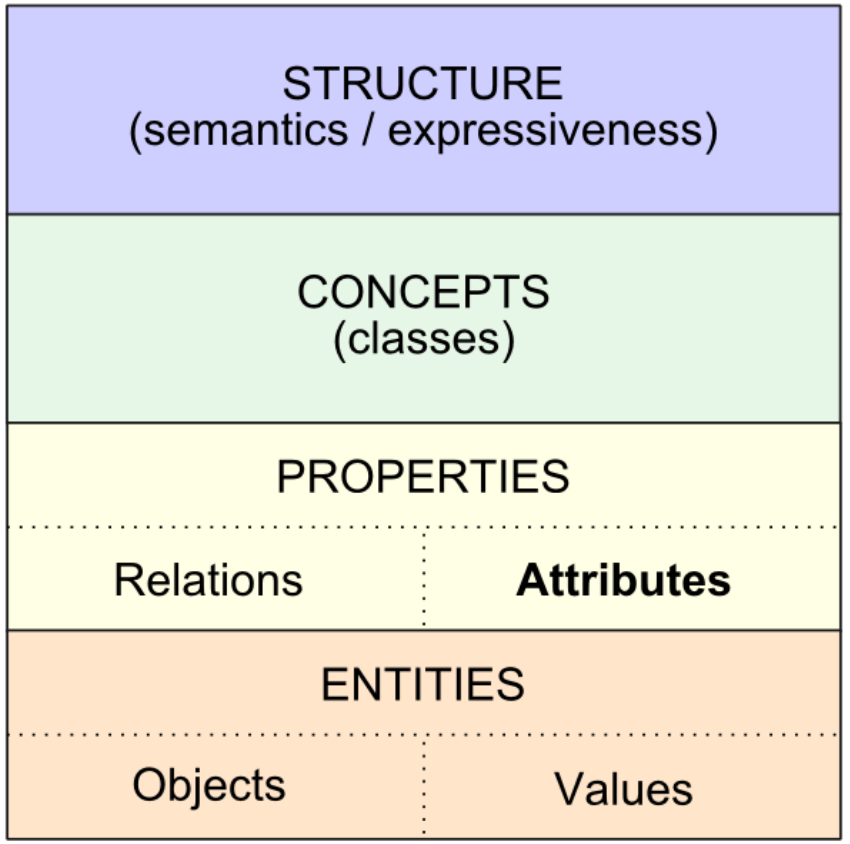
- Interdisciplinary discussions!
- Busse J., Humm B., Lubbert C., Moelter F., Reibold A., Rewald M., Schlüter V., Seiler B., Tegtmeier E., Zeh T. **Actually, What Does "Ontology" Mean**? A Term Coined by Philosophy in the Light of Different Scientific Disciplines / Journal of Computing and Information Technology. 23(29), 2015.
  - Participants:
    - Philosopher
    - Psychologist
    - Information scientist

Philosophy	Computer Science
Substance	Class, entity, concept
Accidents	Attribute, property
Fact	Fact, statement, proposition, relation
Ontology	Base-ontology, foundational ontology

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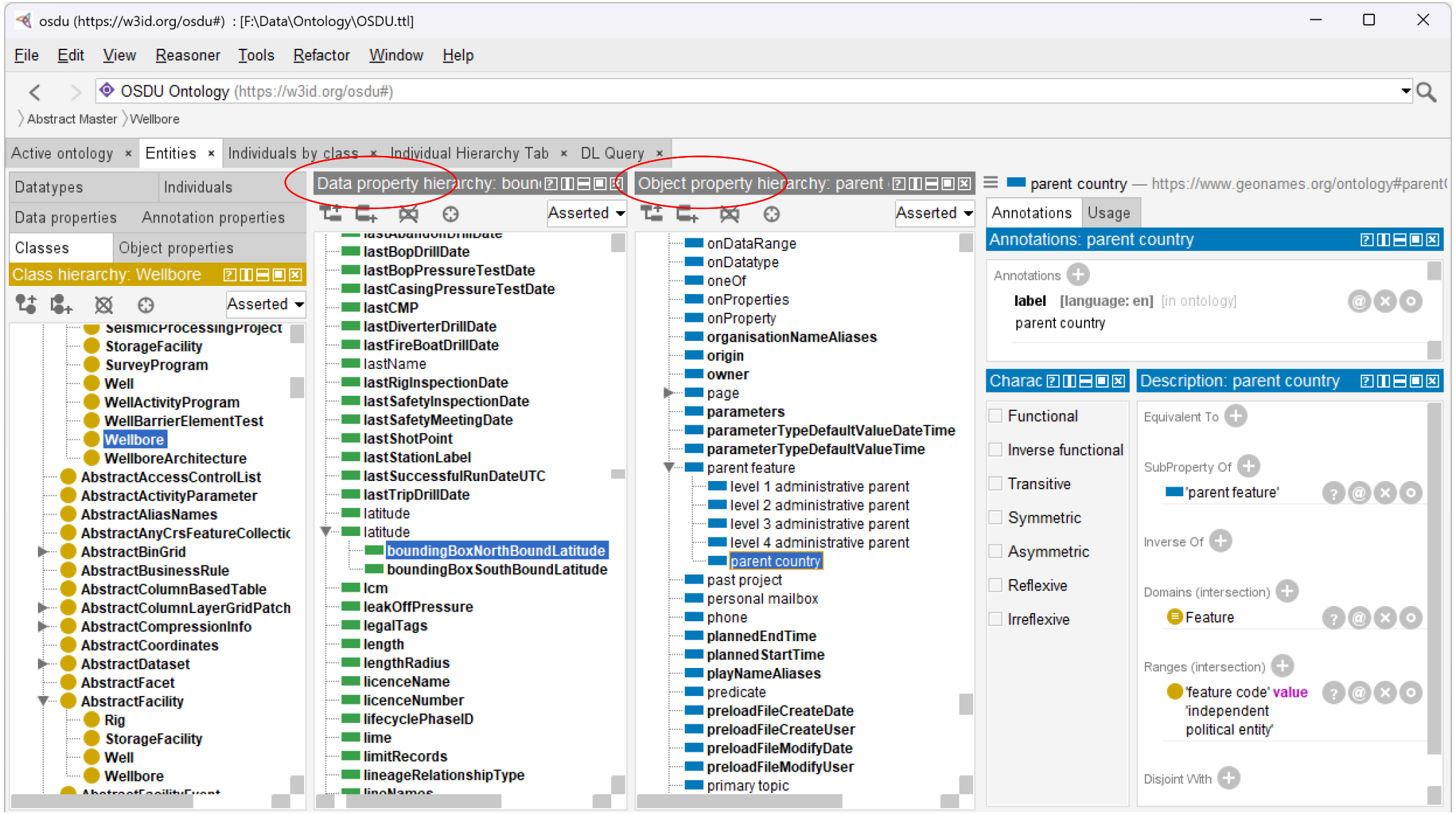
Main question: attribute or relationship?

- An attribute as a value object and as a reference
- Depends on:
  - Abstraction level
  - Level of specification details
  - Data access layer
- Differences In the terminology of ontological engineering
  - A very common use of the term “Property”
  - Attribute does not have Identifier, instead it is stored directly as some datatyped variable inside entity
    - Conceptual and Practical Distinctions in the Attributes Ontology (http://www.mkbergman.com/1842/conceptual-and-practical-distinctions-in-the-attributes-ontology/)
  - Example – Protégé (see next slide)



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Data and object properties in Protégé



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Interesting case of standardization of terminology

- ISO 5127:2017 Information and documentation — Foundation and vocabulary (http://www.iso.org/obp/ui#iso:std:iso:5127:ed-2:v1:en)

3.1.1.01 – object
anything perceivable or conceivable
Note 1 to entry: *Objects* may be material (e.g. an engine, a sheet of paper, a diamond), immaterial (e.g. conversion ratio, a project plan) or imagined (e.g. a unicorn).
[SOURCE:ISO 1087-1:2000, definition 3.1.1]
Note 2 to entry: See also *"entity"* (3.1.13.27); and ISO/IEC 27000:2016, definition 2.55.

3.1.1.60 – material object
physical object
object (3.1.1.01) which has physical extension in space and time
Note 1 to entry: See also *"naturafact"* (3.2.1.01), *"artefact (1) <man-made object>* (3.2.1.02); ISO 21127:2014, classes *"E 18"* and *"E 19"*.

3.1.1.03 – property
distinguishing feature of a *material object* (3.1.1.60)
[SOURCE:http://iso.i-term.dk/login.php, modified]
Note 1 to entry: See also ISO 21127:2014, definition 3.14.

3.1.1.02 – concept
unit of *knowledge* (3.1.1.17) created by a unique combination of *characteristics* (3.1.1.04)
Note 1 to entry: *Concepts* are not necessarily bound to particular languages. They are, however, influenced by the social or cultural background, which often leads to different categorizations.
[SOURCE:ISO 1087-1:2000, definition 3.2.10]
Note 2 to entry: *Concepts* represent *objects* (3.1.1.01).
Note 3 to entry: See also ISO 25964-1:2011, definition 2.11.
Note 4 to entry: See also *"class"* (3.8.5.03), *"conceptual object"* (3.1.1.61), *"mentefact"* (3.2.1.03).

3.1.1.61 – conceptual object
object (3.1.1.01) which is the product of the mental activity of applying abstraction in human mind activities
Note 1 to entry: See also ISO 21127:2014, classes *"E 28"* and *"E 65"*.

3.1.1.04 – characteristic
abstraction of a *property* (3.1.1.03) of an *object* (3.1.1.01) or of a *set* (3.1.1.09) of objects
Note 1 to entry: Characteristics are used for describing *concepts* (3.1.1.02).
[SOURCE:ISO 1087-1:2000, definition 3.2.4]

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Some comparisons of terminology differences

UMBEL/AO Terminology	Terminology Used Elsewhere	
Entity type	concept kind set	collection type class
Entity	object instance exemplar element	member record individual dependent variable
Attribute	property predicate relationship feature facet	dimension characteristic field header independent variables

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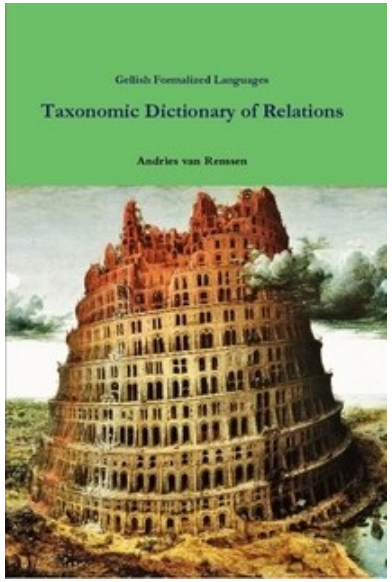
Conceiving and understanding: concepts and meanings

- Seiler T.B. **Begreifen und Verstehen**: ein Buch über Begriffe und Bedeutungen. Darmstädter Schriften zur allgemeinen Wissenschaft. Verlag Allgemeine Wissenschaft, Mühltal, 2001.
  - Seiler discusses a great variety of concept theories in philosophy and psychology and concludes with his own theory which extends the concept understanding of Piaget's structure-genetic approach
  - In his theory, Seiler describes concepts as cognitive structures whose development in human mind is constructive and adaptive
  - Twelve aspects!
    - They are briefly described in the following twelve subsections and used to review the adequacy of the mathematizations of FCA
    - In each subsection, the first paragraph concisely summarizes Seiler's understanding of the corresponding aspect; then related notions and relationships from FCA are discussed and partly concretized by at least one example. The connections between Seiler's concept theory and FCA which come apparent in this way are far from being exhaustive. But they already show an astonishing multitude of correspondences between both theories which may be taken as arguments for the adequacy of the discussed mathematizations

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Controlled language and semantic relations

- Andries van Renssen A. Taxonomic Dictionary of Relations, 2016 (http://www.lulu.com/shop/andries-van-renssen/taxonomic-dictionary-of-relations/paperback/product-22534783.html)
  - This is a taxonomic dictionary, which means that the defined concepts are arranged in a strict subtype-supertype hierarchy, also called a *taxonomy*
  - The dictionary defines kinds of relations that are usually not found in conventional dictionaries.
    - A number of the kinds of relations are also defined in various ISO and non-ISO standards
  - More than **5000** relations!



- Implementation in Gellish:
  - Gellish Syntax and Contextual Facts (http://github.com/AndriesSHP/Gellish/blob/master/Data/Gellish%20Syntax%20and%20Contextual%20Facts%20-%20Oct2017.pdf)

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Main classes of relations

- Entities are linked together in “relations”,
  at the level of both instances (individuals) and types (classes)

- Three groups of relations are distinguished
  Examples from BFO reference

1. Instance-level relations:
   - Your heart (instance-level) **continuant_part_of** your body **at** t
     - **Boldface** indicates a label for an **instance-level relation**!
   - Your heart beating (instance-level) **has_participant** your heart

2. Type-level relations:
   - human heart ***continuant_part_of*** human body
   - human heart beating process ***has_occurrent_part*** beat profile

3. Instance-type relations:
   - John’s heart *instance_of* human heart

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Mereology and holarchy

- Parthood relations (or part-whole relationships)

- The axioms of **Minimal Extensional Mereology** (Simons P. Parts: A study in ontology. Clarendon Press, 1987):
  - Antisymmetry: If x part of y, then if y part of x, then $x = y$.
  - Transitivity: If x part of y, and y part_of z, then x **part_of** z.
  - Weak Supplementation: If x **part_of** y & not $x = y$, then there is some z such that (z **part_of** y and z has no part in common with x).
  - Unique Product: If x and y have a part in common, then there is some unique z such that for all w (w is part of z if and only if (w is part of x and w is part of y)).

  - Additional relation "proper_part_of" is defined as x **proper_part_of** $y \equiv x$ **part_of** y & not $x = y$

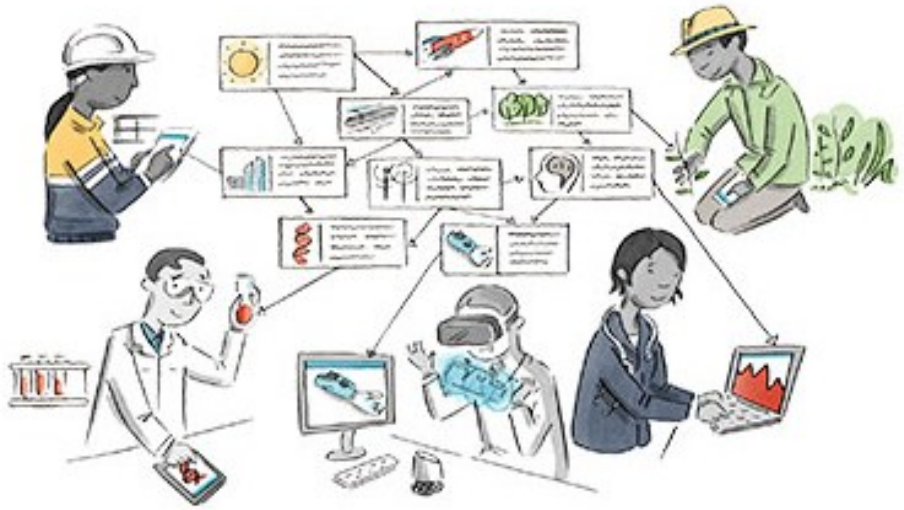
- For us more important: Barton A., Toyoshima F., Vieu L., Fabry P., Ethier J.-F. The **Mereological Structure of Informational Entities**. Formal Ontology in Information Systems // 11th International Conference (FOIS 2020), pp.201-215, 2020 (http://hal.science/hal-03041523/document)

- **Holarchy** is a system of organization where entities, called holons, are both self-contained wholes and parts of larger wholes

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Why we need ontology?

- We need to talk formally about domains:
  - Human ↔ Human
  - Computer ↔ Computer
  - Human ↔ Computer
- Applications:
  - Knowledge representation
  - Knowledge exchange and merging
  - Knowledge bases construction
  - Logic reasoning
  - Explanation of decisions
  - Knowledge gaps detection
- *Statistical Machine Learning* is a tool to “guessing” an answer
  - We need interpretation, explanation, harmonization, and tutoring!
- Ontology engineering is a very broad topic, with forays into many industrial sectors…



© 2021 https://blog.metaphacts.com/visual-ontology-modeling-for-domain-experts-and-business-users-with-metaphactory

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Natural language, modelling, and grounding

- Descriptions and representations (Notational complexity ↔ Conceptual complexity!)
  - "A thought once uttered is untrue"
    - Tyutchev F. Silentium. 1833 (Vladimir Nabokov's rhymed version)
  - "The map is not the territory"
    - Yudkowsky E. Map and Territory. Machine Intelligence Research Institute, 2018. 231 p.
  - "Essentially, all models are wrong, but some are useful"
    - Box G. Science and Statistics // Journal of the American Statistical Association, 71 (356), pp. 791-799.
  - "Practice is the sole criterion of truth"
    - Hu Fuming. Guangming Daily. 1978-05-11
    - cf. Marx K. Theses on Feuerbach. 1845 → Lenin V. Materialism and Empirio-criticism. 1908 → Mao Zedong. On Practice, 1937.
      - Important qualification by Lenin: the criterion of practice "can never, in the nature of things, either confirm or refute any human idea completely"; it is sufficiently "indefinite" to prevent knowledge from becoming absolute, yet "sufficiently definite" to counter idealism and agnosticism
  - "Action cannot remove ignorance; but knowledge disperses it as light disperses darkness"
    - Shankara A. Atma Bodha (Advaita Vedanta treatise). Early-to-mid 8th century
- Physical/Chemical/… ontology and technology domain
  - Terminology systems and their merging
- Cyber-Physical Systems (CPSs) and Internet of Things (IoT)
  - Robotics

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Temporal and geospatial relations

- Precedence relation and events
  - Temporal logics and time axis
- 4D life cycle of objects
  - New objects
  - New attributes
  - Object ↔ Attribute transformation
    - Level of details!
- Consequences:
  - *Static* vs *dynamic* data structures and **VERSIONING** of object-attribute data
  - 4D context is sufficient for temporal and geospatial relations – see below
- It's a very deep "rabbit hole"…
  - See Imaguire G. On the Ontology of Relations. Disputatio, 4(34), 2012, pp. 689-711

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Natural languages in the land of LLMs

- Problem: context, discourse, narrative
  - Remember “literary criticism” as a humans’ activity!

- Natural language as:
  - User interface (UI)
  - Universal programming language

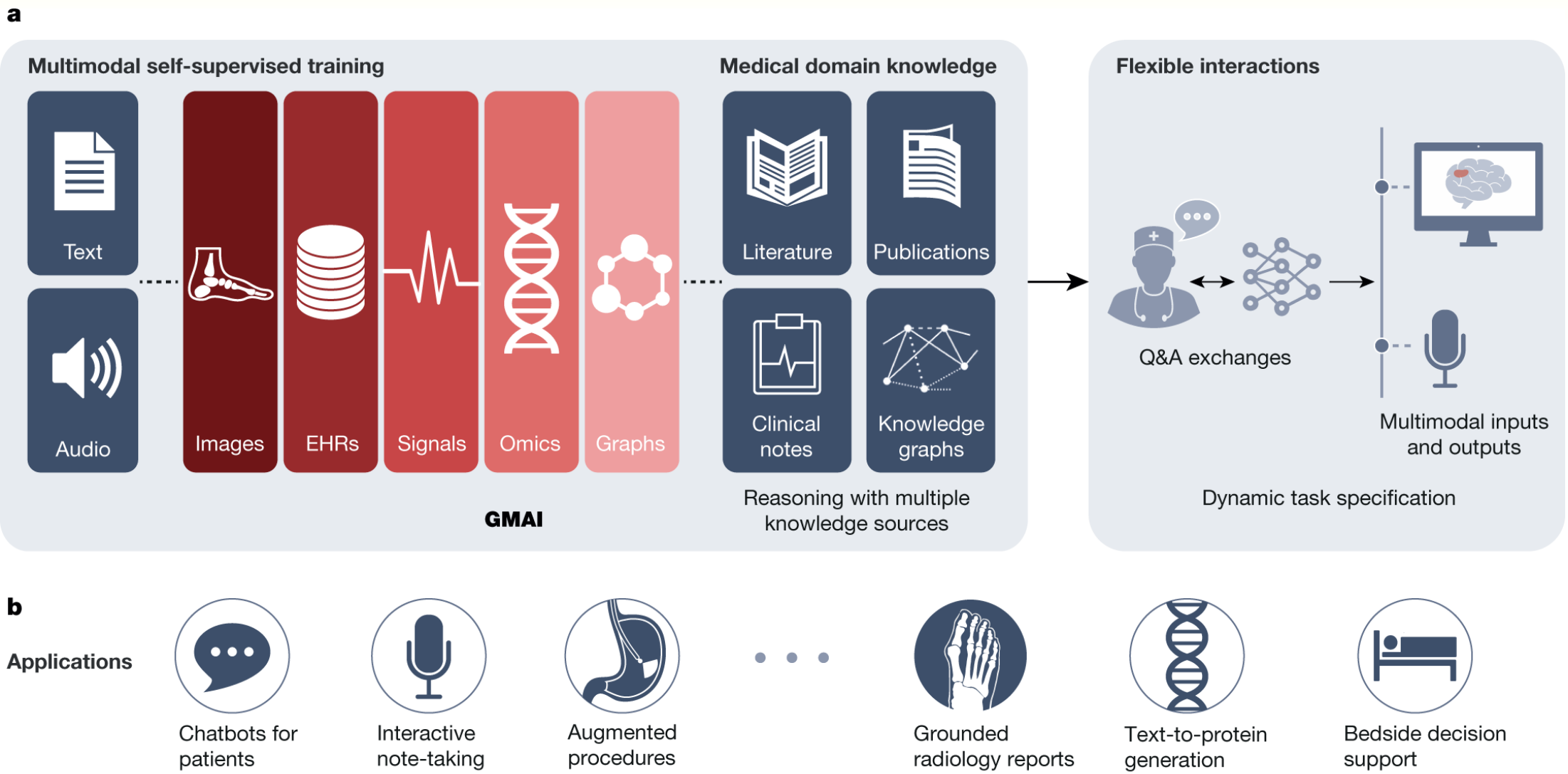
  - [old] Manaris B. Natural Language Processing: A Human-Computer Interaction Perspective. Advances in Computers, 47, 1998, pp. 1-66.
  - Hall E. Conversational Design. 2nd ed. 2026 (http://www.mulebooks.com/conversational-design)

  - Pirozelli P., Câmara I. Natural Language at a Crossroads: Formal and Probabilistic Approaches in Philosophy And Computer Science. Manuscrito, 45(2), 2022 (http://www.scielo.br/j/man/a/8CcCj7HdRDXmmyZXpPZFp7t/)

  - Vulich R. Frege’s Sharpness Requirement and Natural Language (http://cah.ucf.edu/fpr/article/freges-sharpness-requirement-and-natural-language/)

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Example of “Generalist medical AI”



(C) Foundation models for generalist medical artificial intelligence. 2023 (https://www.nature.com/articles/s41586-023-05881-4)

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Medical knowledge formalization – thesauri and ontologies (1)

- Fundamentals of Interoperability:
  - **HL7** Reference Information Model (ANSI/HL7 V3 RIM) (http://www.hl7.org/implement/standards/rim.cfm)
    - Arden Syntax (Health Level Seven Arden Syntax for Medical Logic Systems, Version 3) (http://ardensyntax.com)

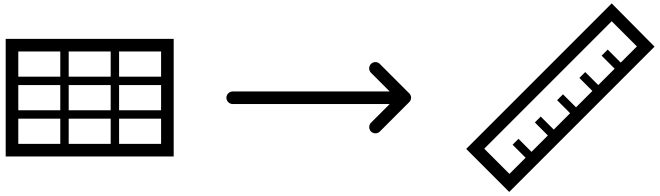
- Core ontologies and thesauri:
  - **Unified Medical Language System** (UMLS®) (http://www.nlm.nih.gov/research/umls/)
    - **Metathesaurus**: Terms and codes from CPT, ICD-10-CM, LOINC, MeSH, RxNorm, SNOMED CT, etc. (hierarchies, definitions, and other relationships and attributes)
    - **Semantic Network**: Broad categories (semantic types) and their relationships (semantic relations)
    - **SPECIALIST Lexicon and Lexical Tools**: A large syntactic lexicon of biomedical and general English and tools for normalizing strings, generating lexical variants, and creating indexes
  - **SNOMED International – SNOMED Clinical Terms** (SNOMED CT) (http://www.snomed.org)
    - The most comprehensive multilingual thesaurus of medical terminology
  - **Medical Subject Headings** (MeSH) (http://www.nlm.nih.gov/mesh/meshhome.html)
  - **Medical Dictionary for Regulatory Activities** (MedDRA) (http://www.meddra.org)
  - ...

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Medical knowledge formalization – thesauri and ontologies (2)

- Some open medical ontologies:
  - **International Classification of Diseases** (ICD) (http://www.who.int/classifications/classification-of-diseases)
    - + ICF, ICHI, …
    - The Disease Ontology (http://disease-ontology.org)
  - **RxNorm** (http://www.nlm.nih.gov/research/umls/rxnorm/)
    - Normalized drug names
    - RxNav is a browser for several drug information sources, including RxNorm, RxTerms and NDF-RT (http://mor.nlm.nih.gov/RxNav/)
  - **LOINC** (http://loinc.org)
    - Medical tests, measurements, and observations
      - [ru] Федеральный справочник лабораторных исследований (**ФСЛИ**). Справочник лабораторных тестов (http://nsi.rosminzdrav.ru/#!/refbook/1.2.643.5.1.13.13.11.1080)
      - [ru] Справочник лабораторных тестов и услуг (**ЛАТЕУС**) (http://spbmiac.ru/ehlektronnoe-zdravookhranenie/informacionno-spravochnye-sistemy/spravochnik-lateus/)
  - Hymenoptera **Anatomy Ontology** Portal (http://api.hymao.org/projects/32/public/site/hymglossary/home/)
- Some collections:
  - Open Biological and Biomedical Ontology (OBO) Foundry (http://obofoundry.org)

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THE CONCEPT OF “MEASUREMENT”

- ✓ Representational measurement theory
- ✓ Context of value in object-attribute data representation
- ✓ Scales
- ✓ ISO standards
- ✓ Ontologies of units of Measure

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Foundations of measurements

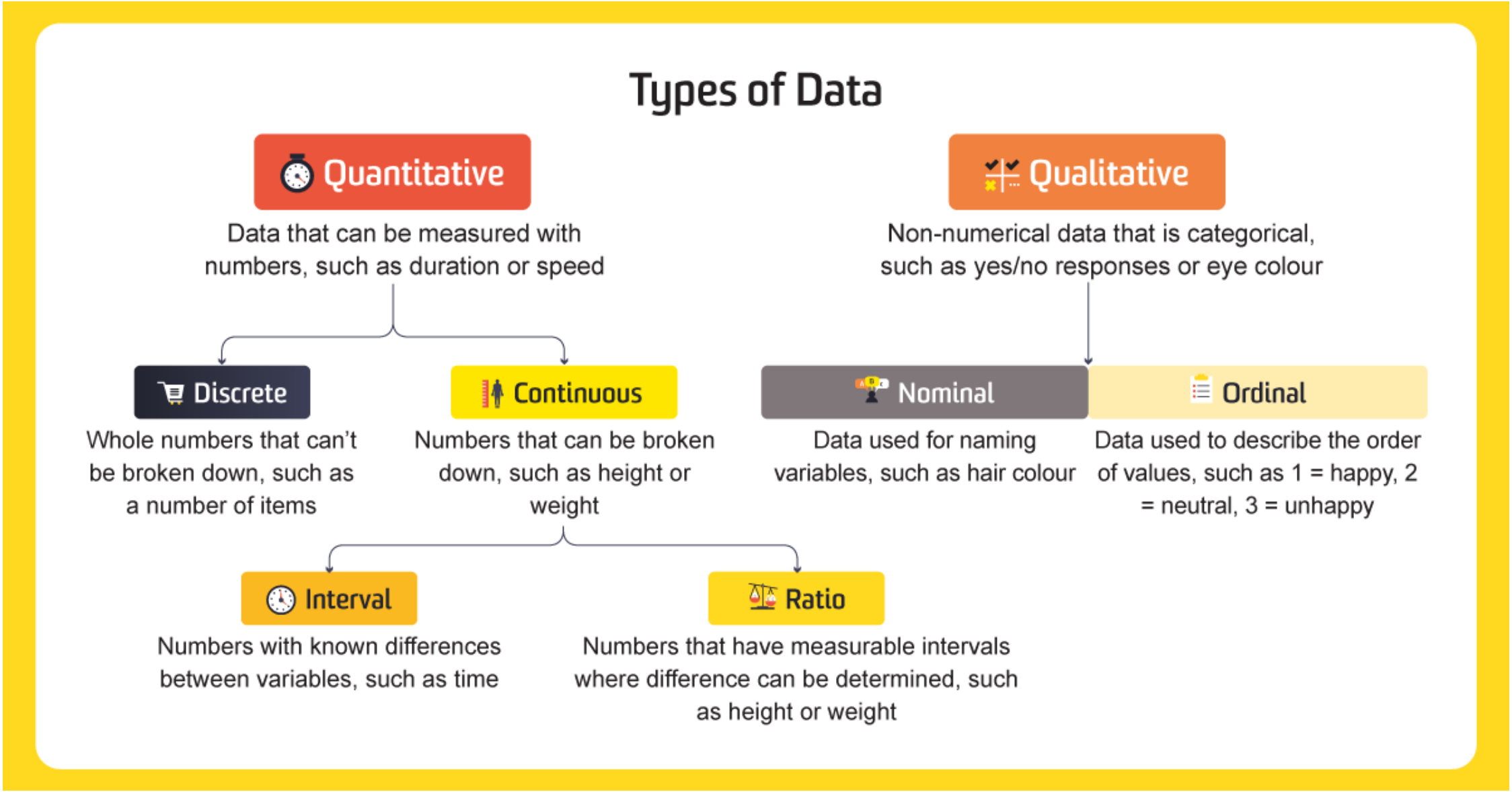
- "**Attribute value**" as a result of a measurement!
  - Each measurement has some **context** – a rich description of the circumstances of a measure
    - Repeated measures design – a research design that involves multiple measures of the same variable taken on the same or matched subjects either under different conditions or over two or more time periods → statistics
      - Accuracy is how close a given set of measurements (observations or readings) are to their true value
      - Precision is how close the measurements are to each other

- **Measurement theory** is the study of how numbers are assigned to objects and phenomena

- **Representational measurement theory** (RMT)
  - Michell J. Representational measurement theory: Is its number up? / Theory & Psychology, 31(1), 2021, pp. 3-23. (http://journals.sagepub.com/doi/10.1177/0959354320930817)
  - Benoit E. New scale classification within the representational theory of measurement / Journal of Physics: Conference Series, 459, 2013 (http://iopscience.iop.org/article/10.1088/1742-6596/459/1/012004/pdf)
  - Heilmann C. A New Interpretation of the Representational Theory of Measurement / Philosophy of Science, 82(5), 2015, pp.787-797 (http://philsci-archive.pitt.edu/11009/1/RTM-Heilmann-PSA.pdf)

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Data and scales



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Main kinds of measurement scales

- Types of data & the scales of measurement (http://studyonline.unsw.edu.au/blog/types-of-data)
  - Properties of Measurement
    - **Identity**: Identity refers to each value having a unique meaning
    - **Magnitude**: Magnitude means that the values have an ordered relationship to one another, so there is a specific order to the variables
    - **Equal intervals**: Equal intervals mean that data points along the scale are equal, so the difference between data points one and two will be the same as the difference between data points five and six
    - **A minimum value of zero**: A minimum value of zero means the scale has a true zero point
      - Degrees, for example, can fall below zero and still have meaning, but if you weigh nothing, you don't exist

- Four kinds of measurement scales:

Scale	Basis empirical operation	Mathematical group structure	Admissible transformation
Nominal	determination of equality	Permutation group	$y = f(x)$, f is a bijection
Ordinal	+ determination of greater or less	Isotonic group	$y = f(x)$, f is a monotonic increasing function
Interval	+ determination of equality of intervals	General linear group	$y = ax + b,\ a > 0$
Ratio	+ determination of equality of ratio	Similarity group	$y = ax,\ a > 0$

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Four main kinds of scales

The Four Scales of Measurement



Nominal Scale

Used for naming variables in no particular order

For example, eye colour



Ordinal Scale

Used for variables in ranked order, but the difference between is not determined

For example, #1 happy, #2 neutral, #3 unhappy



Interval Scale

Used for numerical variables with known equal intervals of the same distance

For example, time



Ratio Scale

Used for variables on a scale that have measurable intervals

For example, weight

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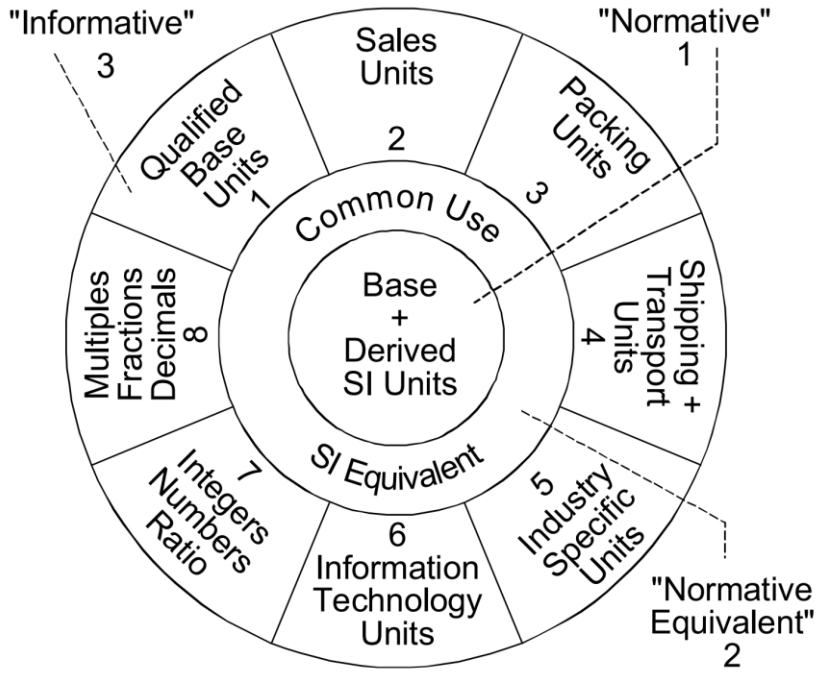
Main ISO standard for quantities and units

- ISO **80000**-1:2022 Quantities and units — Part 1: **General** (http://www.iso.org/standard/76921.html)
- ISO 80000-2:2019 Quantities and units — Part 2: **Mathematics** (http://www.iso.org/standard/64973.html)
- ISO 80000-3:2019 Quantities and units — Part 3: **Space and time** (http://www.iso.org/standard/64974.html)
- ISO 80000-4:2019 Quantities and units — Part 4: **Mechanics** (http://www.iso.org/standard/64975.html)
- …
- ISO 80000-12:2019 Quantities and units — Part 12: **Condensed matter physics** (http://www.iso.org/standard/63480.html)
- IEC 80000-13:2008 Quantities and units — Part 13: **Information science and technology** (http://www.iso.org/standard/31898.html)
  - (+ Amd 1:2025) for most of the published parts

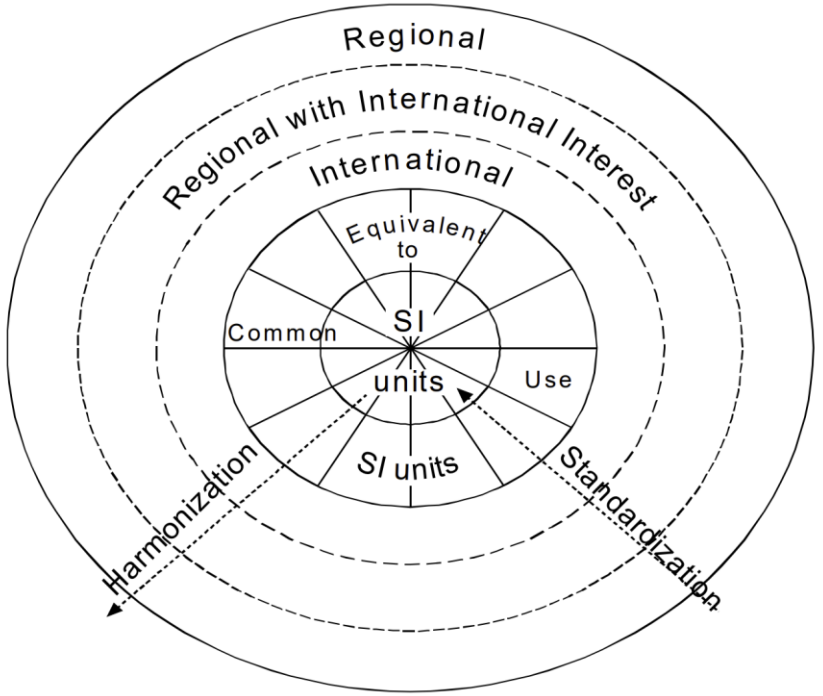
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UN Centre for Trade Facilitation and Electronic Business

- UN/CEFACT Common Code for units of measure – Recommendation 20 (http://unece.org/trade/documents/2021/06/uncefact-rec20-0)
  - Actual revision 17, 2021 (1640 code elements in normative Annex I)
  - Unit of measure: Particular quantity, defined and adopted by convention, with which other quantities of the same kind are compared in order to express their magnitudes relative to that quantity
  - Units of measure schema components:



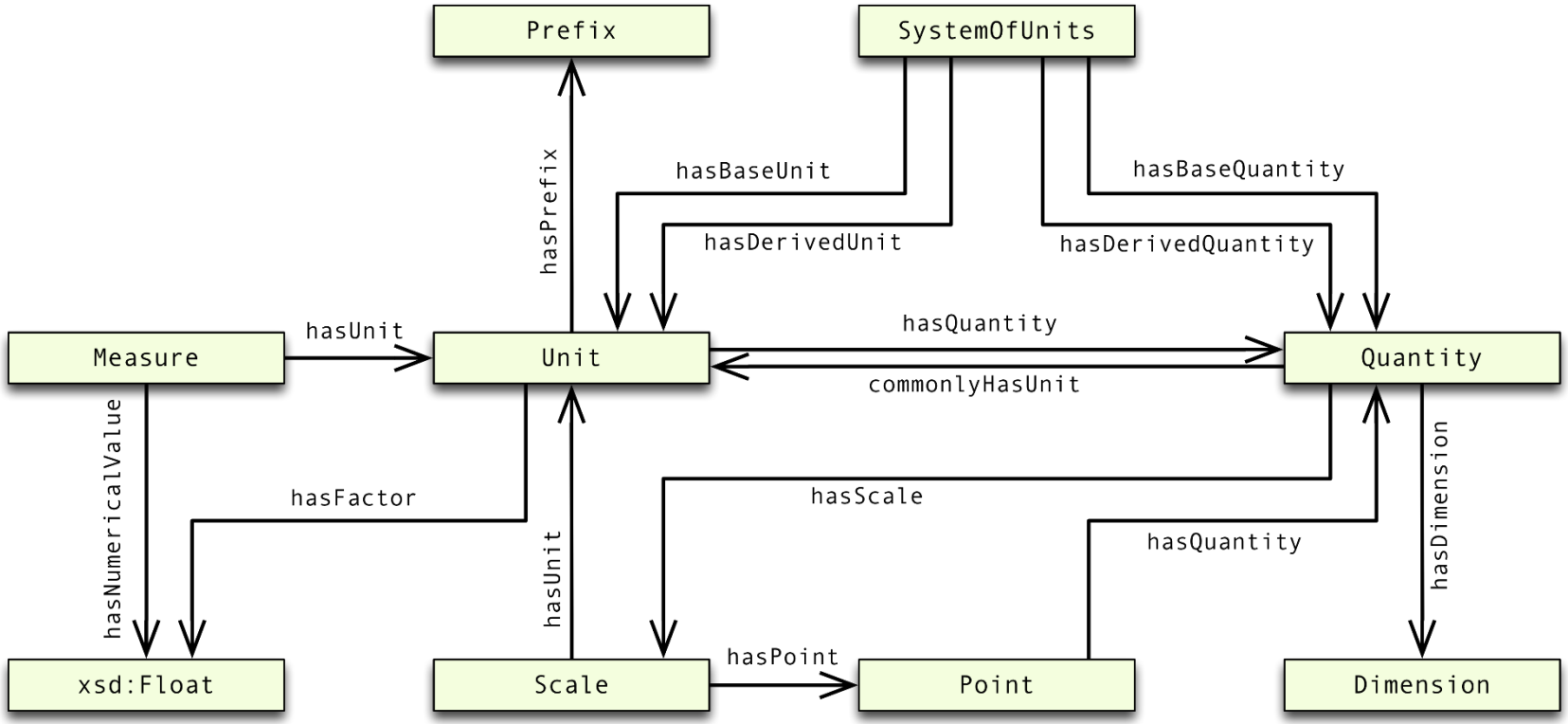
  - Units of measure "onion skin" levels of standardization:



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Ontologies of units of Measure – OM

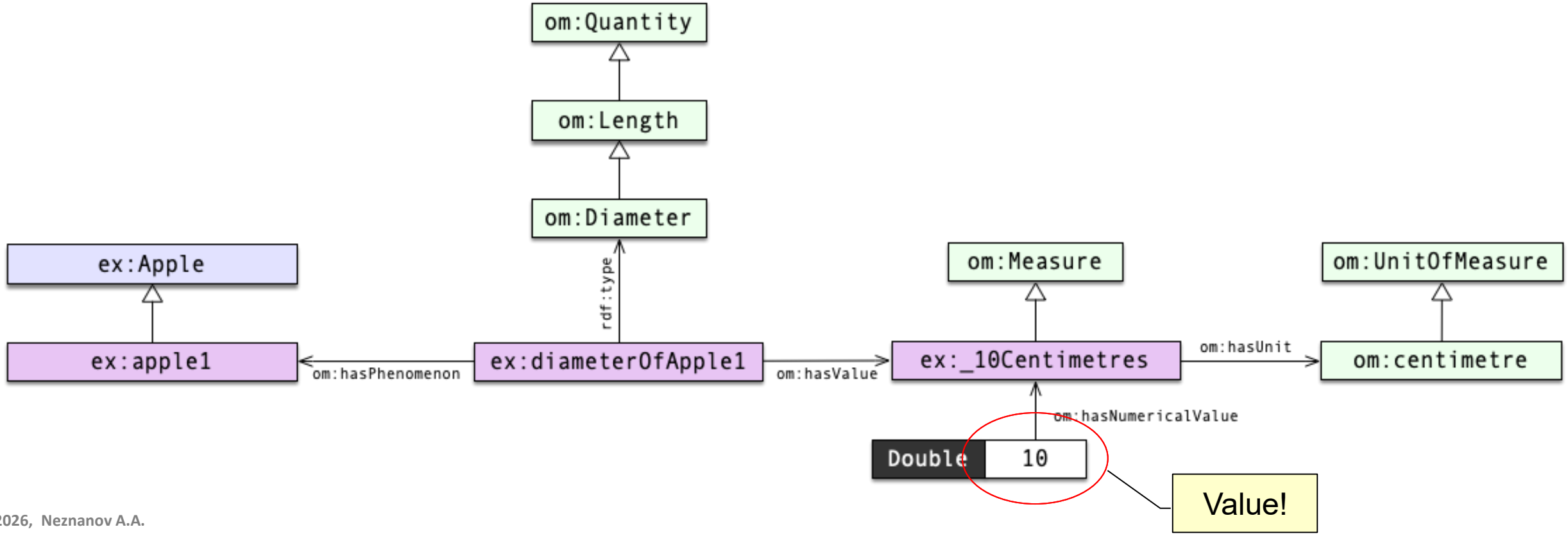
- [GitHub] HajoRijgersberg/OM (http://github.com/HajoRijgersberg/OM)
  - Current version – 2.0.55
  - Classes:814, individuals: 2,246, properties: 34



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Example of measurement (OM): the diameter of an apple

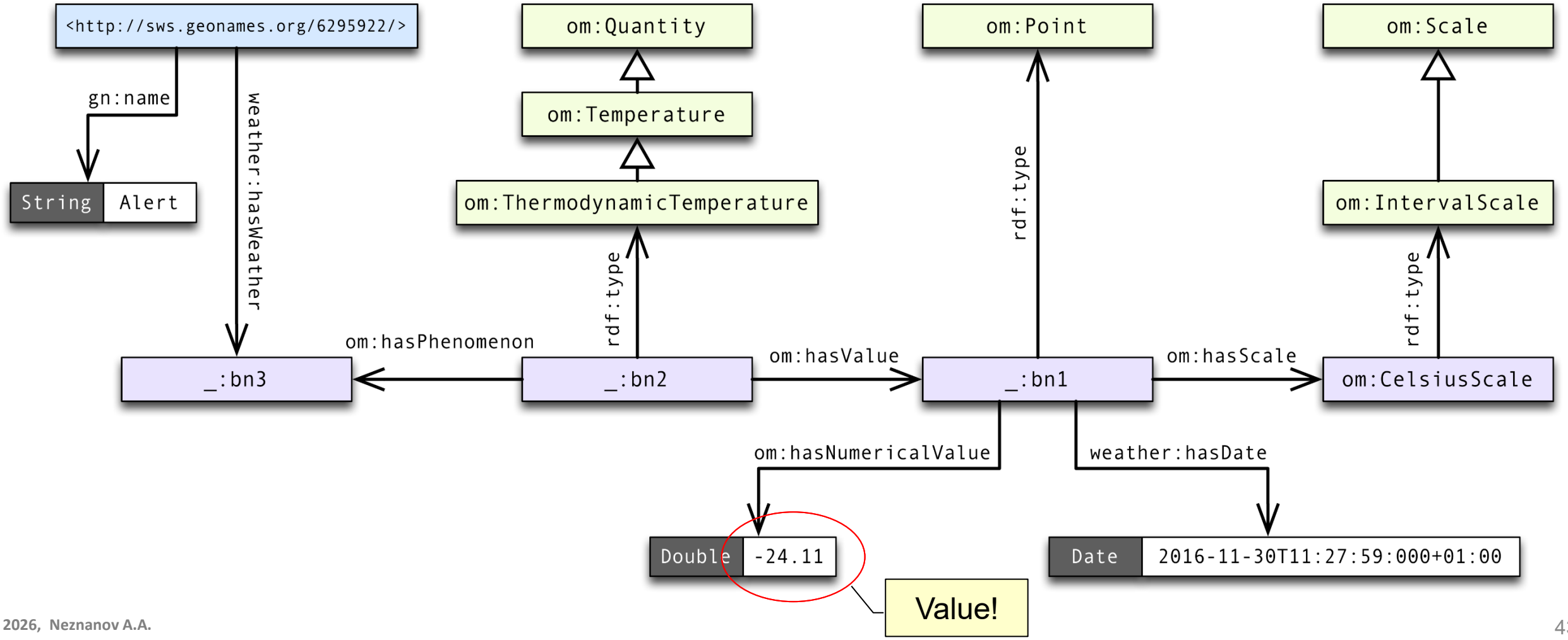
- Representation of the “concrete act of measurement” with **references** (and grounding!) to:
  - Ontology of units of Measure («centimeter», «diameter»)
  - Ontology of the subject area («apple»)



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Example of measurement (OM): temperature

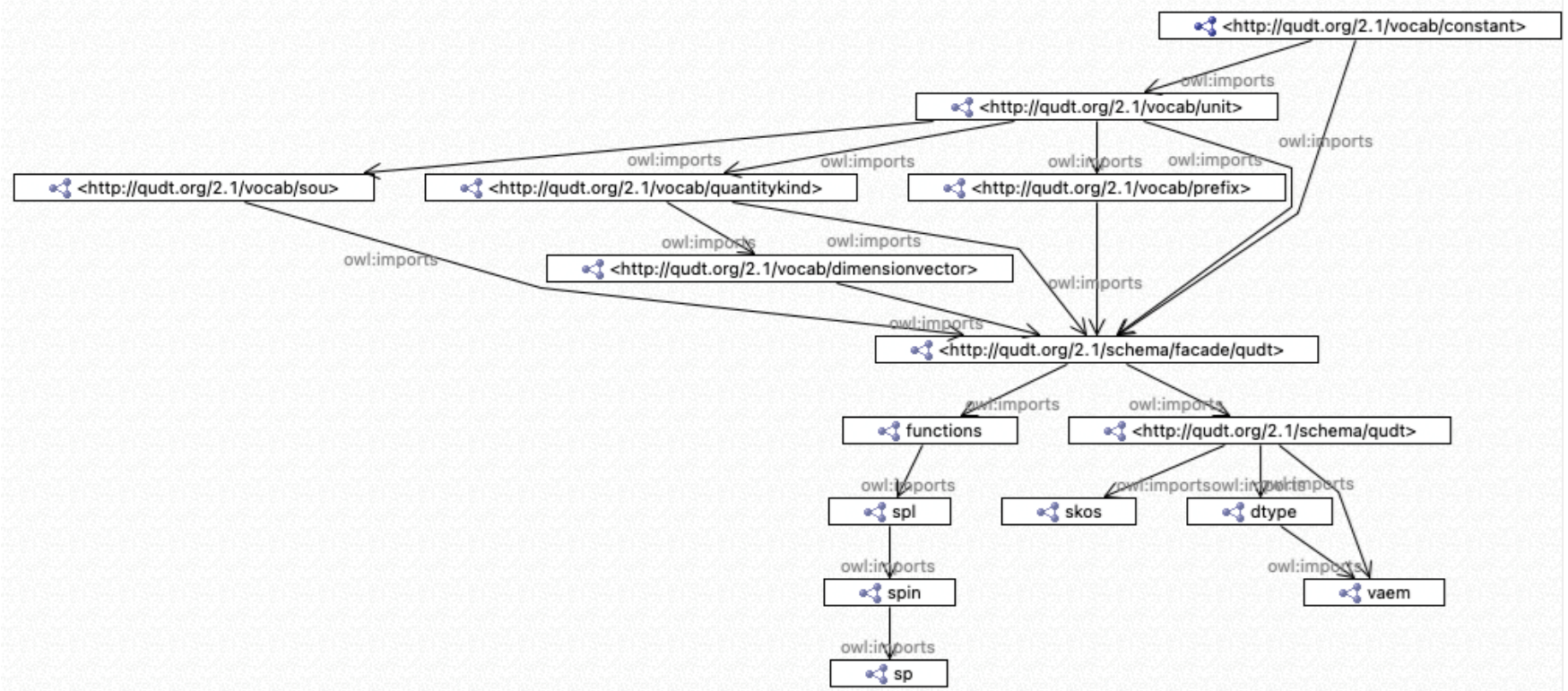
- The temperature at Alert, Canada on November 30th, 2016 at 11:28 AM



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Ontologies of units of Measure – QUDT

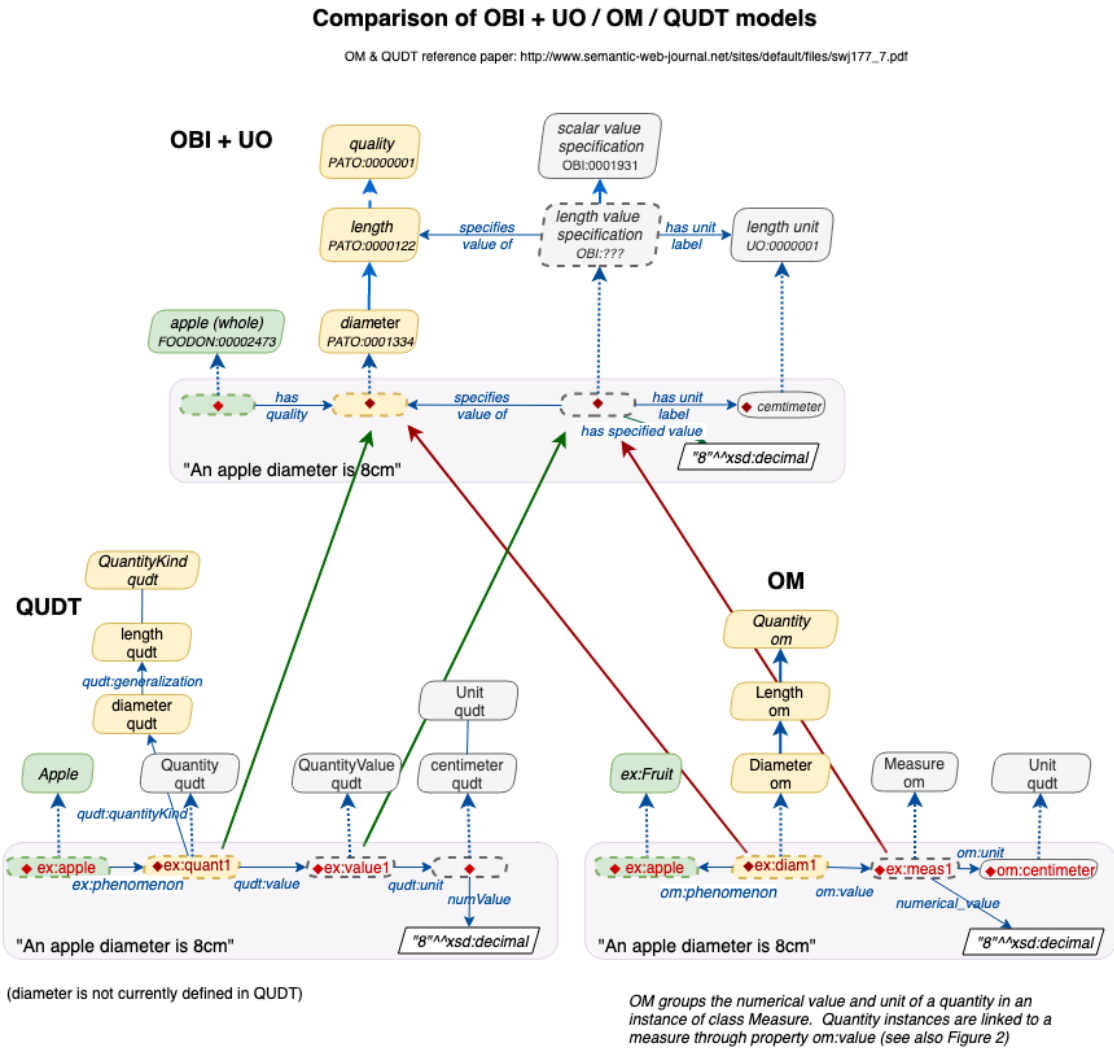
- QUDT - **Q**uantities, **U**nits, **D**imensions and Data **T**ypes Ontologies
  - QUDT Units Version 2.1 (http://www.qudt.org/doc/DOC_VOCAB-UNITS.html)
    - Base ontology – classes: 133, individuals: 84, properties: 248
    - + 2509 instances



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Ontologies of units of Measure – approaches and comparing

- LinkML – How to model quantities and measurements (http://linkml.io/linkml/howtos/model-measurements.html)



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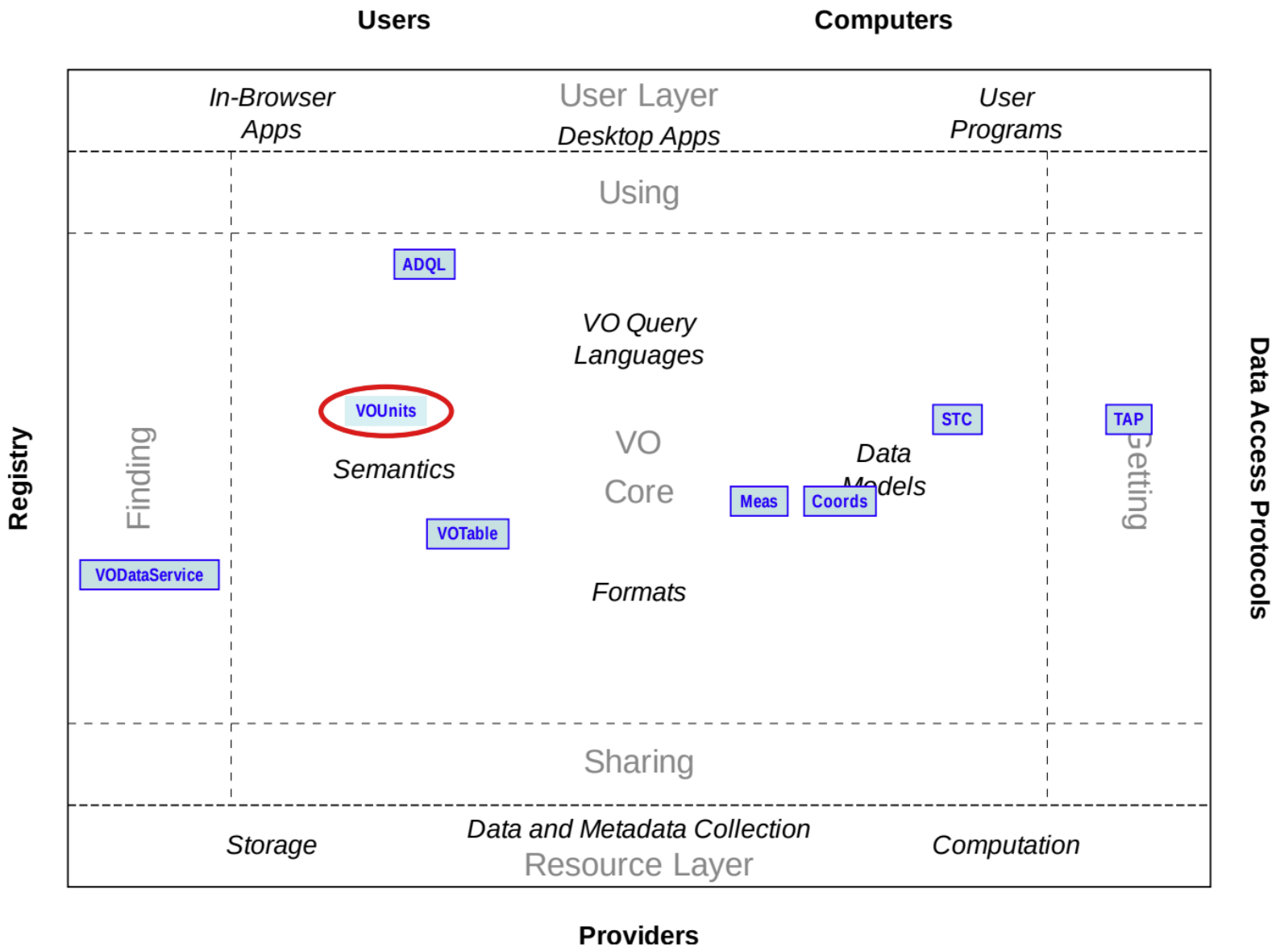
Ontologies of units of Measure – Specific symbols

- Unit designations from International Virtual Observatory Alliance
  - **Units in the VO** (http://www.ivoa.net/documents/VOUnits/index.html)
    - Version 1.1, IVOA Recommendation 2023-12-15
    - Very convenient standard, agreed with BIPM, ISO/IEC, IAU
  - Example:

m (metre)	**g** (gram)	**J** (joule)	**Wb** (weber)
s (second)	**rad** (radian)	**W** (watt)	**T** (tesla)
A (ampere)	**sr** (steradian)	**C** (coulomb)	**H** (henry)
K (kelvin)	**Hz** (hertz)	**V** (volt)	**lm** (lumen)
mol (mole)	**N** (newton)	**S** (siemens)	**lx** (lux)
cd (candela)	**Pa** (pascal)	**F** (farad)	**Ohm** (ohm)

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VOUnits as a core building block in the VO:

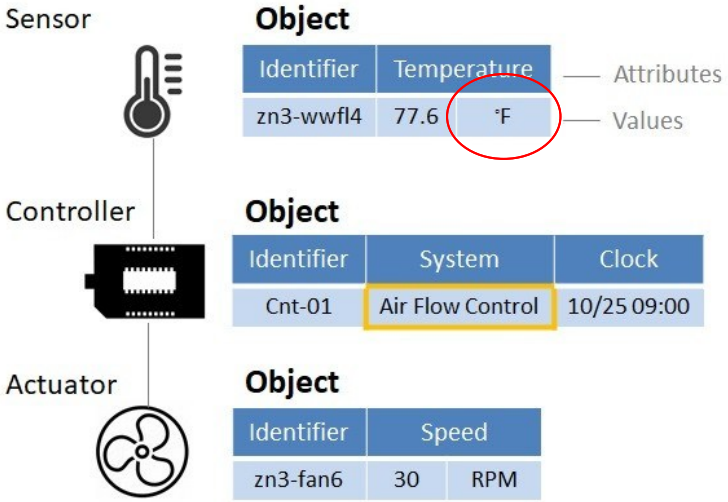


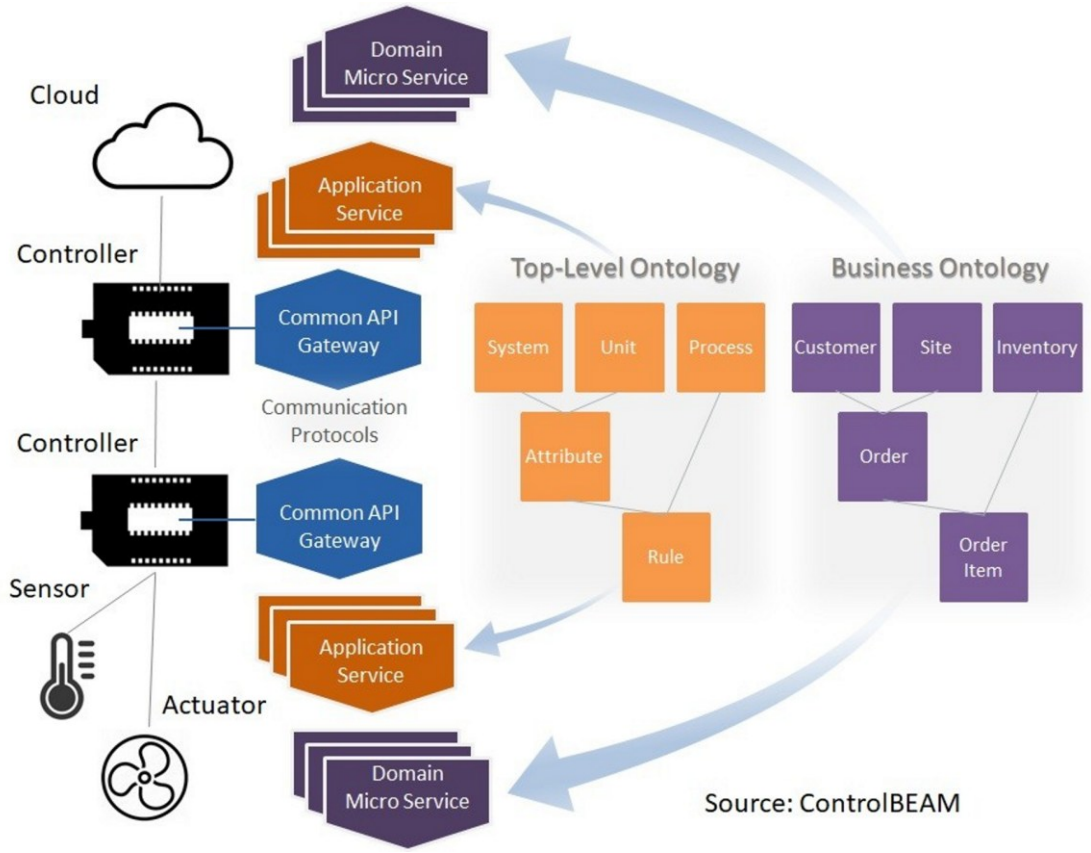
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More general semantic interoperability

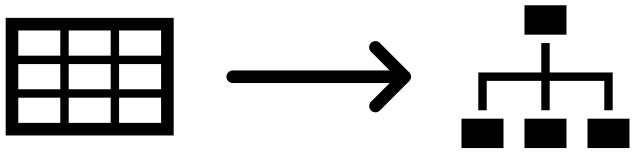
- Cross-industry semantic interoperability. Part 4: The intersection of business and device ontologies. 2017 (http://embeddedcomputing.com/technology/iot/cross-industry-semantic-interoperability-part-four-the-intersection-of-business-and-device-ontologies)

System Attribute	
System	Attribute
Air Flow Control	Temperature (of Sensor)
Air Flow Control	Clock (of Controller)
Air Flow Control	Speed (of Actuator)
Refrigeration	Temperature (of Sensor)
Refrigeration	Switch (of Controller)
Refrigeration	Open Level (of Valve)





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FORMALIZATION OF CONCEPTS

- ✓ Partially Ordered Sets (posets)
- ✓ Formal Concept Analysis
  - ✓ Objects and attributes
  - ✓ Formal context and formal concept
  - ✓ Extension → extent, intension → intent
  - ✓ Formal concept lattice and line diagrams
- ✓ Implications in context and association rules

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Formal Concept Analysis

- Historically, **Formal Concept Analysis** (FCA) is a branch of applied lattice theory that appeared in 1980's
  - Barbut M., Monjardet B. Ordre et classification: algèbre et combinatoire. Hachette, 1970.
  - Davey B.A., Priestley H.A. Introduction to lattices and order. Cambridge University Press, 1990.
  - **Ganter B., Wille R. Formal concept analysis. Springer, 1999.**
  - Kuznetsov S.O. On stability of a formal concept / Annals of Mathematics and Artificial Intelligence, 49(1–4), 2007, pp. 101-115.
  - Belohlávek R. Relational data, formal concept analysis, and graded attributes / Handbook of research on fuzzy information processing in databases. IGI Global, 2008, pp. 462-489.
  - Poelmans J., Ignatov D.I., Kuznetsov S.O., Dedene G. Formal concept analysis in knowledge processing: a survey on applications / Expert Systems with Applications 40(16), 2013, pp. 6538-6560.
  - Poelmans J., Ignatov D.I., Kuznetsov S., Dedene G. Fuzzy and rough formal concept analysis: a survey / International Journal of General Systems, 43(2), 2014, pp. 105-134.
  - Codocedo V., Napoli A. Formal concept analysis and information retrieval – a survey // Proceedings of the 13th international conference on formal concept analysis (ICFCA). LNCS, vol 9113, Springer, 2015, pp. 61-77.
  - **Ganter B., Obiedkov S.A. Conceptual exploration. Springer, 2016.**
  - Belohlavek R., Trnecka M. Basic level of concepts in formal concept analysis: formalization and utilization / International Journal of General Systems, 49(7), 2020, pp. 689-706.
  - Ferré S., Huchard M., Kaytoue M., Kuznetsov S.O., Napoli A. Formal Concept Analysis: From Knowledge Discovery to Knowledge Processing / A Guided Tour of Artificial Intelligence Research. Volume II: AI Algorithms. Springer, 2020.
  - Roscoe S., Khatri M., Voshall A., Batra S., Kaur S., Deogun J. Formal Concept Analysis Applications in Bioinformatics / ACM Computing Surveys, 55(8), 2022, pp. 1-40.

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Formal context

- Formal context – triple $\langle G, M, I \rangle$, where:
  - G – a set of objects
  - M – a set of attributes
  - $I \subseteq G \times M$ – binary relation between G and M

- The statement $(g, m) \in I$ (also denoted by gIm) is interpreted as "object g **has attribute** m"
  - Remember algebraic definition of "graph"!

- Cross-table representation:

I	y_1	y_2	y_3	y_4
x_1	×	×	×	×
x_2	×		×	×
x_3		×	×	×
x_4		×	×	×
x_5	×			

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Formal concept

- With $A \subseteq G$ and $B \subseteq M$ in formal context $\langle G, M, I \rangle$:
- Operator $(\cdot)'$ define a **Galois connection** between the powersets $\left(2^G, \subseteq\right)$ and $\left(2^M, \subseteq\right)$:
  - $A' = \{m \in M | \forall g \in A : gIm\}$
  - $B' = \{g \in G | \forall m \in B : gIm\}$
    - Three properties: (1) $Z_1 \subseteq Z_2 \Rightarrow Z_1' \supseteq Z_2'$, (2) $Z \subseteq Z''$, (3) $Z''' = Z'$
- Formal concept – pair (A, B), such that $A' = B \ \& \ B' = A$ (*duality*!)
  - Set A is called the "**extent**" (to the *set of instances* of the concept)
  - Set B is called the "**intent**" (corresponds to the *description* of the concept)

- Concepts are partially ordered by $(A_1, B_1) \leq (A_2, B_2) \Leftrightarrow A_1 \subseteq A_2 \ (\Leftrightarrow B_2 \subseteq B_1)$
- With respect to this partial order, the set of all formal concepts forms a complete lattice called the "**concept lattice**" of $\langle G, M, I \rangle$

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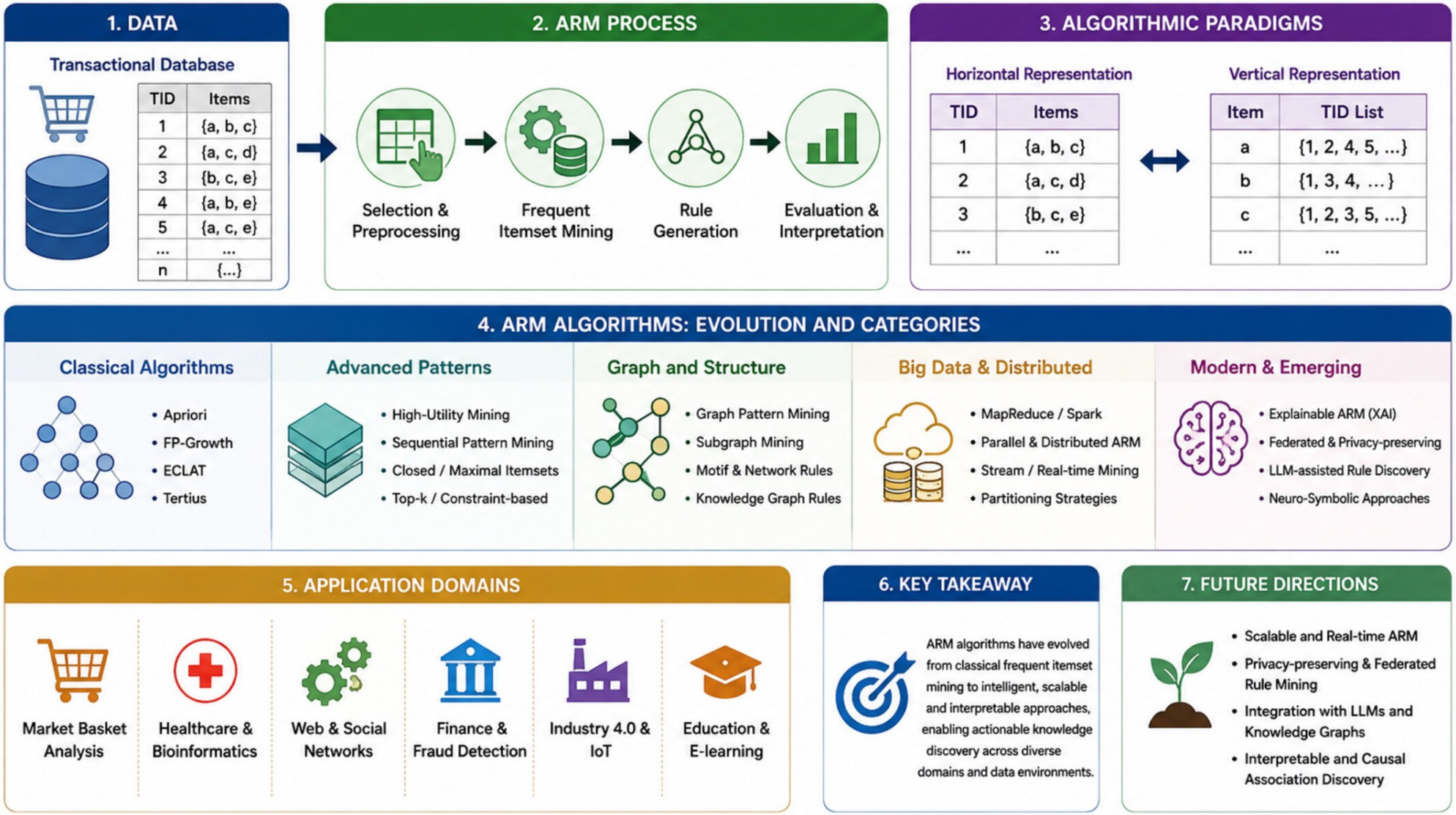
Implications and association rules

- **Attribute implication** $A \Rightarrow B$ holds if $A' \subseteq B'$
- **Association rule** $A \to B$ is valid if:
  - Its "**support**" $s = \frac{|(A \cup B)'|}{|G|} = \frac{|(A' \cap B')|}{|G|} \geq \sigma_s$, $\quad s = P(A \cup B) = \frac{freq(A,B)}{N}$
  - Its "**confidence**" $c = \frac{|(A \cup B)'|}{|A'|} = \frac{|(A' \cap B')|}{|A'|} \geq \sigma_c$, $\quad c = P(B|A) = \frac{freq(A,B)}{freq(A)}$
  - Where σ_s and σ_c are user defined thresholds
  - Additionally, Its "**lift**" $l = \frac{|G| \times |(A \cup B)'|}{|A'| \times |B'|}$, $\quad l = \frac{P(A \cup B)}{P(A) \times P(B)} = \frac{s(A \cup B)}{s(A) \times s(B)}$
- Attribute implication is an association rule with $c = 1$
- Implications obey Armstrong rules (see below in RM topic)
- Apriori ++ (next slide)



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Vertical and Horizontal Association Rule Mining Algorithms

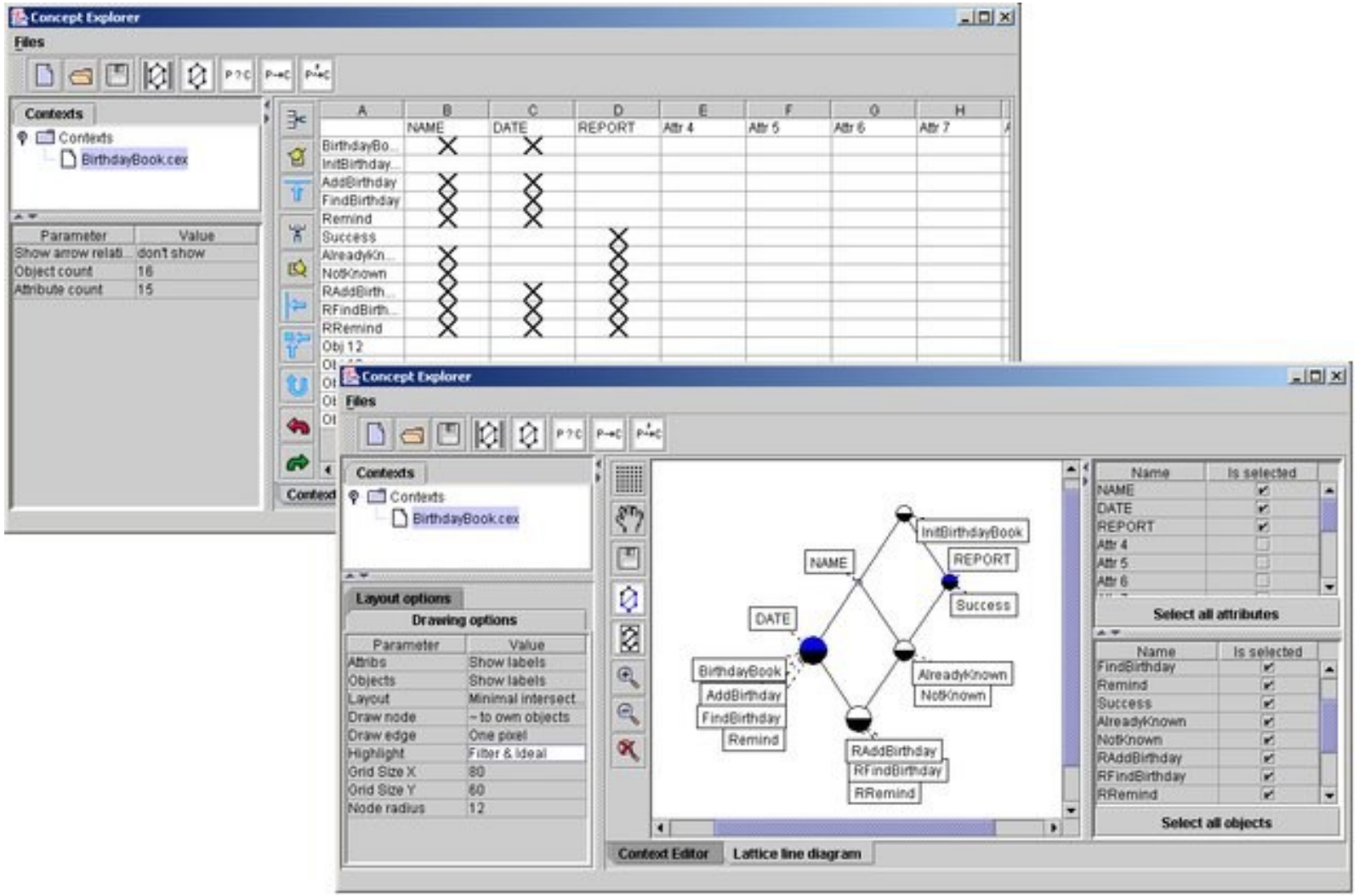


(C) Anand, H. S., and S. S. Vinod Chandra. A Comprehensive Review on Vertical and Horizontal Association Rule Mining Algorithms." Wiley Interdisciplinary Reviews: Data Mining and Knowledge Discovery, 16(3): e70129, 2026. http://doi.org/10.1002/widm.70129

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Line (Hasse) diagrams (1) – ConExp

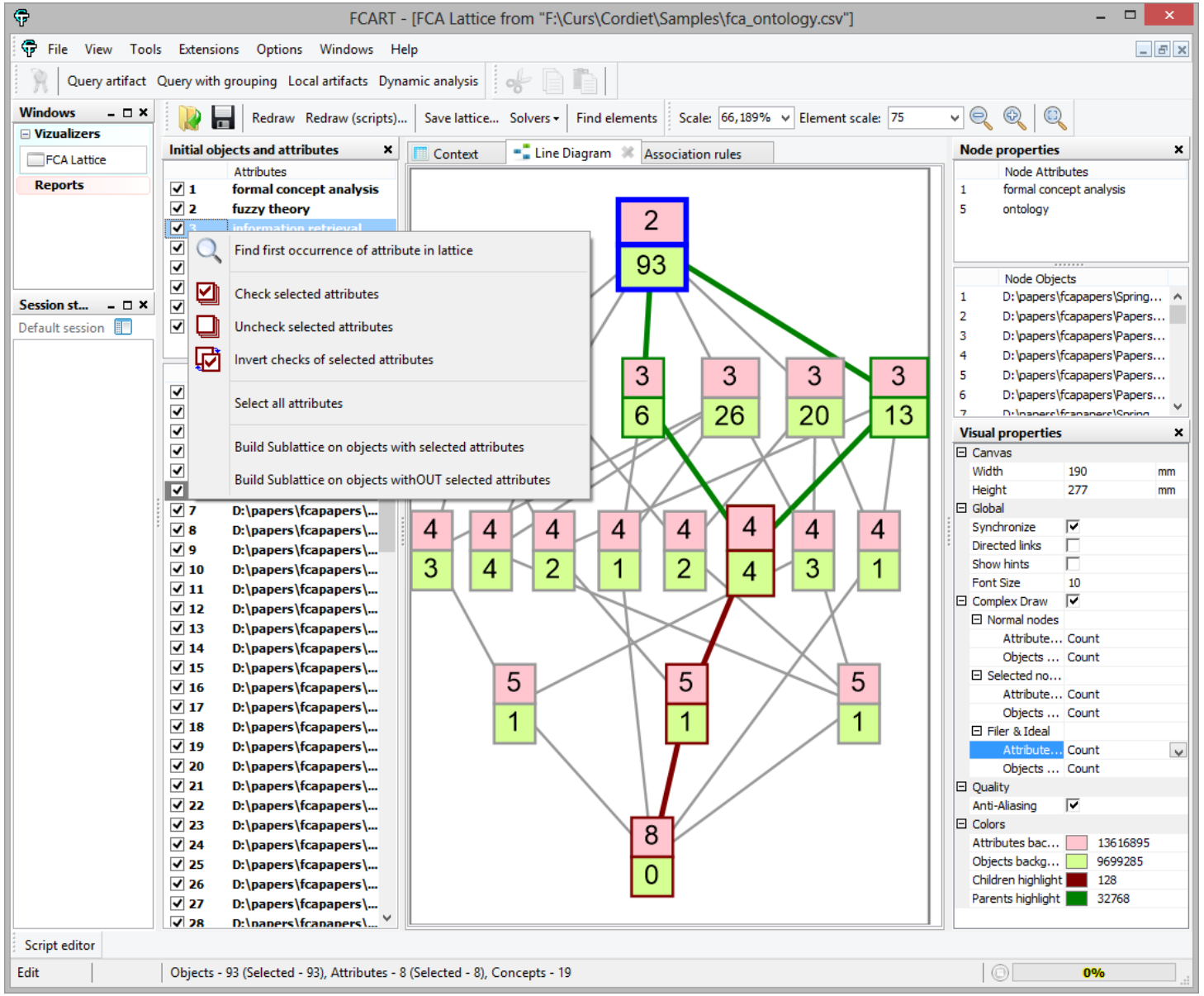
- Canonical representation in The Concept Explorer (ConExp) by Serhiy Yevtushenko



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Line (Hasse) diagrams (2) – FCART

- FCART by Alexey Neznanov



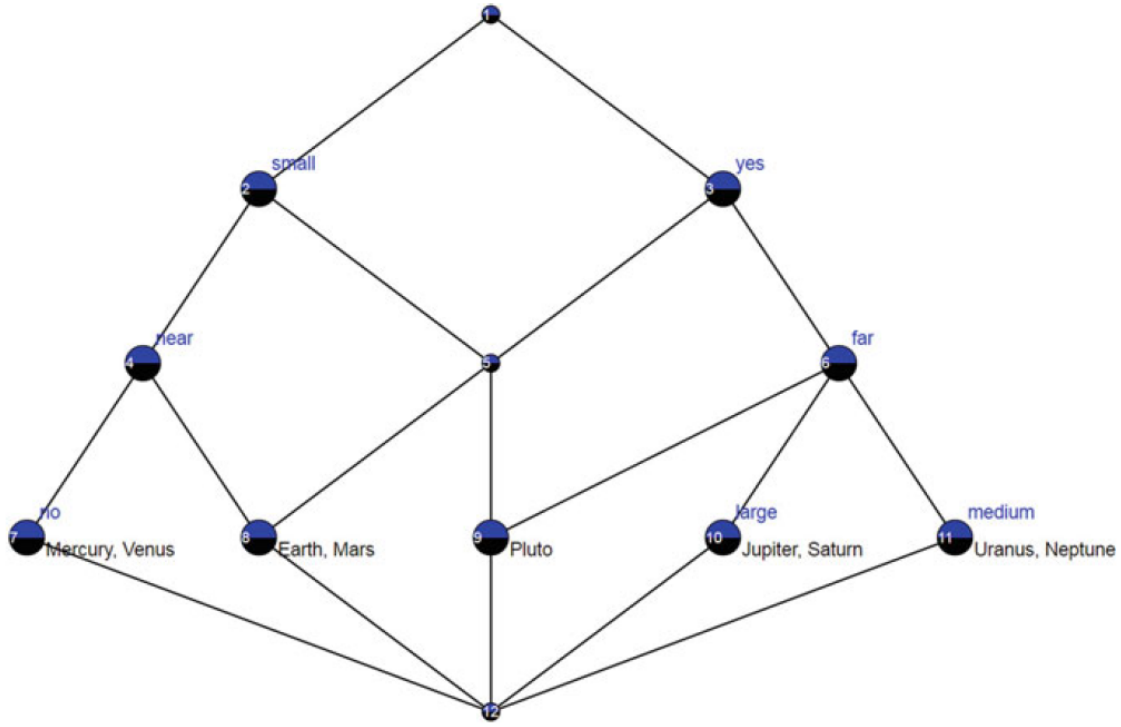
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Binary attributes and scaling

- Example from Davey and Priestley (1990)
  - Planets and their properties

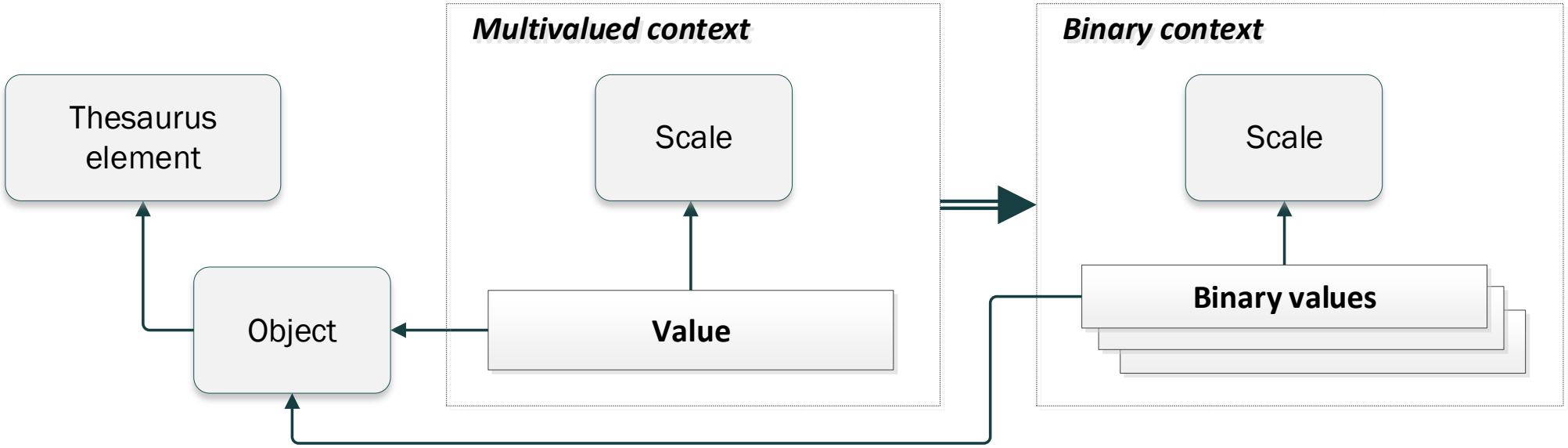
- Scaling of attributes:
  - Size: diameter, m => (“small”, “medium”, “large”)
  - Distance to Sun: distance, m => (“near”, “far”)
  - Moon(s): predicate, Boolean => (“yes”, “no”)

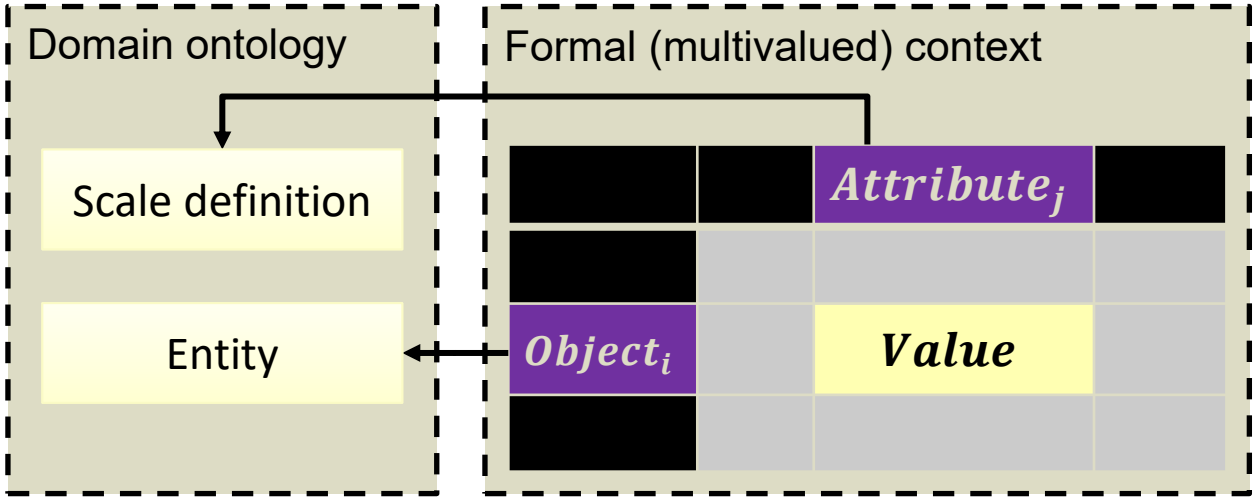
Planets	Size			Distance to Sun		Moon(s)	
	small	medium	large	near	far	yes	no
Jupiter			x		x	x	
Mars	x			x		x	
Mercury	x			x			x
Neptune		x			x	x	
Pluto	x				x	x	
Saturn			x		x	x	
Earth	x			x		x	
Uranus		x			x	x	
Venus	x			x			x



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Scaling, ontologies, and context tracking

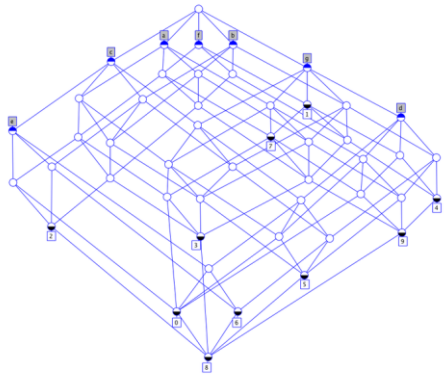




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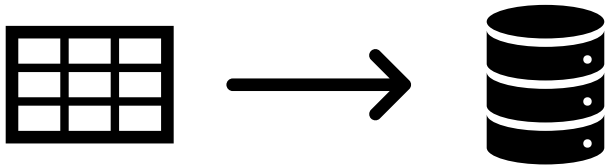
Exceptions and sources

- Priss U. Formal Concept Analysis Homepage (http://www.upriss.org.uk/fca/fca.html)
- Kaytoue M. An Introduction to Formal Concept Analysis. 2013 (http://perso.liris.cnrs.fr/mehdi.kaytoue/teaching/fca/fca-29-10-slides.pdf)



- Fresh studies:
  - Priss U. Alternative Diagrams for Concept Lattices. Conceptual Knowledge Structures (CONCEPTS 2026). Lecture Notes in Computer Science, vol 16811, Springer, 2026
  - Boissier F., Sen M., Rychkova I. Large Language Model and Formal Concept Analysis: a comparative study for Topic Modeling. arXiv:2602.01933, 2026.

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ENTITIES AND DATA MODELS

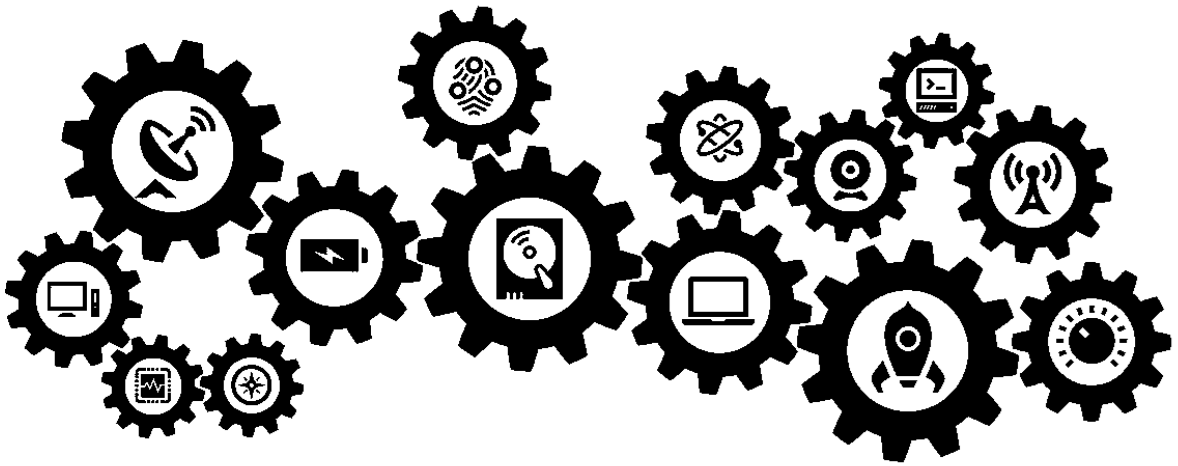
- ✓ Formal Concept Analysis
  - ✓ Objects and attributes
  - ✓ Formal context and formal concept
  - ✓ Extension → extent, intension → intent
  - ✓ Formal concept lattice and line diagrams
- ✓ Implications in context and association rules

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Main tasks of data processing

- Information system (IS) as a data processing facility
  - Relevant knowledge about data → adequate data processing
  - Workflows and pipelines (another big lecture)

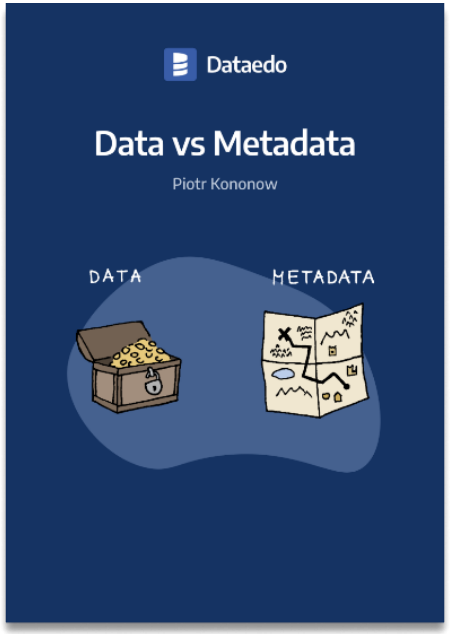
- Six main tasks:
  1. **Input/Output**
  2. **Format conversion** (sic!)
  3. **Telecommunication**
  4. **Storage**
  5. **Search (data retrieval)**
  6. **Manipulation**
     - Including “Data analysis”!



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Data and metadata

- *The knowledge* appears at the **metadata** level
  - Data about data ☺
  - What is Metadata (with examples) (http://dataedo.com/kb/data-glossary/what-is-metadata)
  - Dataedo – Data vs Metadata (http://landing.dataedo.com/data-vs-metadata)
- Metadata management
  - Full chapter in DMBoK (Chapter 12: Metadata Management)!
- Metadata formalization and standardization
- Metadata about metadata (infinite hierarchy)
  - Where can we stop? Metadata, **metametadata**, **metametametadata**, ...
  - The problem of harmonization of *ontologies* and *ontics* (interpretation)
  - The problem of interoperability
- Metadata and meta-modelling: modelling, notations, architecture
  - Model, metamodel, metametamodel, …
    - See UML meta-model, BPMN metamodel, SysML v2 metamodel (Kernel Modeling Language – KerML)



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Object-oriented analysis and design

- **Object** – an arbitrary entity for which the boundaries, state and behavior are easy to define
- **Attribute** – an object property
  - The set of attribute values defines the object state
- **Method** – a variant of object behavior
  - The set of methods defines the object behavior
- **Interface of object** – the external representation of the object, i.e., all its attributes and methods
  - The interface description defines the object boundaries by the very fact of attributes and methods inclusion or non-inclusion in the interface
- **Class** – a description of objects of the same type, i.e. their common interface
  - A class can be represented as a template that describes the possible objects' states and behavior
  - The object that satisfies this pattern is called an instance of a class
  - All instances of a class have the same interface
  - The class declaration formalizes the boundaries, state and behavior discussed in the object definition above
  - ☑ In OO programming languages a class is a data type

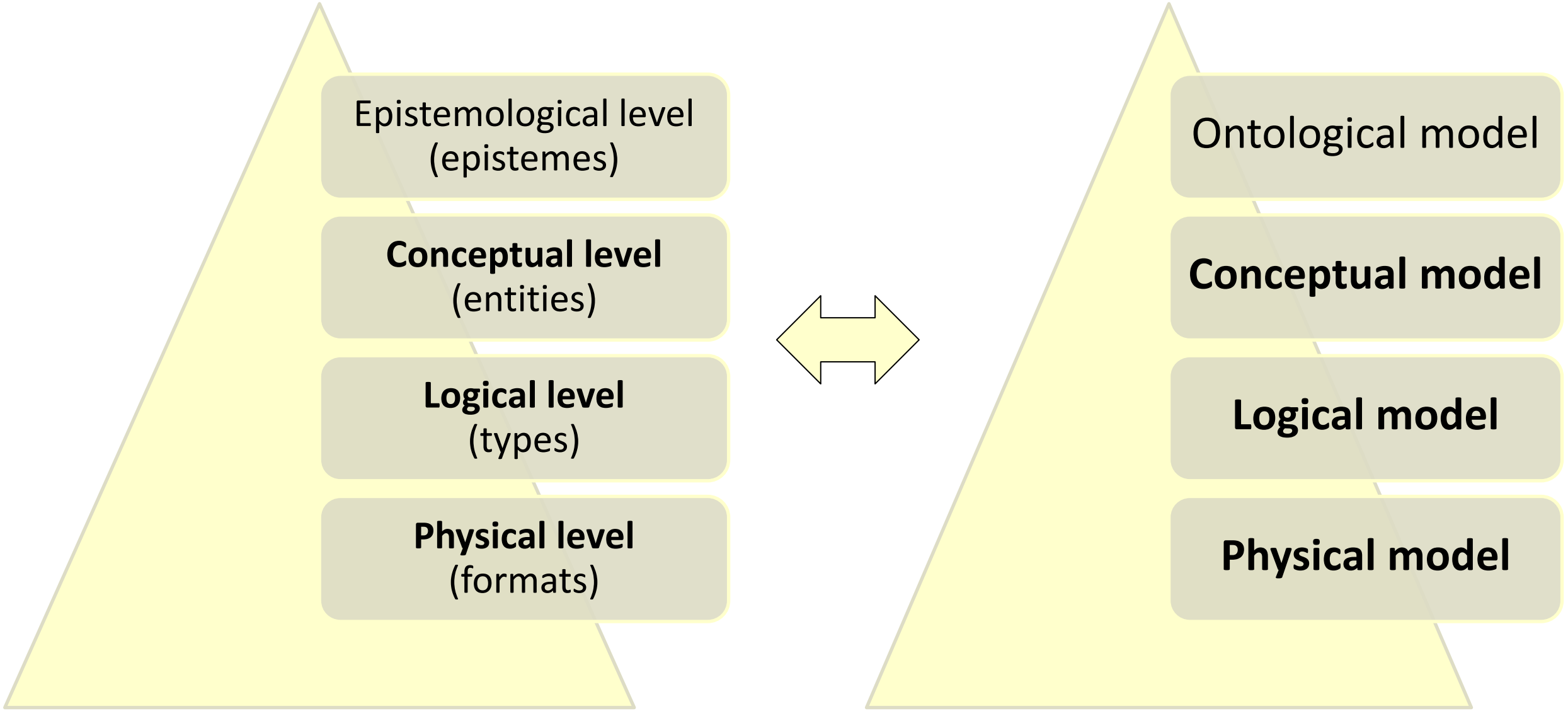
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Basic types of relationships in OO-analysis

- Relationships between objects and classes
  - **Instantiation** (“instance-of”) – relationship between a class instance and a class
- Relationships between classes
  - **Generalization** (“is-a”) – relationship between a successor and an ancestor (oriented from a successor to an ancestor)
    - In the opposite direction – **specialization** (from an ancestor to a successor)
    - ☑ **Inheritance** – a mechanism for implementing a generalization/specialization relationship
  - **Realization/implementation** – relationship between a class and an interface/abstract class
- Relationships between objects (usage)
  - **Association** – arbitrary “using” relationship between two or more objects in which the objects have their own lifetime and there is no “owner”
    - Binary associations can be unidirectional and bidirectional (symmetrical)
  - **Aggregation** (“part-of”) – relationship between a part and a whole (weak containment)
  - **Composition** – aggregation, when a part is integral (one “part” cannot belong to more than one “whole” and the lifetime of the “part” coincides with the lifetime of the “whole”) (strong containment)

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Three levels of data models + knowledge background



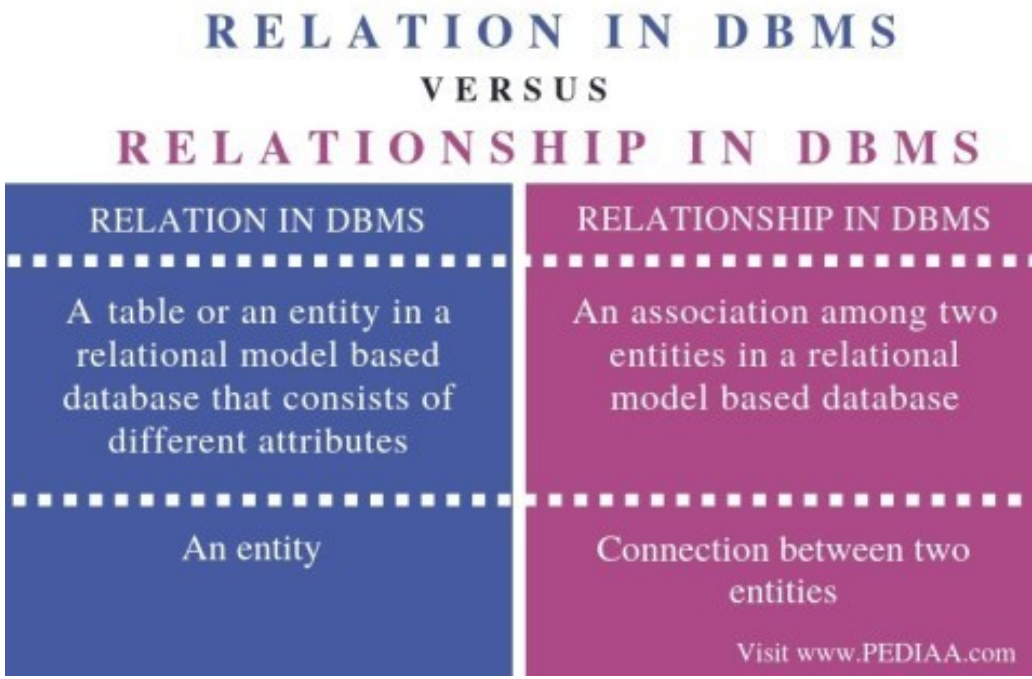
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The note about relation(ship)

- Complex word usage!
  - Historically and traditionally:
    - Britannica: “Relations” and “Relationship“ (http://www.britannica.com/dictionary/eb/qa/relations-and-relationship)
    - BBC: Relative / relation – relationship (http://www.bbc.co.uk/worldservice/learningenglish/grammar/learnit/learnitv235.shtml)
    - Discussion (http://english.stackexchange.com/questions/15208/relation-versus-relationship)

- The main difference is that relationship is broader than relation
  - Both can mean “the way in which two or more concepts, objects, or people are connected”
  - In addition, relationship can mean “the state of being connected”

- Later we discuss additional senses!
  - ☑“Relation” in relational algebra



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Entity-relationship model

- Peter Pin-Shan Chen, 1976
- **Entity-Relationship Diagrams** – visual language for conceptual data modelling
  - Standard abbreviations – **ER** and **ERD**
  - Both terms “entity” and “relationship” are incomprehensible and need clarification
    - Wikipedia is a very bad source here!
  - ☑ ER is a static model
    - We will discuss further both time consideration and special notations (UML, BPMN, etc.)
  - A huge number of extensions and variations:
    - Semantics extensions
    - Various “syntax sugar” for working with large diagrams
    - A plenty of visual images variations
    - Confusion with IDEF1X
      - See ISO 31320-2:2012 Information technology — Modeling Languages — Part 2: Syntax and Semantics for IDEF1X97 (IDEFobject) (http://www.iso.org/standard/60614.html)

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ER origins and advancing

- Almost **fifty years** of development
  - Chen P. The Entity-Relationship Model-Toward a Unified View of Data. ACM Transactions on Database Systems, 1, 1976. pp. 9-36 (http://www.csc.lsu.edu/~chen/pdf/erd-5-pages.pdf, http://pdfs.semanticscholar.org/1348/1add6a6ce710af7004577ccb0ea7f92e887d.pdf)

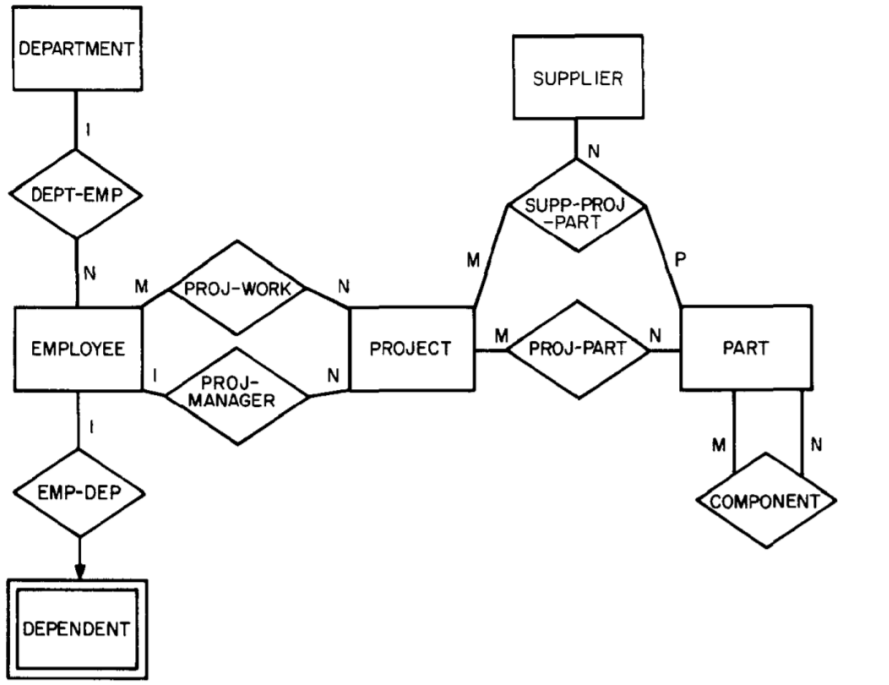
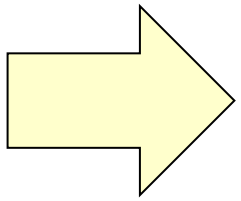
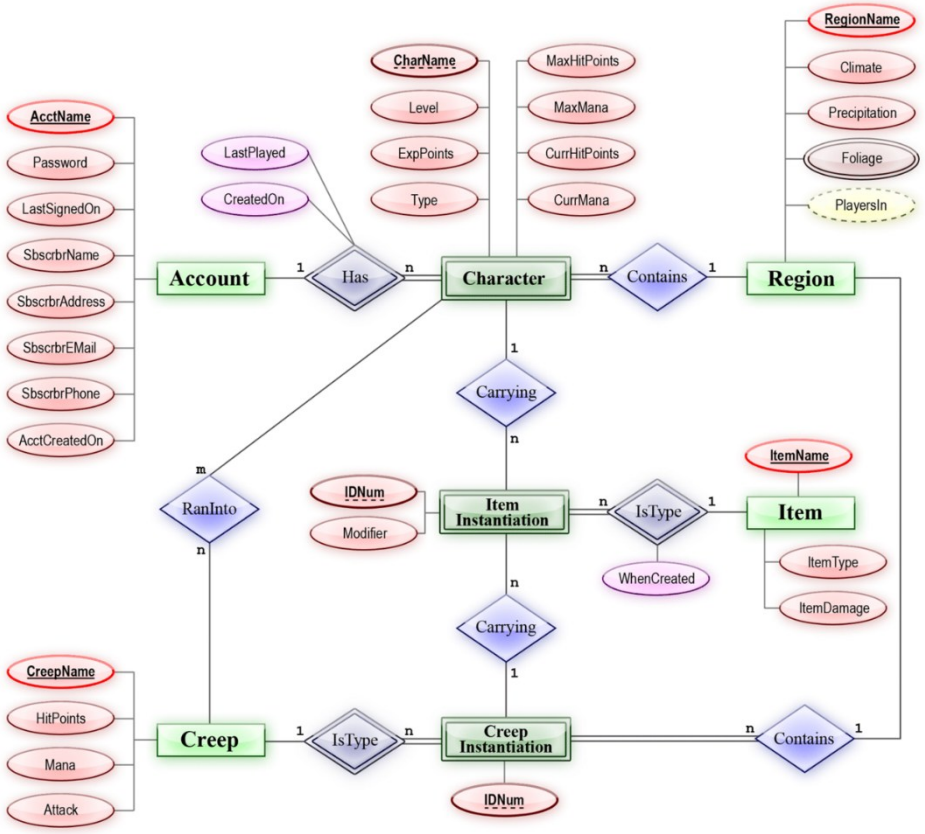


Fig. 11. An entity-relationship diagram for analysis of information in a manufacturing firm

ACM Transactions on Database Systems, Vol. 1, No. 1, March 1976.

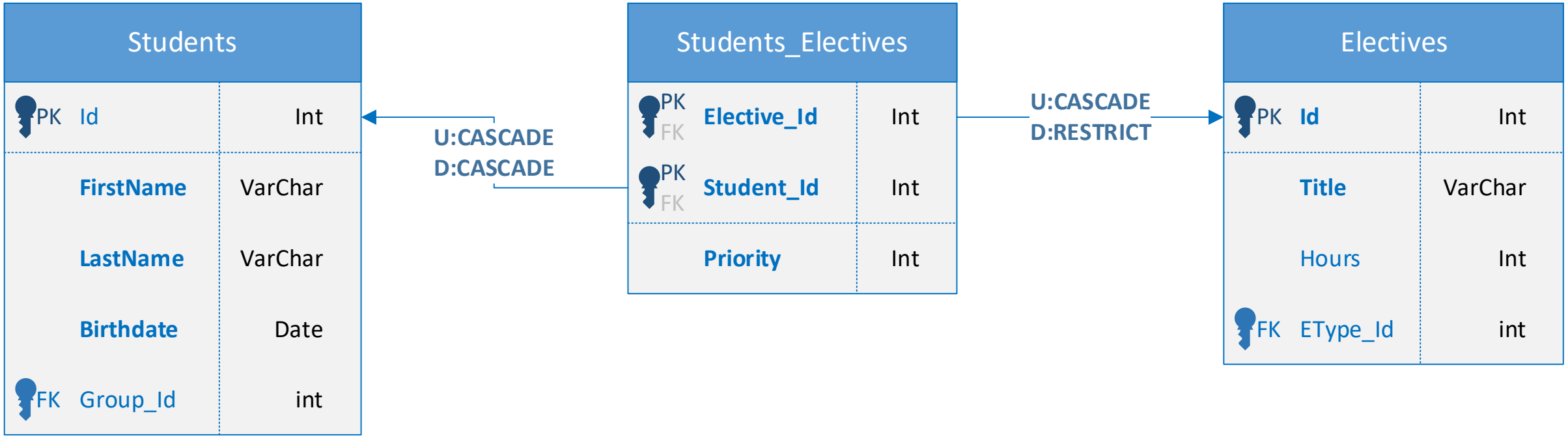




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RS/TR origins and advancing

- Relational schemas or table-reference diagrams
  - Codd E. A Relational Model of Data for Large Shared Data Banks. Communications of the ACM, 13(6), 1970, pp. 377-387. (http://www.seas.upenn.edu/~zives/03f/cis550/codd.pdf)
  - Codd E.F. The relational model for database management: version 2. Addison-Wesley, 1990. 538 p.



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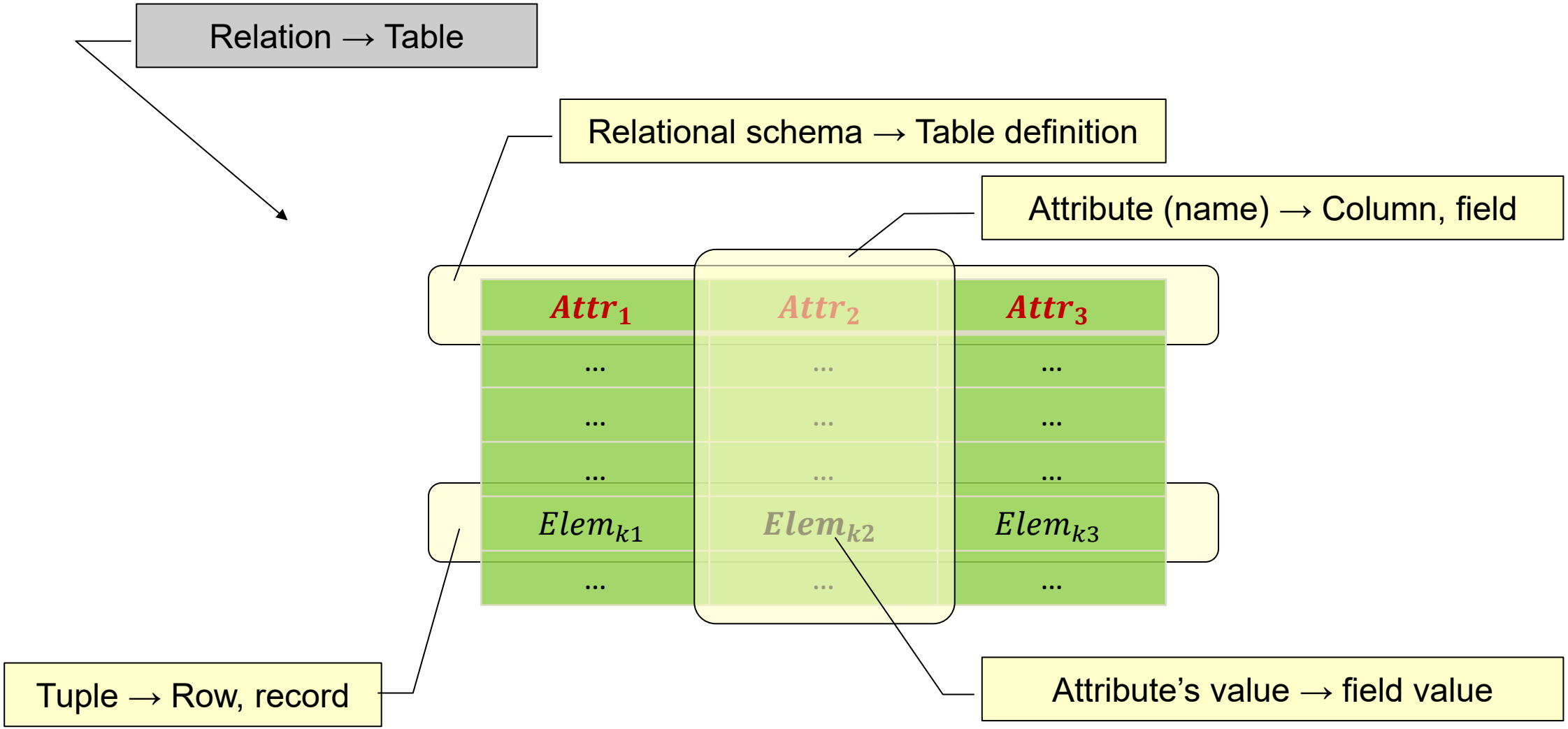
Conceptual and logical levels of classical ER/TR approach

	ER Model	**Relational Model**
Level	Conceptual	Logical (representation or implementation)
Purpose	Shows real-world entities, their characteristics, and the relationships between them.	Shows data about objects in tables, and how the tables relate to each other.
Used by	For anyone who needs to see how the data model will be formed.	Mostly for data engineers and programmers who need to see where data will be stored.
Language and notations	Chen, Martin, UML, and other notations.	Relational/Schema diagrams, SQL.
Main components	Entities, attributes, relationships, arrows	Tables (relations), columns, domains, keys, records
Relationships	Relationships can be easily seen and identified through the use of arrows and symbols.	It is more difficult to determine the relationships between tables.
Mapping	Has the possibility to show cardinality mapping in explicit way.	Not possible to see cardinality mapping (only "foreign keys" with constraints).

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Relation → Table

- Date C.: "a table is not a relation“ (Database in Depth: Relational Theory for Practitioners. O'Reilly, 2005)



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SQL table and their rows VS query and their recordset

- A relation's **tuple** <> a **row** from a query result (output rows of a SQL SELECT statement)
  - Remember PKs and "DISTINCT" keyword!

- A recordset (or result set) is a data structure that consists of a group of records as the result of a query to a data source
  - Recordset is not a set (in set-theoretic sense)!
  - Recordset has local schema!
  - Recordset has ordering!
  - Recordset can have duplicates!

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Dependency mining and relational constraints

- Dependency
  - Functional dependency (FD)
  - Approximate functional dependency (AFD)
  - Multi-valued dependency (MVD)
  - …
- Apparatus of dependencies was introduced by William W. Armstrong (Armstrong's Axioms)
  - Beeri, Fagin и Howard created apparatus of multi-valued dependencies
  - Hugh Darwen formulated General Unification Theorem
- Keys (PKs, FKs)!
- Normalization/**denormalization**!
  - How to build object-attribute dataset from relational model? See further!
- Data mining! …

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Functional dependency mining

- **Functional dependency** (FD) – a relationship between two sets of attributes in a relation
  - $X \to Y$ – X determines Y, i.e. there is a many-to-one relationship between the values of the attribute set X and the values of the attribute set Y
  - X – determinant, Y – dependent set (or simply dependent)

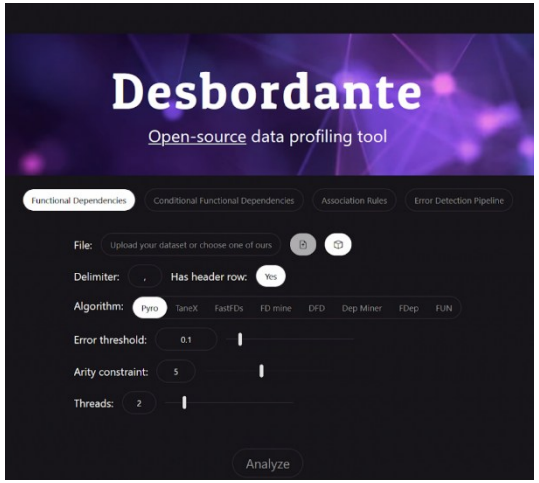
- Armstrong's Axioms:
  - Remember Association Rule Mining!

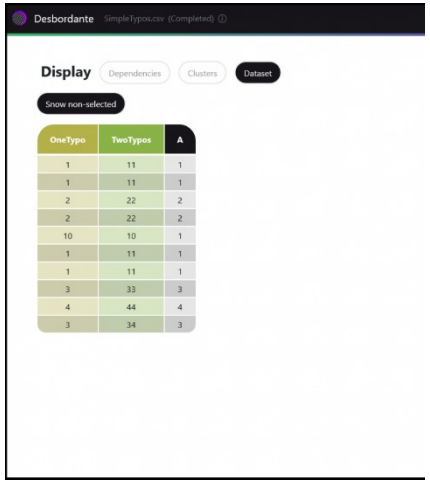
Axiom/inference rule	**Expression**					
Axiom of reflexivity	$X \to X$					
Axiom of augmentation	IF	$X \to Y$	THEN	$XZ \to Y$		
Axiom of transitivity	IF	$X \to Y$	AND	$Y \to Z$	THEN	$X \to Z$
Decomposition rule	IF	$X \to YZ$	THEN	$X \to Y$	AND	$X \to Z$
Union rule	IF	$X \to Y$	AND	$X \to Z$	THEN	$X \to YZ$
Composition rule	IF	$X \to Y$	AND	$Z \to V$	THEN	$XZ \to YV$
Pseudotransitivity rule	IF	$X \to Y$	AND	$YZ \to V$	THEN	$XZ \to V$

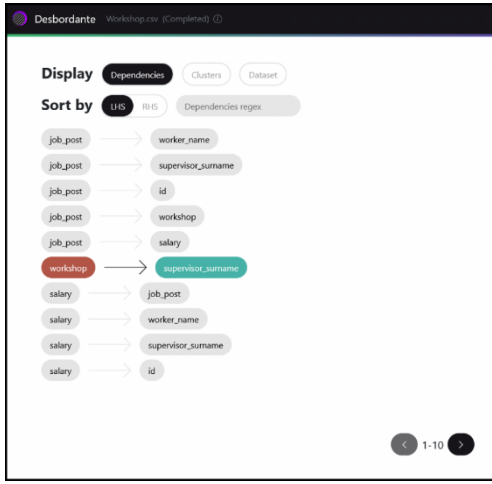
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Dependency mining in practice

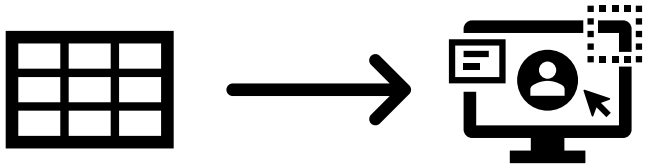
- **Desbordante** – a high-performance data profiler that is capable of discovering and validating many different patterns in data using various algorithms (http://github.com/Desbordante/desbordante-core)
- Data patterns:
  - Functional dependencies, both exact and approximate (discovery and validation)
  - Conditional functional dependencies (discovery)
  - Metric functional dependencies (validation)
  - Fuzzy algebraic constraints (discovery)
  - Unique column combinations (discovery and validation)
  - Association rules (discovery)
  - …







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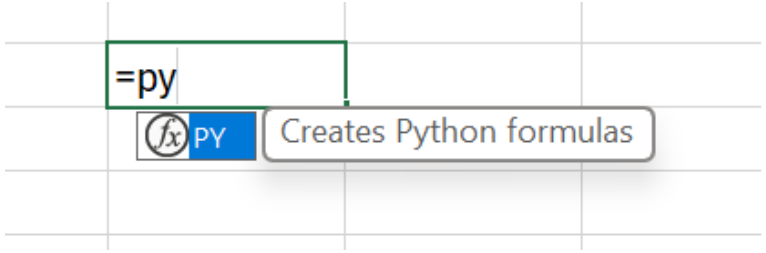
OBJECT-ATTRIBUTE DATA IN HCI AND UI

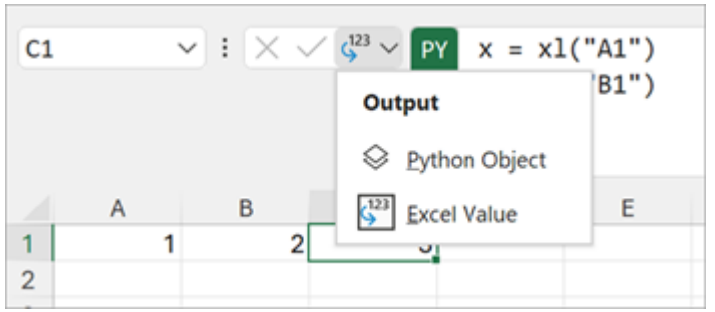
- ✓ HCI and UI
- ✓ Tables and spreadsheets
  - ✓ Pivot tables
- ✓ Rows and columns in UI
  - ✓ Implicit and explicit ordering

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Tables in human-computer interaction

- **Spreadsheet** – electronic table with formulas and hyperlinks
- **Lotus 1-2-3** was arguably the application which made the IBM PC a success
- **Microsoft Excel** – industry standard
  - Hierarchy: Workbook → Worksheet → Range → Cell
  - Formula language
  - Now with Python ☺
    - (http://support.microsoft.com/en-gb/office/get-started-with-python-in-excel-a33fbcbe-065b-41d3-82cf-23d05397f53d)

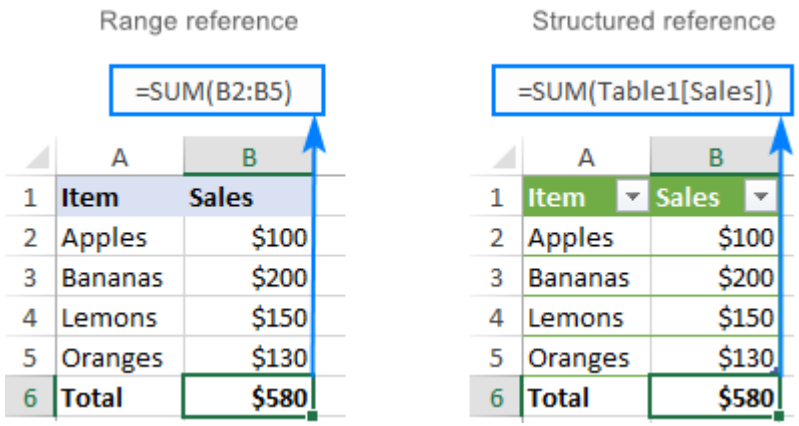




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Tables in Microsoft Excel

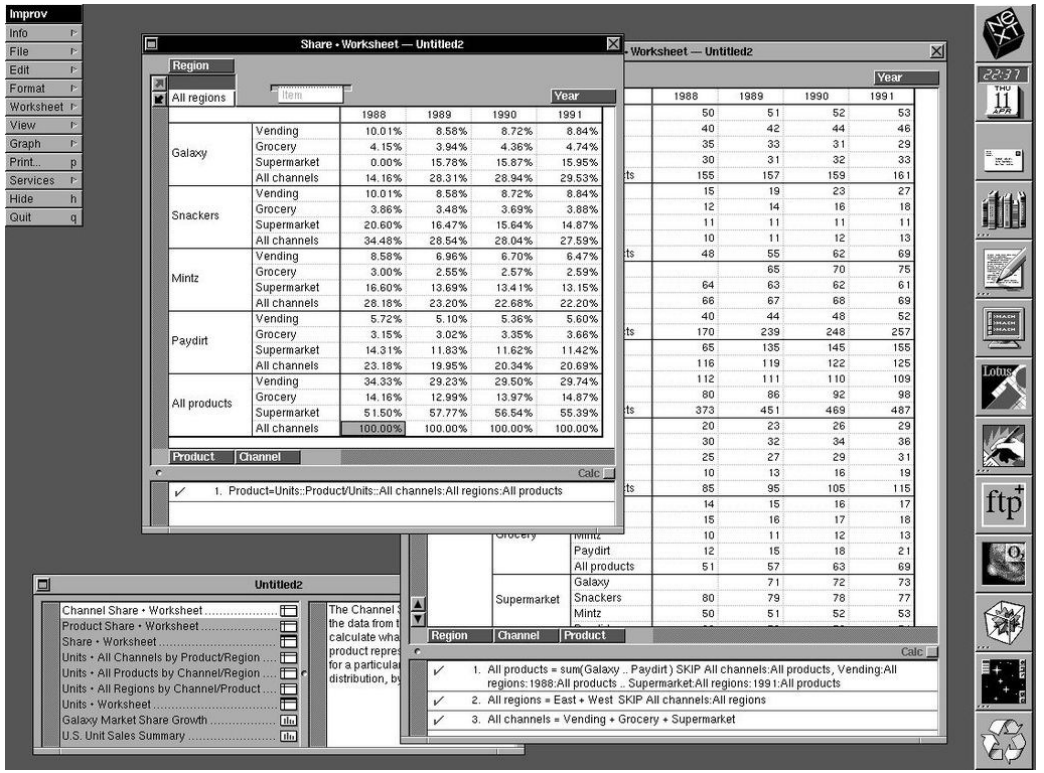
- 2003 – Lists
- **2007 – Tables**
  - With new access language (http://support.microsoft.com/en-gb/office/using-structured-references-with-excel-tables-f5ed2452-2337-4f71-bed3-c8ae6d2b276e)
    - Example: `=DeptSales[[#Totals],[Sales Amount]:[Commission Amount]]`
  - Table is the successor of List!
    - VBA Tables and ListObjects (http://excelofftthegrid.com/vba-excel-tables-code/)
- 2013 – New query engine in Power Query
- 2016 – New generation of Data Model (internal DBMS)
  - + Tables access



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Pivot tables – beginning

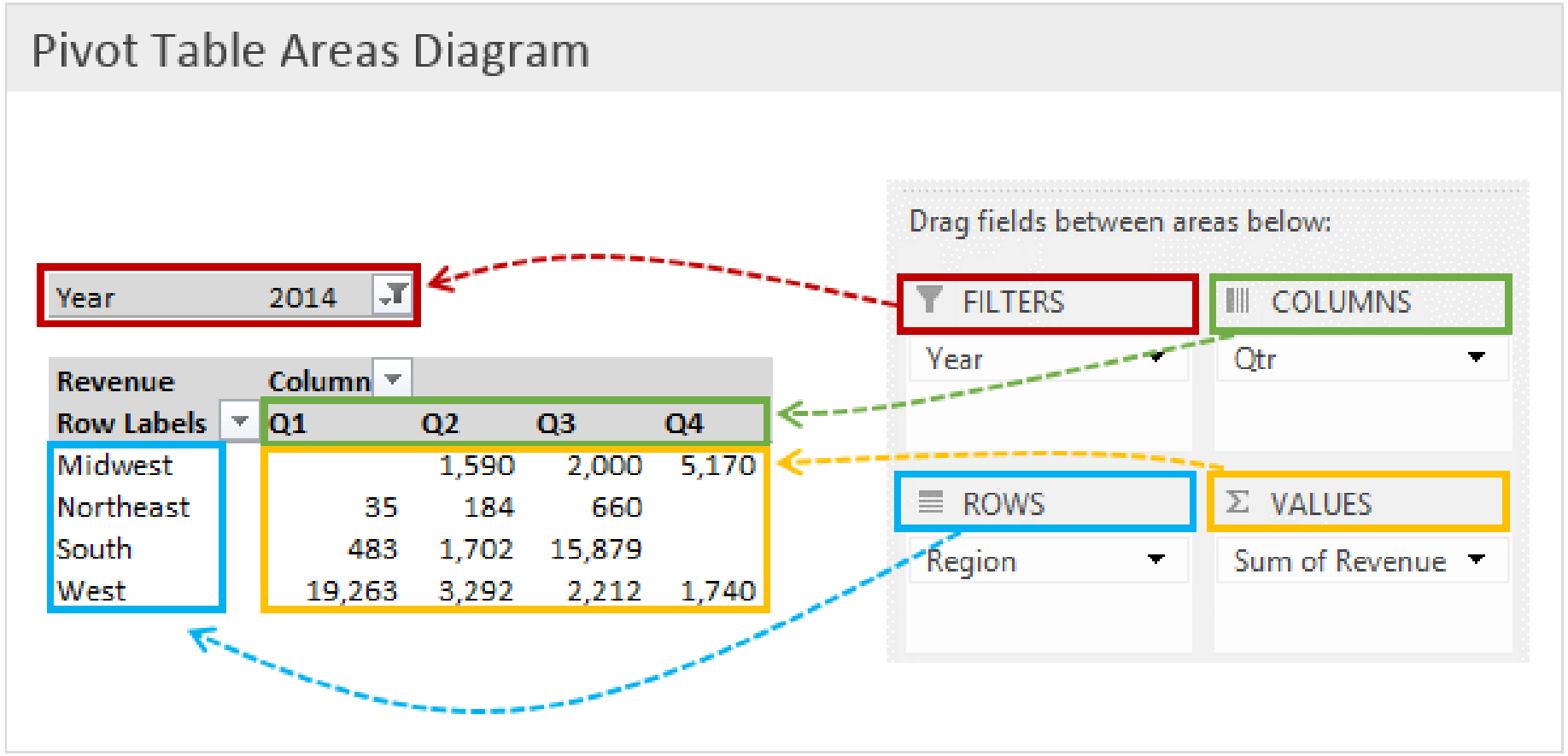
- 1986 – Pito Salas from Lotus Corp – father of pivot tables
- First implementations:
  - 1991 – Lotus Improv (http://en.wikipedia.org/wiki/Lotus_Improv)
  - 1993 – Microsoft Excel 5.x (http://winworldpc.com/product/microsoft-excel/5x)



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Excel Pivot Tutorial

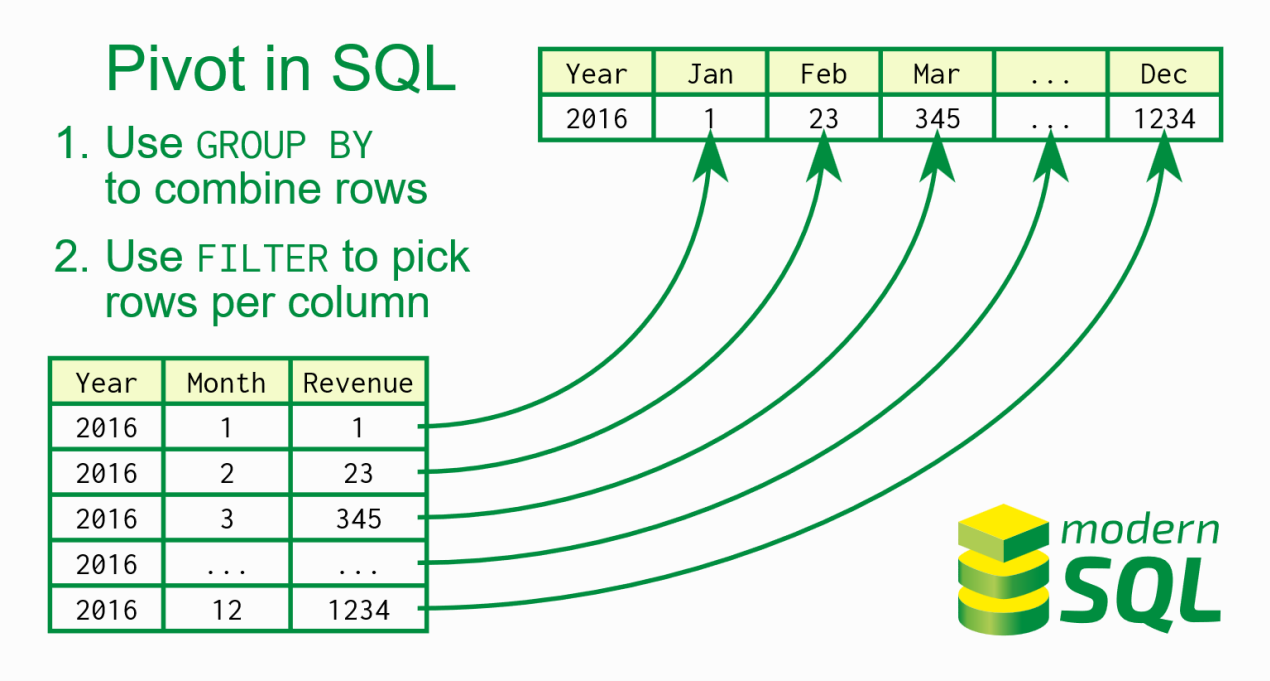
- *How Do Pivot Tables Work?* (http://www.excelcampus.com/pivot-tables/pivot-tables-work/)



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Pivot & SQL (addendum for previous topic)

- In SQL language:
  - *Pivot — Rows to Columns* (http://modern-sql.com/use-case/pivot)

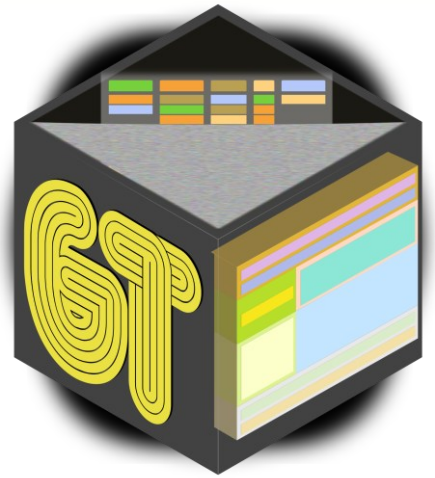


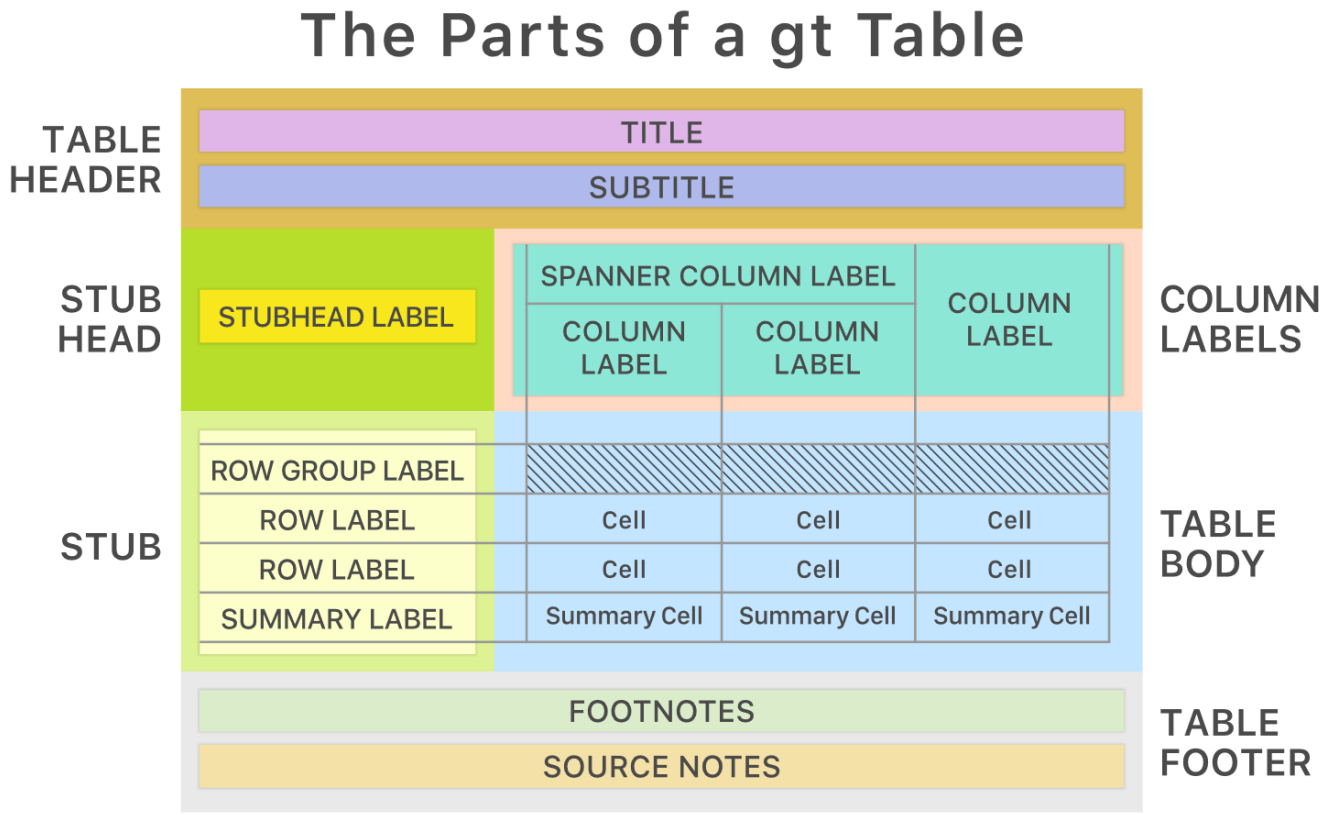
  - *Microsoft SQL Server: FROM – Using PIVOT and UNPIVOT* (http://docs.microsoft.com/en-us/sql/t-sql/queries/from-using-pivot-and-unpivot)

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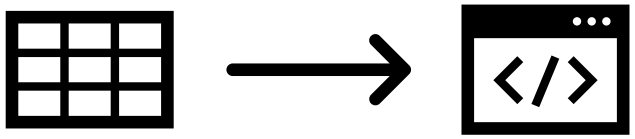
R – gt (A grammar of tables)

- gt package (http://gt.rstudio.com)
  - Constructs a wide variety of useful tables with a cohesive set of table parts:
    - Header, stub, column labels with spanners, body with summary, footer, …





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OBJECT-ATTRIBUTE DATA IN PROGRAMMING

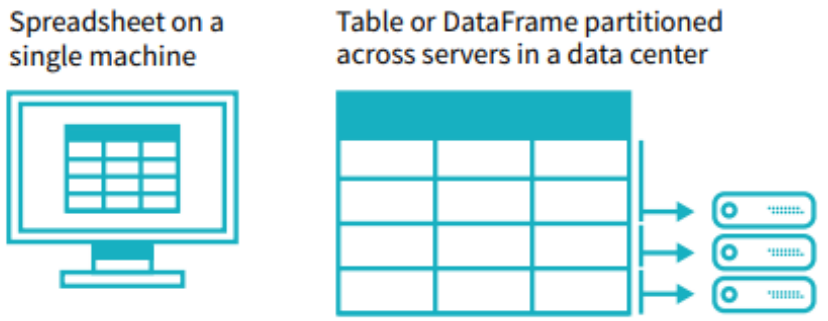
- ✓ Dataframes and tables
- ✓ Dataframes interoperability
- ✓ Rich table construction and visualization

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Dataframe

- A **tabular data structure** common to many data processing libraries:
  - pandas and other Python libraries
  - The Dataframe API in Apache Spark
  - Data frames in the R programming language
  - …

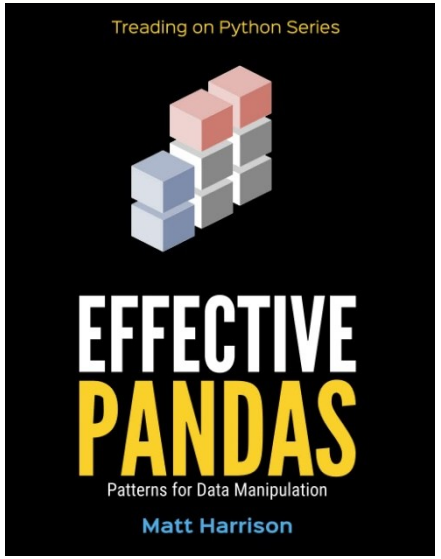
- Databricks – What is a DataFrame? (http://www.databricks.com/glossary/what-are-dataframes)
  - DataFrame contains a blueprint, known as a schema, that defines the name and data type of each column
  - Missing or incomplete values are stored as null values in the DataFrame

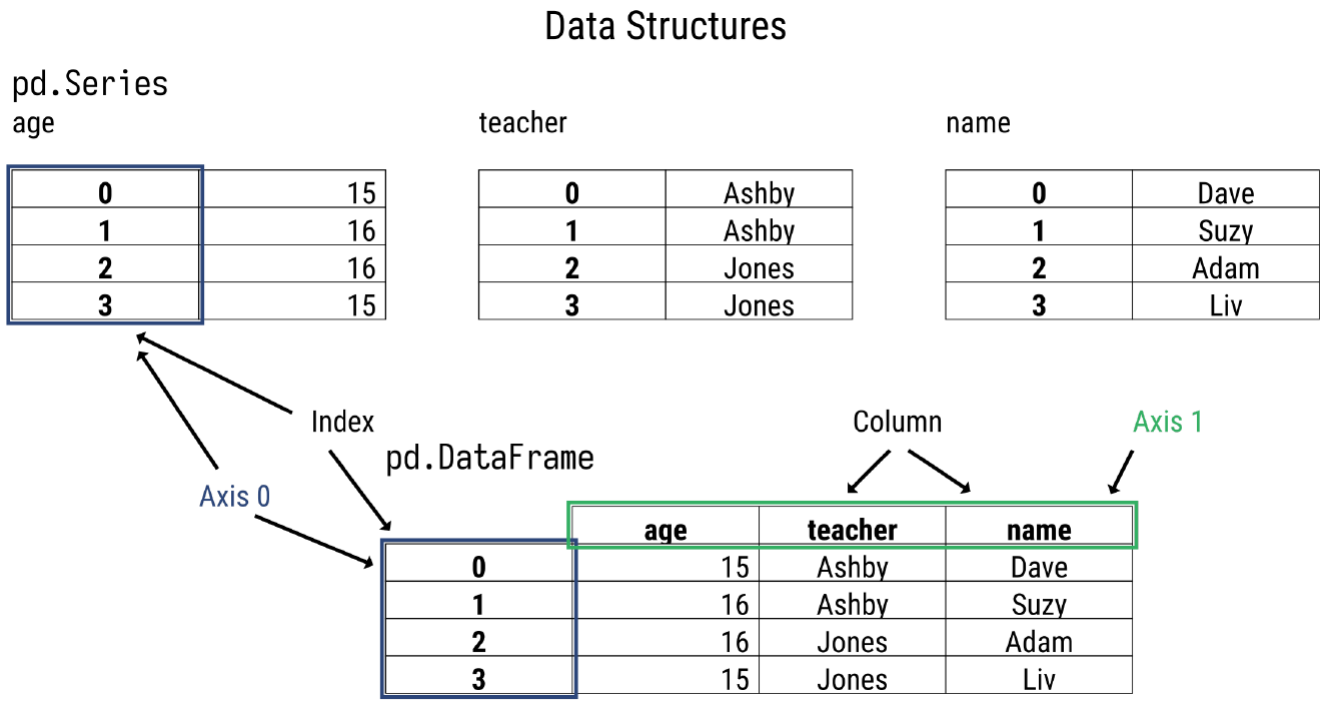


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Python – Pandas

- Harrison M. Effective Pandas. Patterns for Data Manipulation. 2021. 397 p.
  - Series manipulation
  - Creating columns
  - Summary statistics
  - Grouping, pivoting, and cross-tabulation
  - Time series data
  - Visualization
  - Chaining

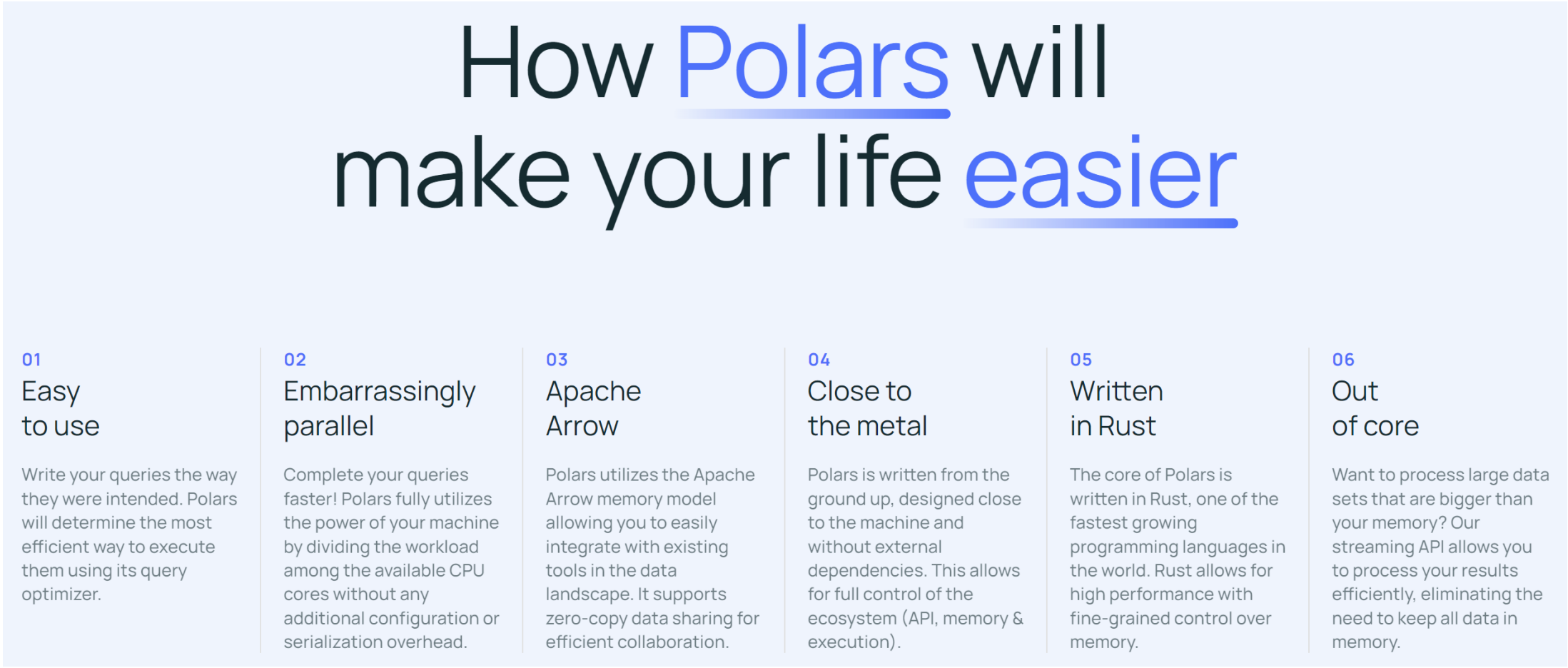




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Python – Polars

- **Polars** – Lightning-fast DataFrame library for Rust and Python (http://www.pola.rs)
  - 2026-10-06: Polars 2.0 (http://pola.rs/posts/release-polars-2/)



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Python dataframes interoperability

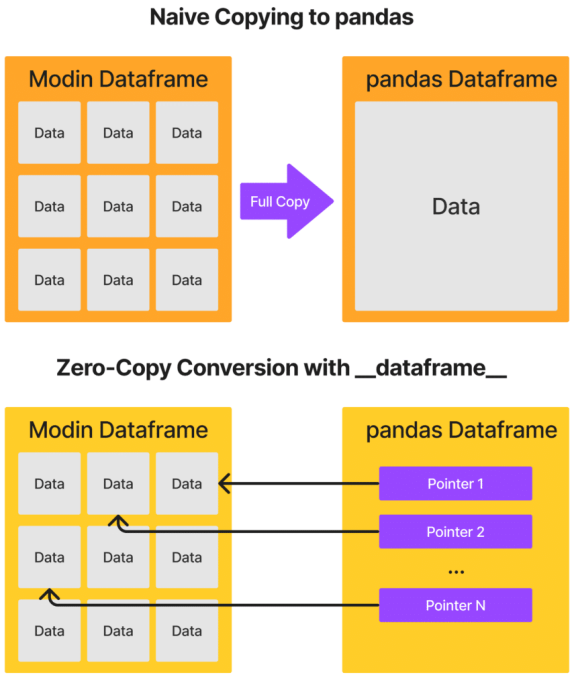
- Consortium for Python Data API Standards (http://data-apis.org)



- Dataframe Interchange Protocol (http://arrow.apache.org/docs/python/interchange_protocol.html)



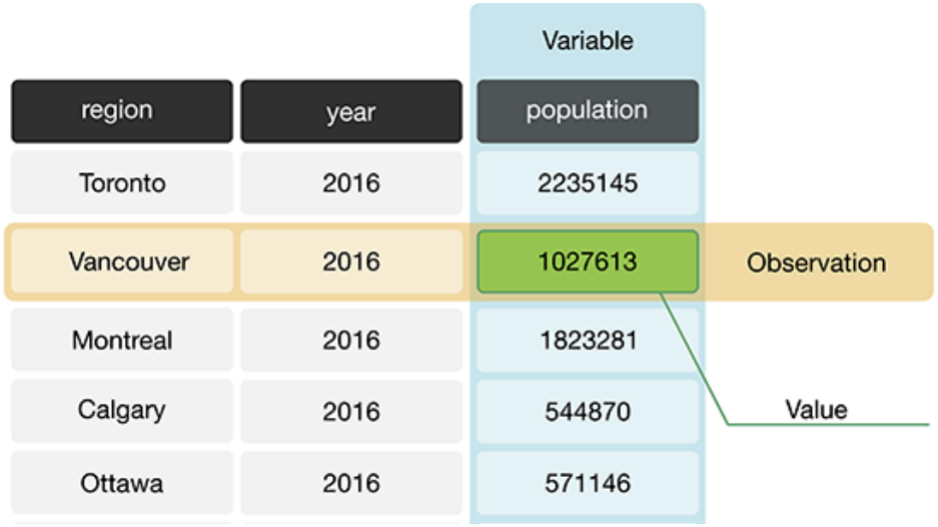
- How the Python Dataframe Interchange Protocol Makes Life Better (http://ponder.io/how-the-python-dataframe-interchange-protocol-makes-life-better/)
  - Vaex (PR merged 10/13/21)
  - cuDF (PR merged 11/17/21)
  - Modin (PR merged 2/25/22)
  - pandas (PR merged 4/27/22)
  - Polars (PR merged 1/30/23)
  - ibis (PR merged 6/16/23)

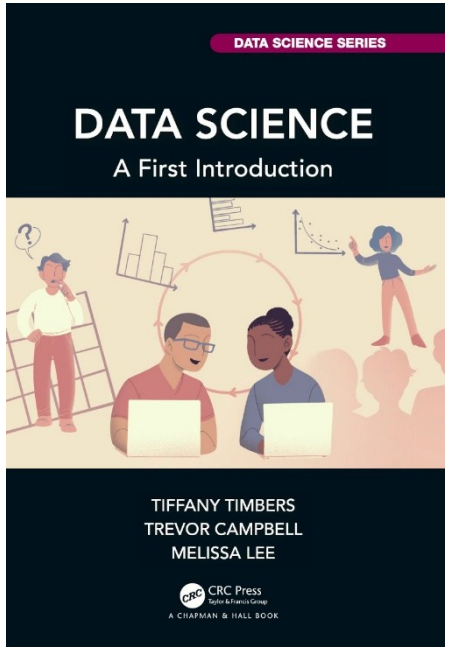


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R – data frame

- Timbers T., Campbell T., Lee M. Data Science. A First Introduction. 2023
  - What is a data frame? (http://datasciencebook.ca/wrangling.html#what-is-a-data-frame)





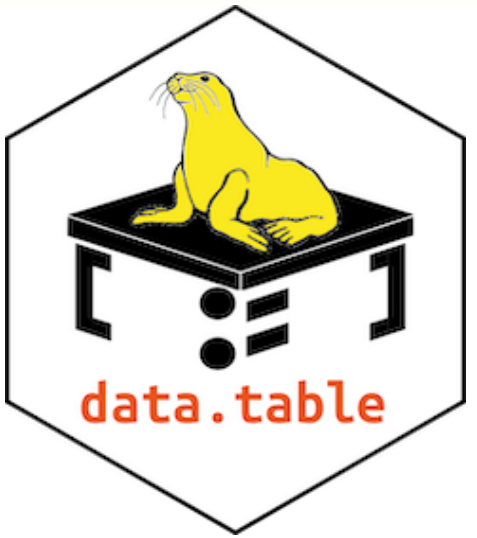
- + Tidyverse – R packages for data science (http://www.tidyverse.org)

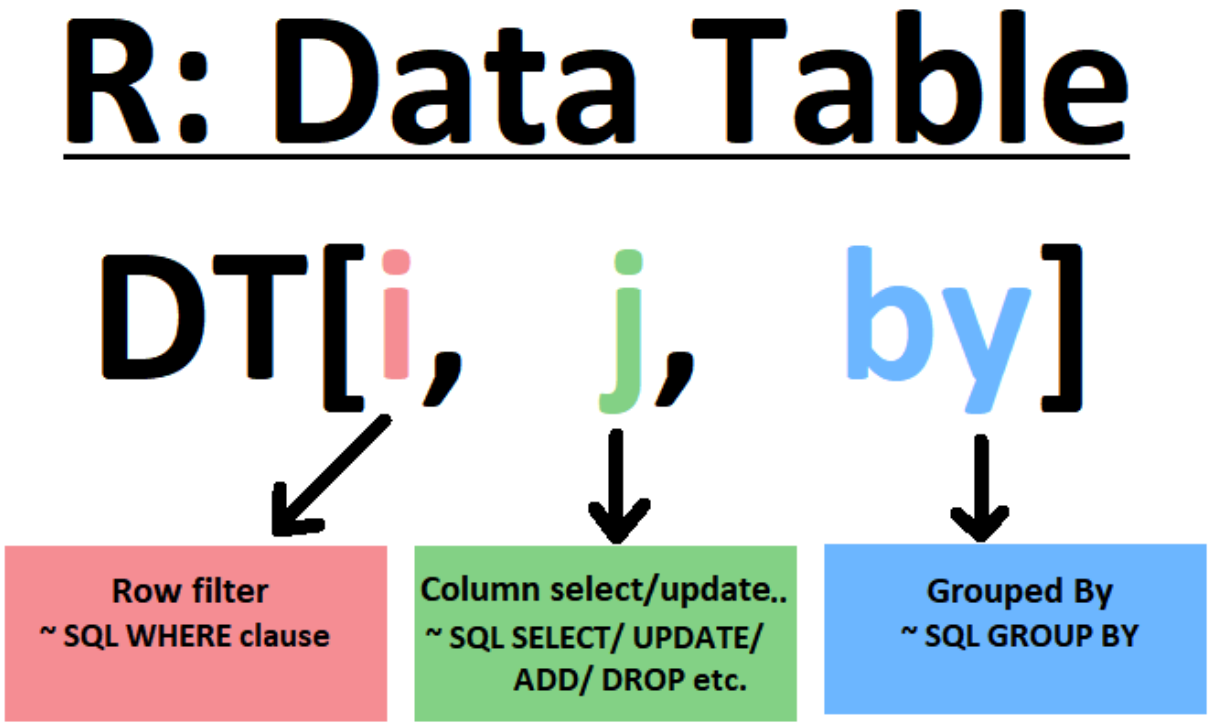


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R – Tables

- `data.table` is an enhanced version of `data.frames`
  - Signorell A. Tables in R – A quick practical overview. 2021 (http://cran.r-project.org/web/packages/DescTools/vignettes/TablesInR.pdf)

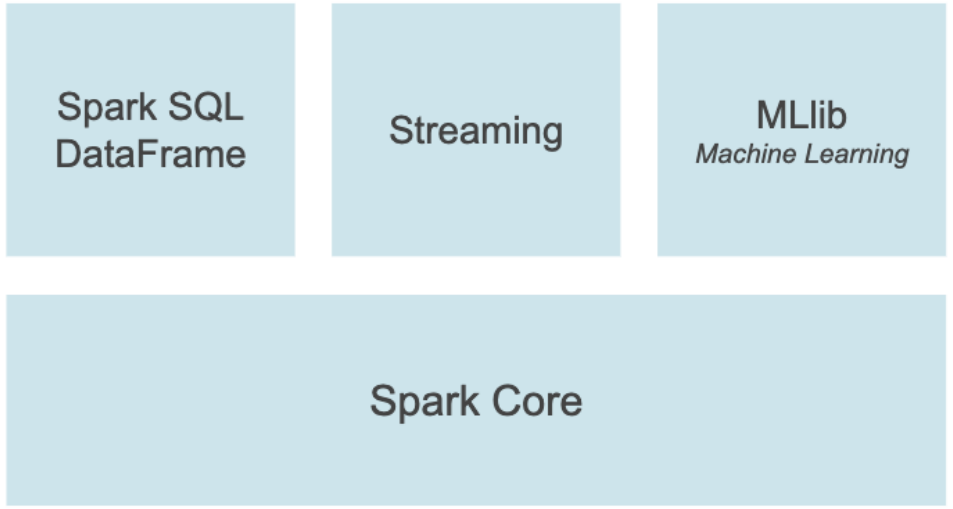




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Spark – DataFrame

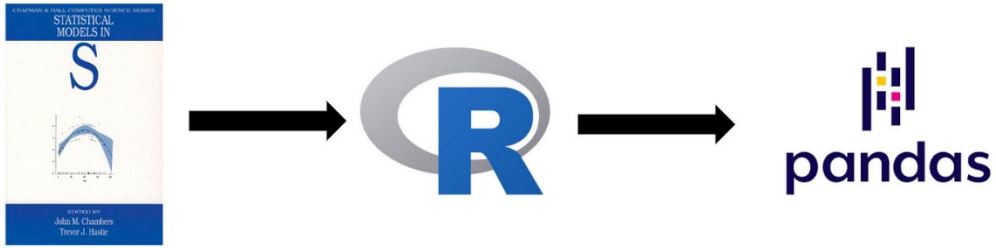
- Apache PySpark – pyspark.sql.DataFrame
  (http://spark.apache.org/docs/3.1.1/api/python/reference/api/pyspark.sql.DataFrame.html)
  - A DataFrame is a Dataset organized into named columns
  - It is conceptually equivalent to a table in a relational database
    or a data frame in R/Python, but with richer optimizations under the hood



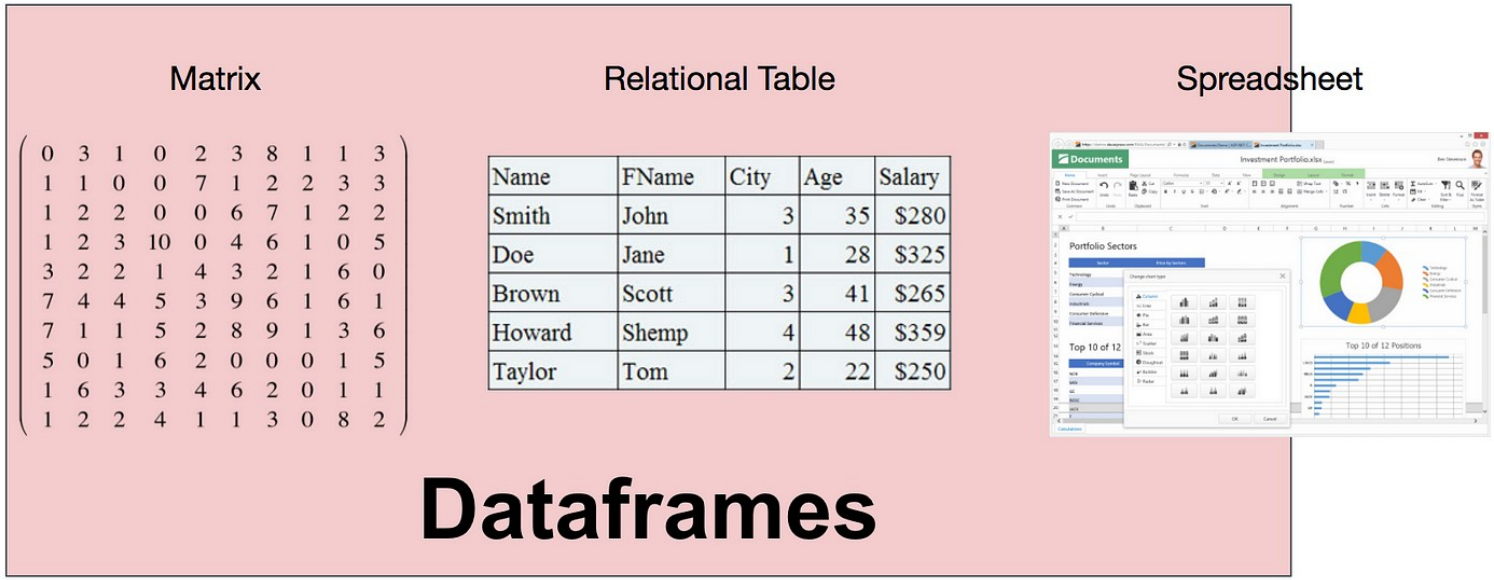
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Terminology drift

- Preventing the Death of the Dataframe (http://towardsdatascience.com/preventing-the-death-of-the-dataframe-8bca1c0f83c8)
  - Dataframes are losing their statistical computing and machine learning roots:



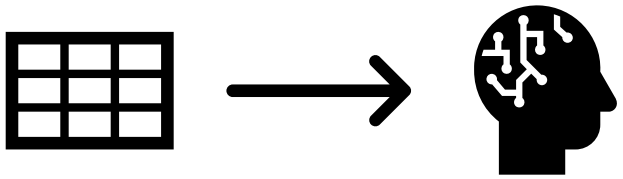
- Dataframe supports:
  - Linear algebra (matrices!)
  - Relational algebra (tables!)
  - Some cell formulae (spreadsheets!)



0	3	1	0	2	3	8	1	1	3
1	1	0	0	7	1	2	2	3	3
1	2	2	0	0	6	7	1	2	2
1	2	3	10	0	4	6	1	0	5
3	2	2	1	4	3	2	1	6	0
7	4	4	5	3	9	6	1	6	1
7	1	1	5	2	8	9	1	3	6
5	0	1	6	2	0	0	0	1	5
1	6	3	3	4	6	2	0	1	1
1	2	2	4	1	1	3	0	8	2

Name	FName	City	Age	Salary
Smith	John	3	35	$280
Doe	Jane	1	28	$325
Brown	Scott	3	41	$265
Howard	Shemp	4	48	$359
Taylor	Tom	2	22	$250

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OBJECT-ATTRIBUTE DATA AND AI

- ✓ Data and decision making
- ✓ AI and AGI
- ✓ Machine leaning : train, test, and validate

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Machine learning – focusing on dataset

- Variables: features and targets
  - **Label** as target value
- Feature engineering
  - Datasets
    - Train, test, validate!

	$Feature_1$	…	$Feature_i$	…	$Target_j$
Obj_1		…	V	…	V
…	…	…	…	…	
Obj_q	V	…		…	
…	…	…	…	…	
Obj_p	V	…	V	…	

Label!

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Train, test, and validation sets – cats and dogs

MLU-EXPLAIN | Introduction | The Split | Train Set | Model | Validation Set | Test Set | Summary

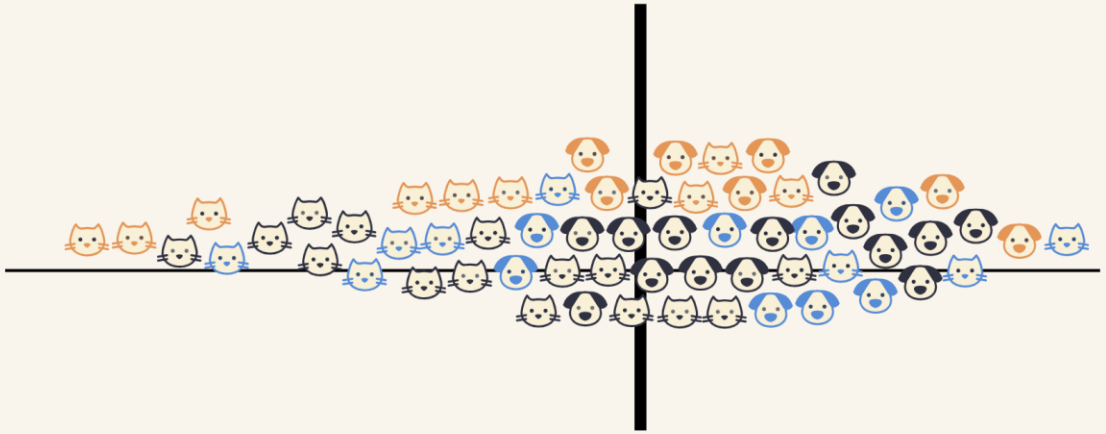
Model Features: None | Weight | Fluffiness | Both

The Testing Set

Once we have used the validation set to determine the algorithm and parameter choices that we would like to use in production, the test set is used to approximate the models's true performance in the wild. It is the final step in evaluating our model's performance on unseen data.

We should never, under any circumstance, look at the test set's performance before selecting a model.

Peeking at our test set performance ahead of time is a form of overfitting, and will likely lead to unreliable performance expectations in production. It should only be checked as the final form of evaluation, after the validation set has been used to identify the best model.



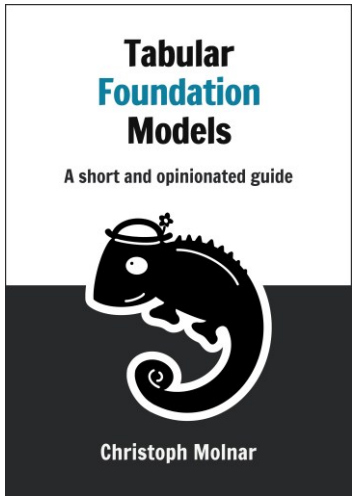
dataset	feature	# cat right	# cat wrong	# dog right	# dog wrong	accuracy
test	weight	7	2	5	2	75.0%
validation	weight	5	3	6	2	68.8%

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(C) 2023, Train,Test, and Validation Sets (https://mlu-explain.github.io/train-test-validation/)

Tabular models → tabular foundation models

- Molnar C. Tabular Foundation Models: A Short and Opinionated Guide (http://tabularfoundationmodels.com)
  - Good introduction



- TabArena: A Living Benchmark for Machine Learning on Tabular Data (http://arxiv.org/abs/2506.16791)
  - TabArena Leaderboard (http://huggingface.co/spaces/TabArena/leaderboard)
    - See next slide



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- TabArena
- 2026-10-16

- Legend:

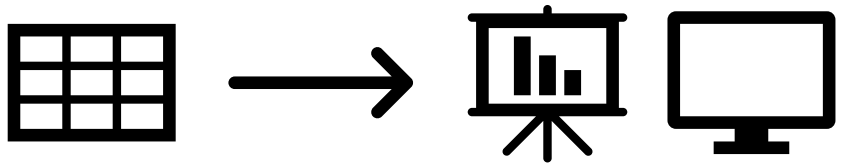
Foundation Model

Neural Network

Tree-based

Type	Model	Overall	What it costs		By Task				By Dataset Size		By Target		
			Fit (s/1K)	Infer (s/1K)	Class.	Regr.	Binary	Multi.	Small	Medium	Balanced	Imbalanced	Extreme
	Kumo-Tabular (default)	1947	2.39	0.253	1904	2343	1879	2072	1873	2396	1894	2148	2059
	LimiX-2 (default) $	1940	30.94	9.028	1915	2220	1910	1983	1884	2286	1926	2016	1858
	Kumo-Tabular-Medium (default)	1889	1.37	0.151	1858	2201	1846	1951	1834	2219	1852	2027	1946
	TabPFN-3.5 (default) $	1853	1.84	0.473	1833	2102	1837	1858	1780	2297	1837	1932	1831
	Kumo-Tabular-Small (default)	1792	0.80	0.092	1764	2077	1752	1855	1754	2032	1763	1902	1851
	Causilo (default) $	1777	0.68	0.104	1760	2006	1740	1894	1752	1962	1755	1870	1826
	TabFM (default) $	1774	16.12	6.653	1765	1970	1801	1686	1745	1978	1788	1784	1621
	TabPFN-3.5-Fast (default) $	1768	0.72	0.152	1770	1911	1775	1787	1709	2108	1757	1829	1736
	Mitra-v2 (default)	1754	57.14	0.878	1745	1946	1738	1812	1713	2005	1740	1826	1692
	EXAONE-Tabular (default) $	1738	2.00	0.580	1744	1865	1741	1794	1698	1978	1692	1901	1857
	TabPFN-3 (default) $	1625	1.29	0.382	1621	1777	1647	1562	1608	1759	1636	1637	1506
	Xiaomi-TabLDM (default)	1583	1.12	0.255	1576	1754	1583	1576	1565	1717	1581	1619	1522
	TabPFN-2.6 (default) $	1577	5.48	0.555	1573	1724	1571	1607	1567	1685	1560	1646	1557
	TabICLv2 (default)	1560	0.75	0.139	1567	1664	1584	1533	1554	1653	1558	1595	1475
	RealTabPFN-2.5 (tuned + ensembled) $	1560	2040.22	8.908	1552	1721	1530	1679	1568	1613	1561	1588	1558
	TabDPT-1.3 (default)	1516	0.57	0.176	1514	1642	1526	1496	1506	1615	1548	1481	1414
	RealMLP (tuned + ensembled)	1474	2950.72	11.975	1462	1642	1462	1483	1454	1602	1457	1541	1553
	TabDPT-Turbo (default)	1430	2.07	0.182	1416	1596	1425	1405	1429	1495	1471	1371	1292
	TabDPT (tuned + ensembled)	1429	4910.38	286.537	1391	1710	1393	1406	1428	1493	1437	1439	1436
	TabM (tuned + ensembled)	1421	2450.13	2.247	1440	1452	1444	1445	1406	1523	1385	1533	1554
	LightGBM (tuned + ensembled)	1409	417.05	2.639	1414	1488	1412	1442	1375	1570	1383	1495	1454
	CatBoost (tuned + ensembled)	1398	1346.21	0.344	1398	1494	1396	1430	1353	1598	1354	1531	1519
	iLTM (tuned + ensembled)	1381	12685.08	464.370	1389	1441	1401	1362	1345	1544	1364	1443	1395
	ModernNCA (tuned + ensembled)	1364	4621.67	8.138	1346	1531	1344	1375	1310	1586	1348	1424	1377
	ChimeraBoost (tuned + ensembled)	1363	518.43	0.469	1372	1418	1389	1325	1294	1639	1307	1524	1425
	XGBoost (tuned + ensembled)	1357	693.49	1.689	1363	1417	1364	1380	1314	1536	1316	1477	1476
	LimiX (default) $	1341	26.46	6.242	1391	1250	1363	1538	1388	1262	1342	1358	1296
	xRFM (tuned + ensembled)	1333	846.89	2.549	1318	1467	1313	1361	1322	1412	1308	1412	1513

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COMMON WORDS ABOUT DATAVIZ (PREWORDS)

- ✓ Common problems
- ✓ Subtopics
- ✓ Some sources and references

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Data Visualization as Art and Technology

- What can be “visualized”?
- How can we systematize visualization methods?
- How to scale rendering methods?
- What tools exist? And where is it applicable?
- What haven't you tried yet? And why…



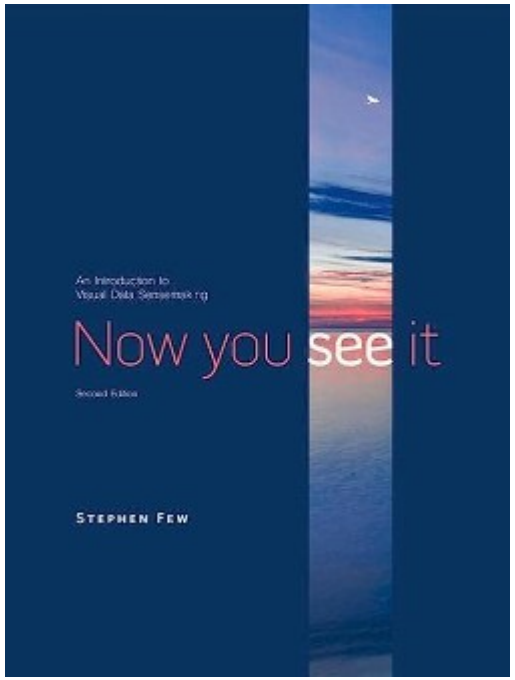
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Some sources

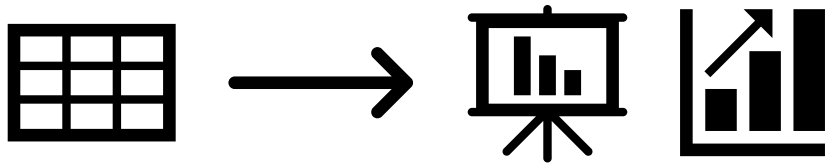
- 47 *Top Data Visualization Books* (http://infogram.com/blog/data-visualization-books/)

- *Tufte E.R. The Visual Display of Quantitative Information*. 2nd *ed. Graphics Press*, 2001. 200 *p*.
  - *THE WORK OF EDWARD TUFTE AND GRAPHICS PRESS* (http://www.edwardtufte.com/tufte/books_vdqi)

- *Few S. Now You See It: An Introduction to Visual Data Sensemaking. Analytics Press*; 2nd *ed.*, 2021. 301 *p*.



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VISUALIZATION OF “TABLES”

- ✓ Object-attribute data representation
  - ✓ Objects, attributes, values, tuples, and tables
- ✓ Statistics
- ✓ Visual design
  - ✓ Tables and charts
  - ✓ Charts classification
- ✓ Methodologies and tools

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Huge terminological inconvenience

- In English, the term **graph** in the field of visualization often means not a structural diagram (see **graph theory**), but a line chart (or even a histogram, etc.)
- In general, it is better to use term **“chart”**
  - **Чарт** – русский жаргонизм, но удобный, особенно в BI

- Although there is an opinion (shared by many researches) that:
  - **Graph** – when a line is drawn (of a function/formula or of continuous data)
  - **Chart** – when other shapes and symbols (e.g. bars) are used to represent data
  - **Plot** – when points are marked on a coordinate system
    - See. http://english.stackexchange.com/questions/43027/whats-the-difference-between-a-graph-a-chart-and-a-plot

- Note that this is very hampering when moving from one subject area to another…

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Do not forget about the visualization of the tables themselves!

- Design systems always describe both tables and charts
  - Example: Tables and charts (http://www.nottingham.ac.uk/brand/visual/tables-and-charts.aspx)

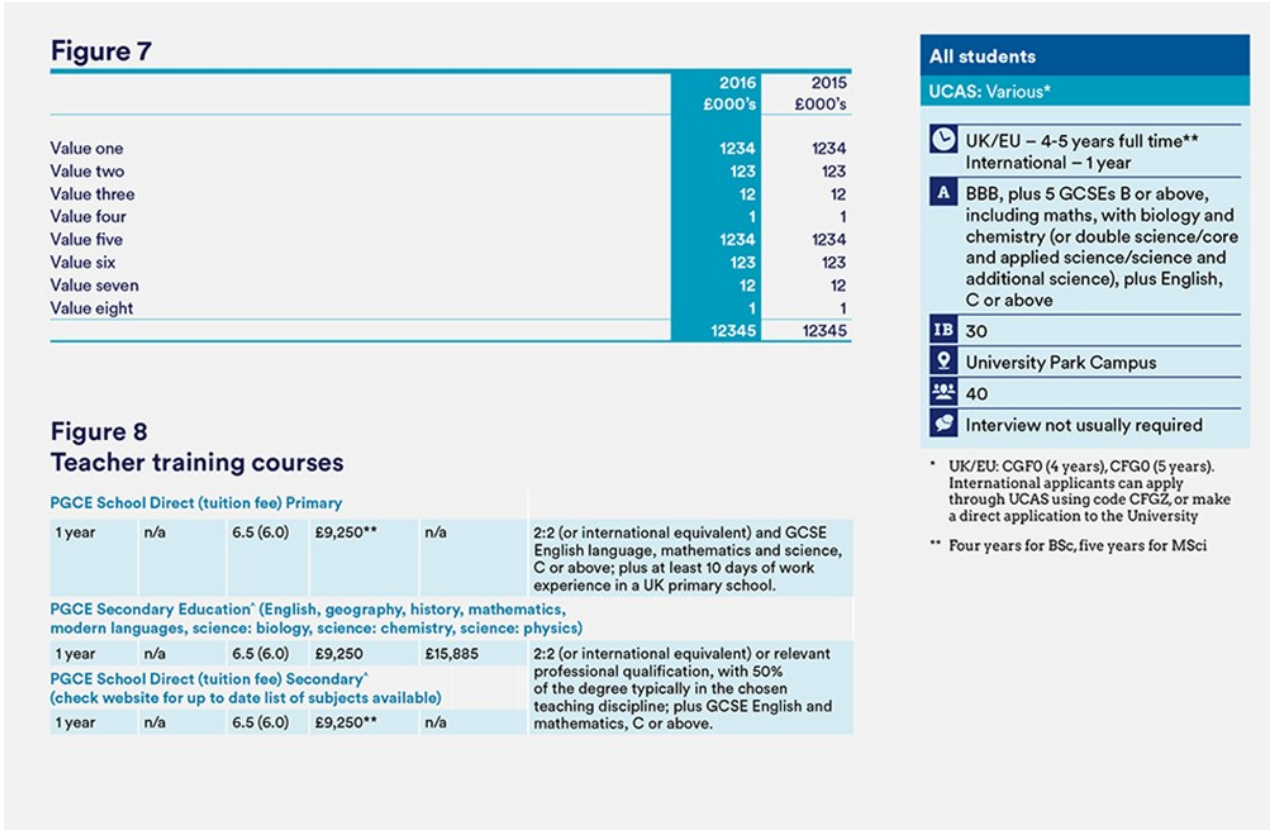


Figure 7

	2016 £000's	2015 £000's
Value one	1234	1234
Value two	123	123
Value three	12	12
Value four	1	1
Value five	1234	1234
Value six	123	123
Value seven	12	12
Value eight	1	1
	12345	12345

Figure 8
Teacher training courses

PGCE School Direct (tuition fee) Primary					
1 year	n/a	6.5 (6.0)	£9,250**	n/a	2:2 (or international equivalent) and GCSE English language, mathematics and science, C or above; plus at least 10 days of work experience in a UK primary school.
PGCE Secondary Education* (English, geography, history, mathematics, modern languages, science: biology, science: chemistry, science: physics)					
1 year	n/a	6.5 (6.0)	£9,250	£15,885	2:2 (or international equivalent) or relevant professional qualification, with 50% of the degree typically in the chosen teaching discipline; plus GCSE English and mathematics, C or above.
PGCE School Direct (tuition fee) Secondary* (check website for up to date list of subjects available)					
1 year	n/a	6.5 (6.0)	£9,250**	n/a	

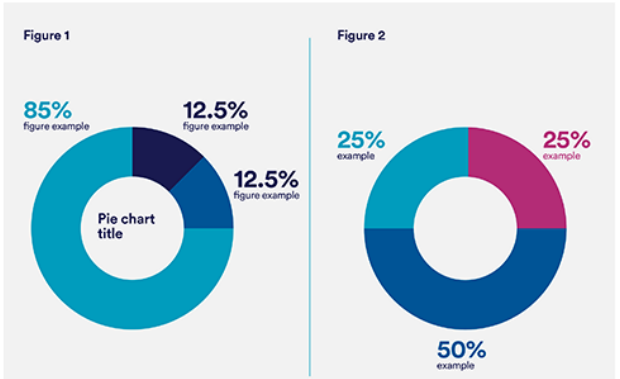
All students

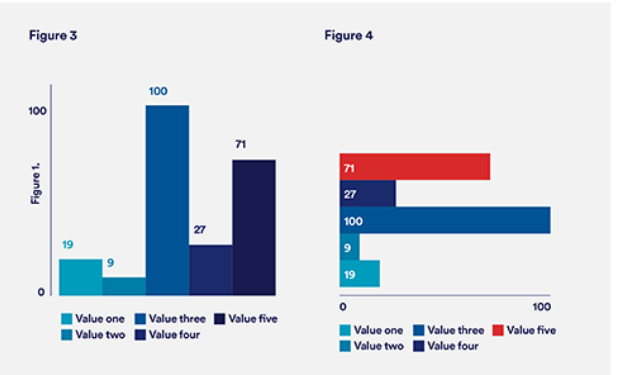
UCAS: Various*

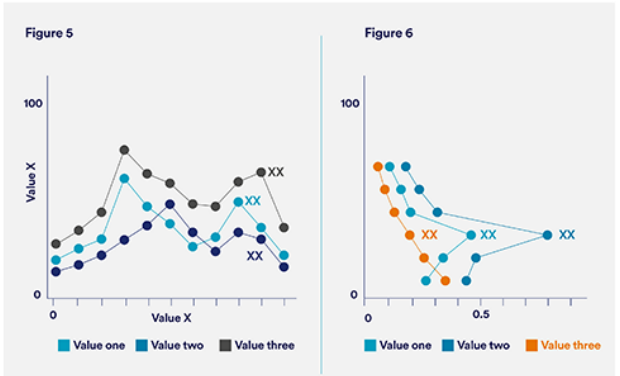
- UK/EU – 4-5 years full time** International – 1 year
- A: BBB, plus 5 GCSEs B or above, including maths, with biology and chemistry (or double science/core and applied science/science and additional science), plus English, C or above
- IB: 30
- University Park Campus
- 40
- Interview not usually required

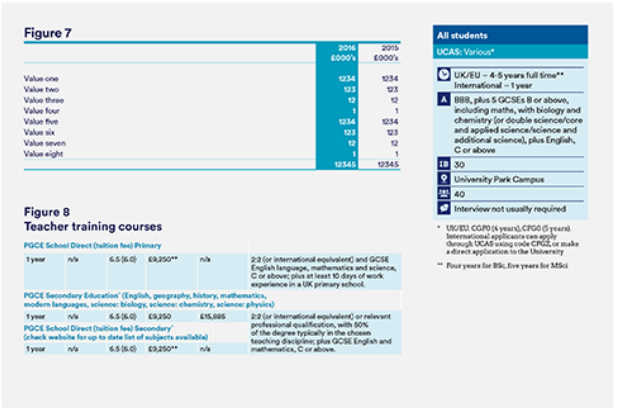
* UK/EU: CGF0 (4 years), CFG0 (5 years). International applicants can apply through UCAS using code CFGZ, or make a direct application to the University

** Four years for BSc, five years for MSci





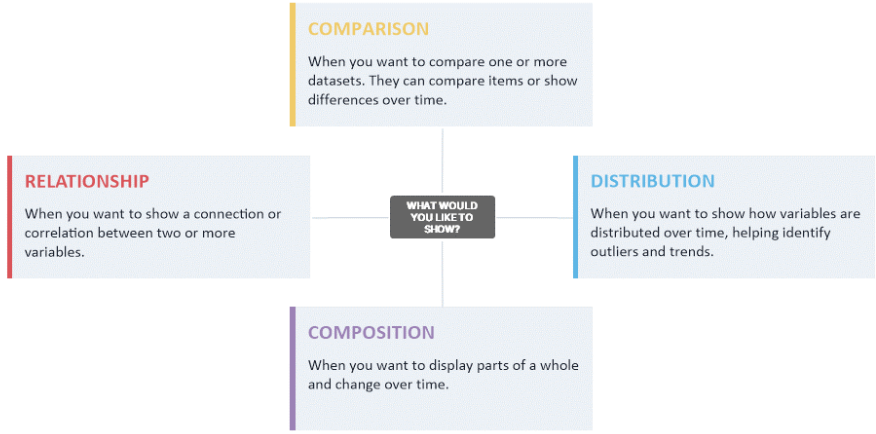


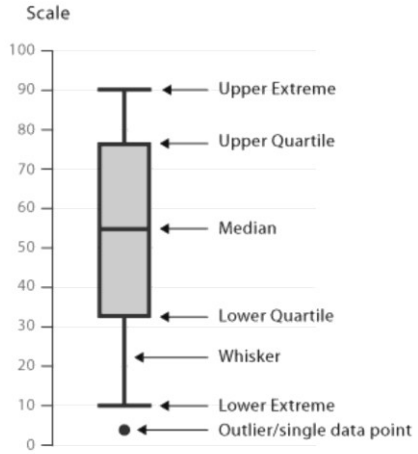


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«Standard» diagrams (charts)

- Classical classification of goals:
  - Comparison
  - Relationship
  - Distribution
  - Composition

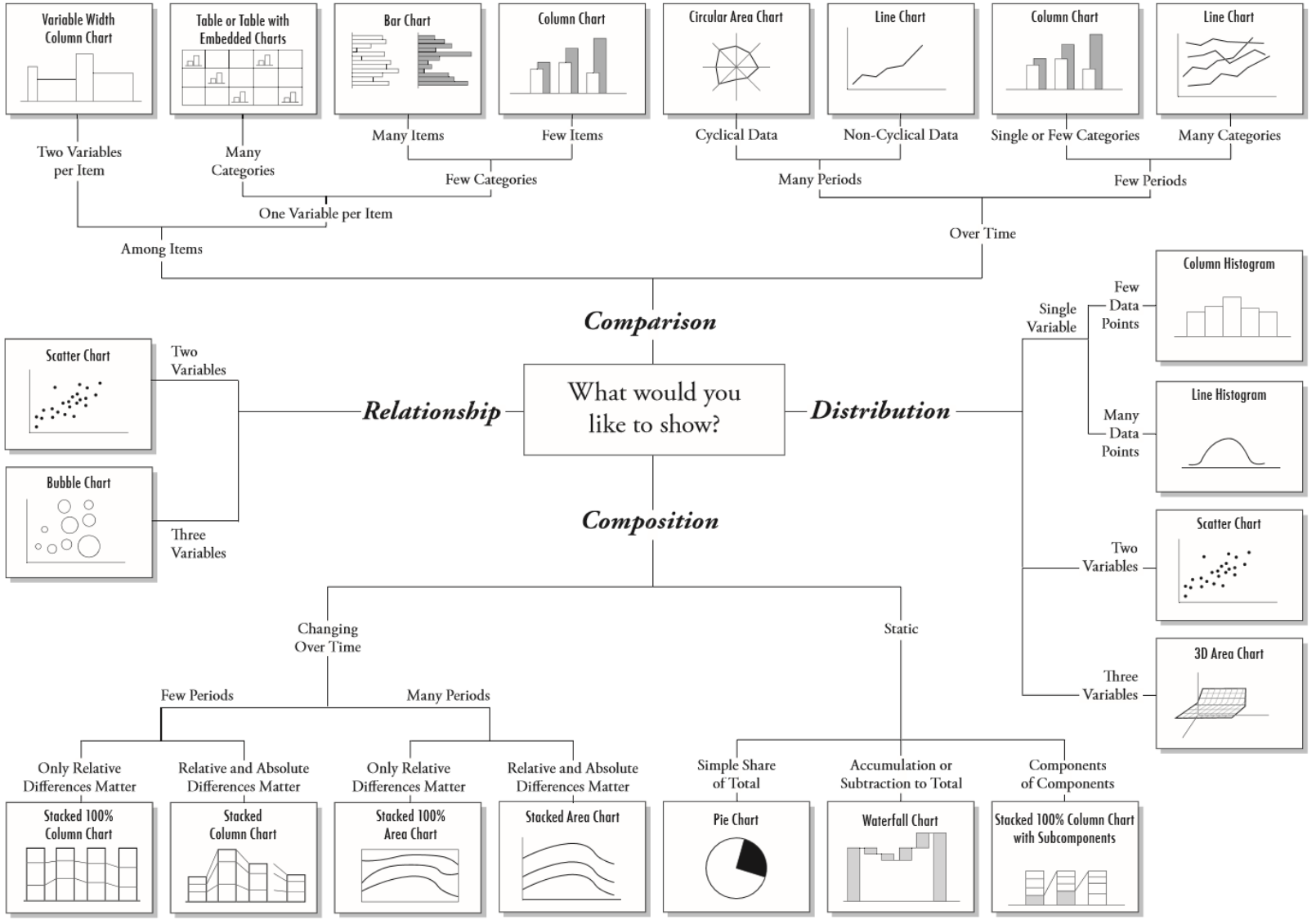




- References:
  - Nuts and Bolts of Chart & Graph Types [infographic] (http://online-behavior.com/analytics/chart-types)
  - A classification of chart types (http://www.excelcharts.com/blog/classification-chart-types/)
  - Hardin M., Hom D, Perez R., Williams L. Which chart or graph is right for you? (http://www.tableau.com/sites/default/files/media/which_chart_v6_final_0.pdf)
  - Ahluwalia H. How to choose the Right Chart for Data Visualization (http://www.linkedin.com/pulse/ultimate-guide-choosing-right-type-chart-your-data-gruntpro)
  - …

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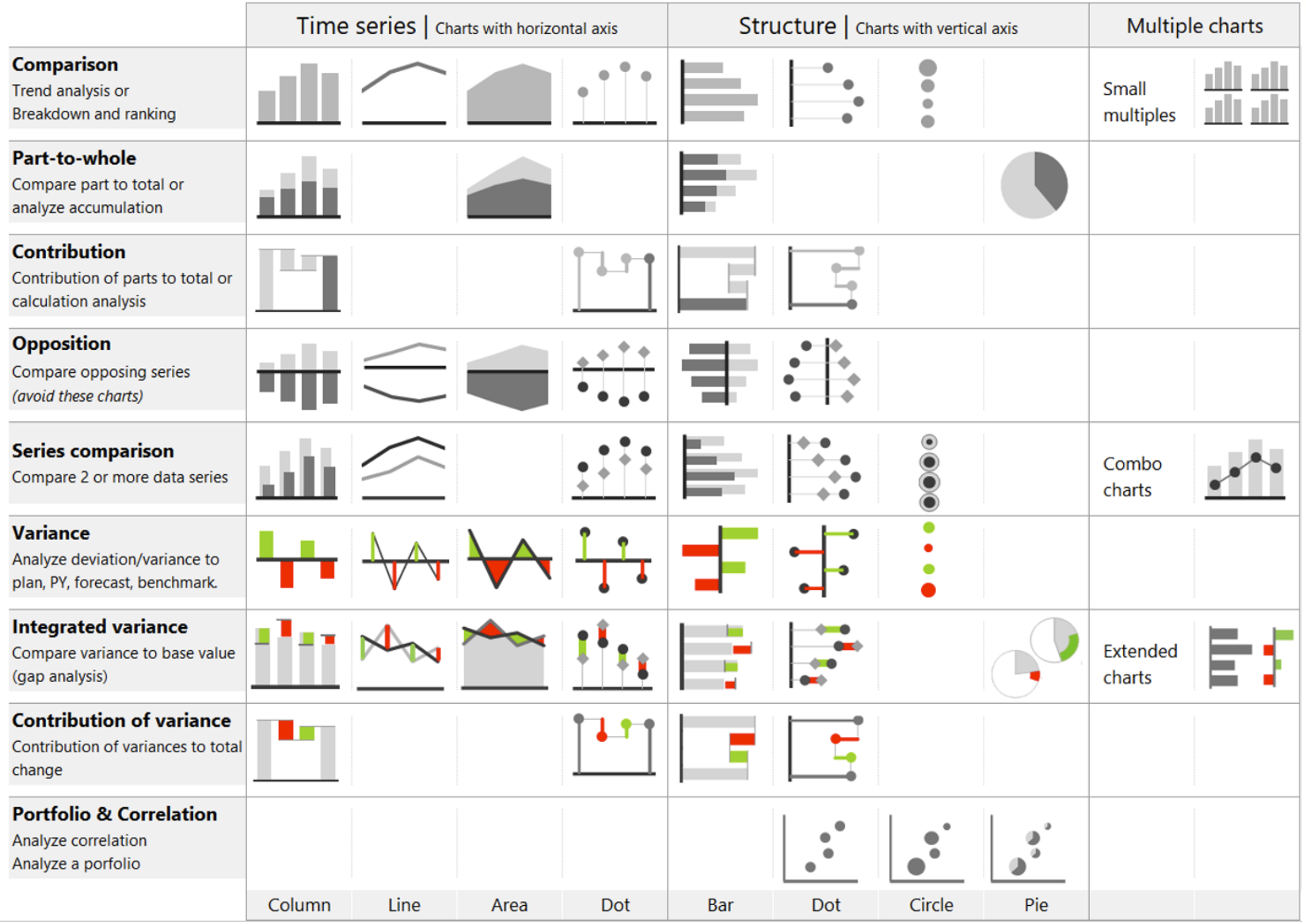
Chart Suggestions – A Thought-Starter



© 2009, A. Abela

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A taxonomy of business charts by Task, Orientation, and Shape

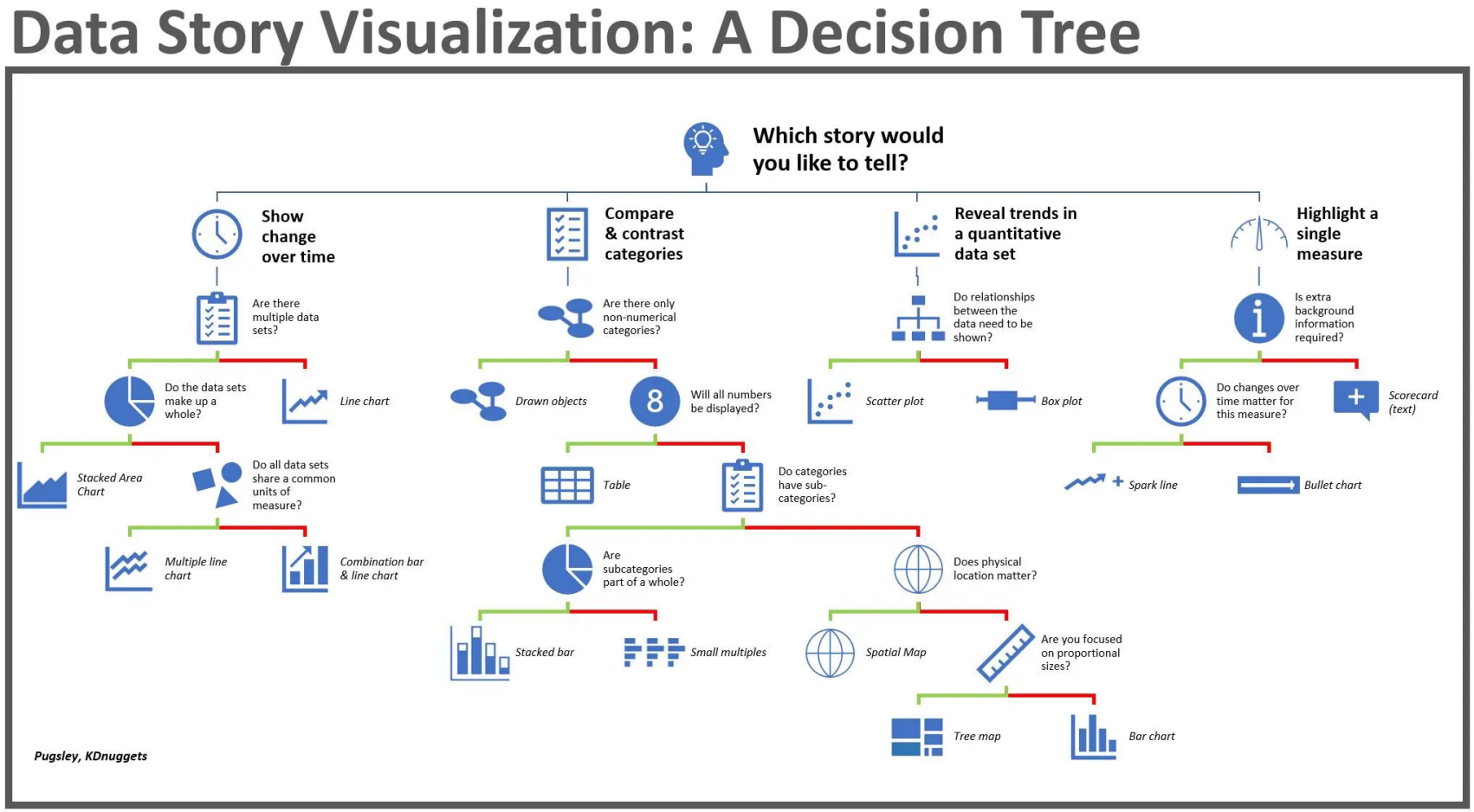


© 2026, Andrej Lapajne, How to choose the correct chart type for your Power BI report (http://zebrabi.com/how-to-choose-the-right-chart-power-bi/)

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Other Data Visualization Chart Cheatsheets

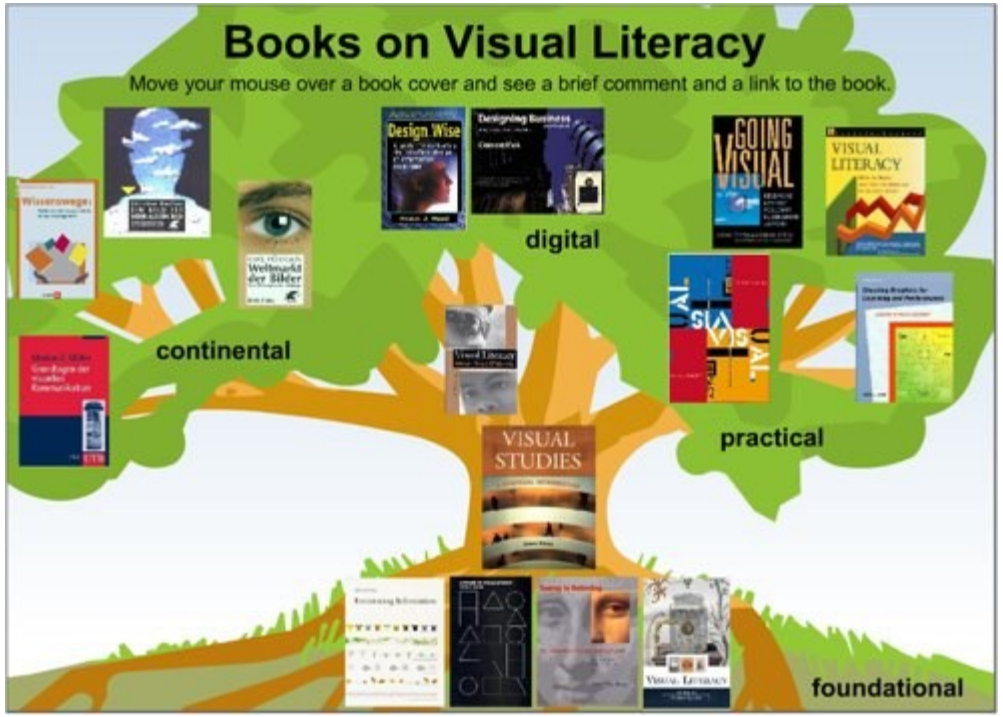
- Satish Chandra Gupta – Data Visualization Chart Cheatsheets (http://www.ml4devs.com/newsletter/006-data-visualization-chart-cheatsheets/)

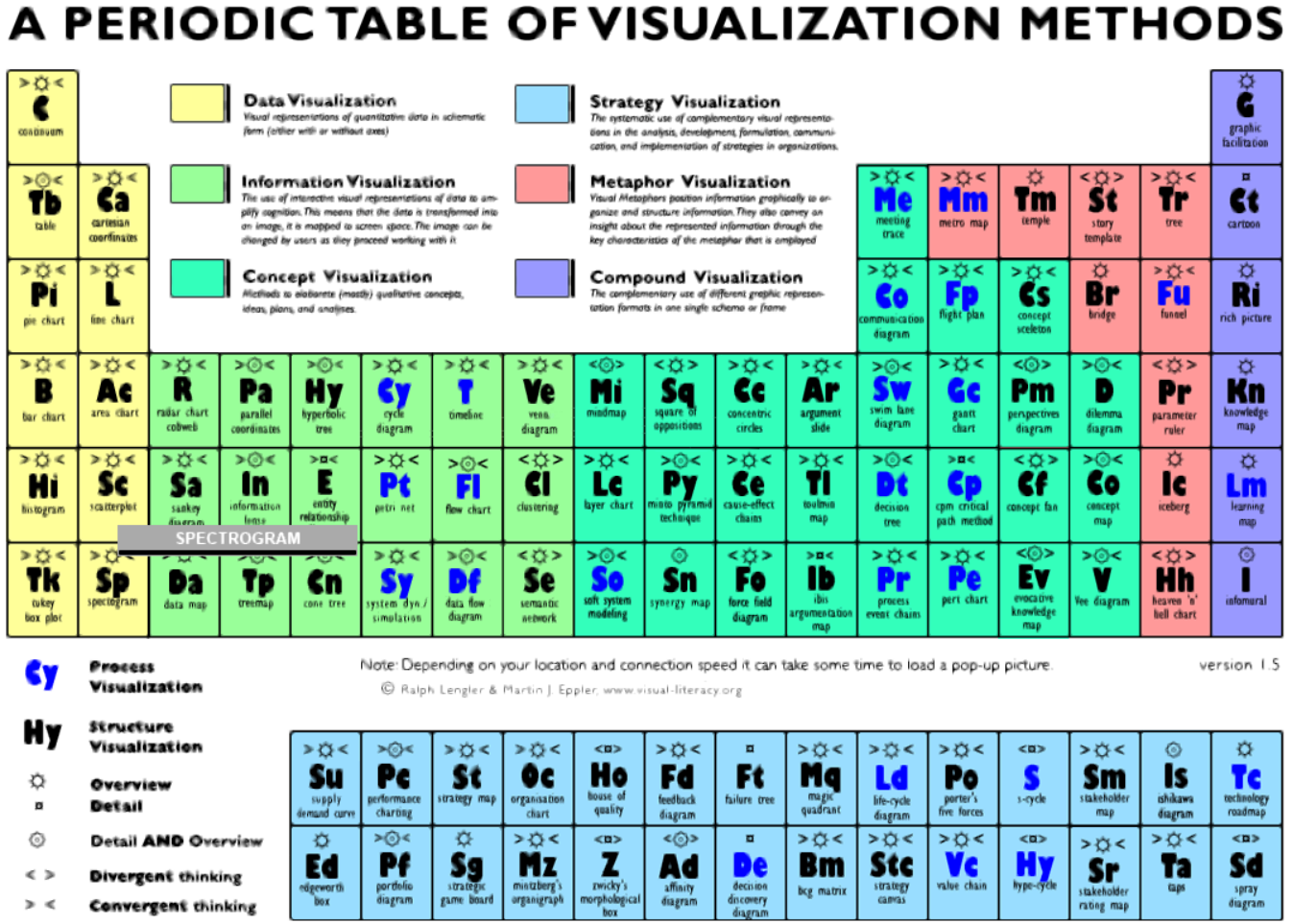


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Visual Literacy Guidelines

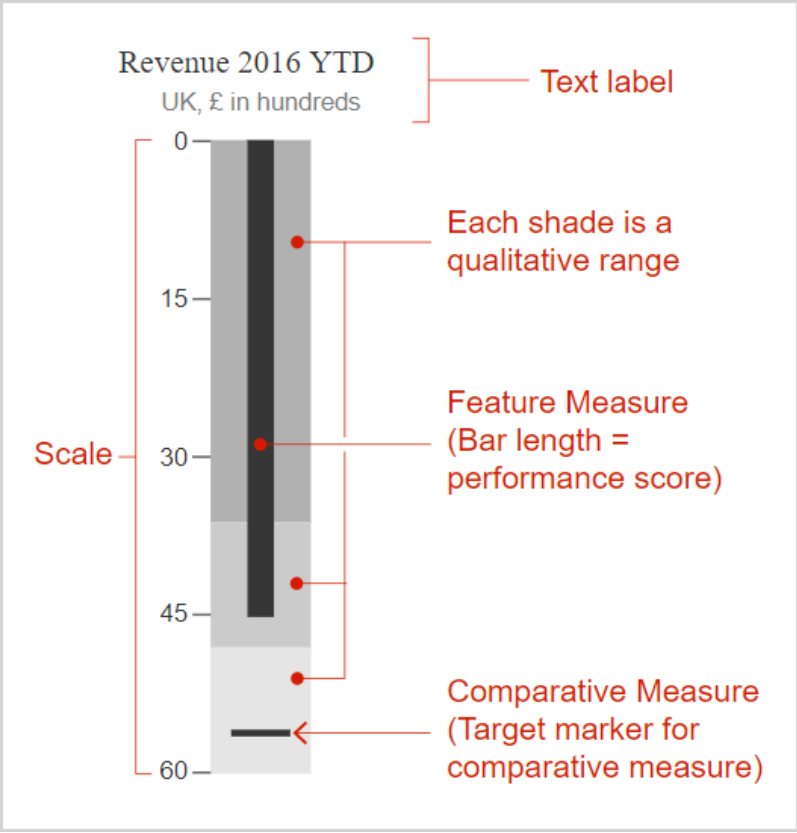
- *Visual Literacy – Visualization Skills Portal* (http://www.visual-literacy.org/maps/)





The Data Visualisation Catalogue

- Catalogue of standard visualizations (http://datavizcatalogue.com)
  - Good description + “anatomy”
  - **Bullet Graph** for example:

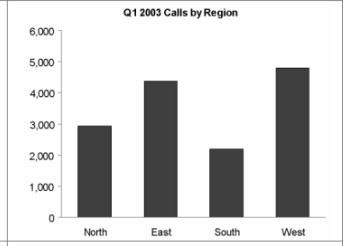
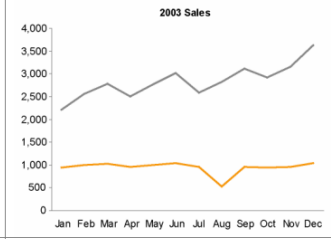
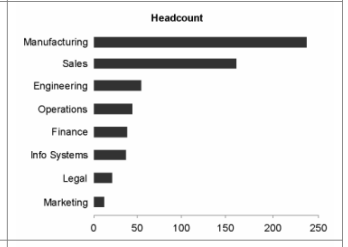
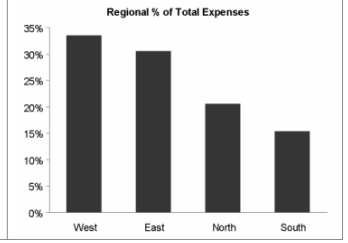


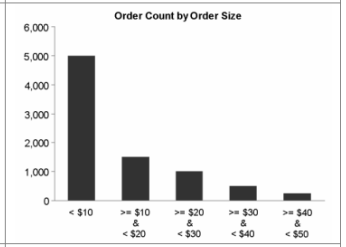
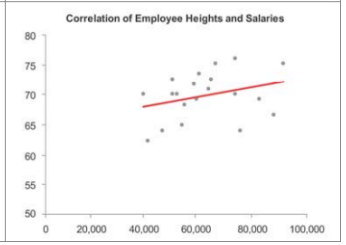


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Attribute measurement scales and chart types

- Few S. Eenie, Meenie, Minie, Moe: Selecting the Right Graph for Your Messag. Perceptual Edge, 2004 (http://www.perceptualedge.com/articles/ie/the_right_graph.pdf)

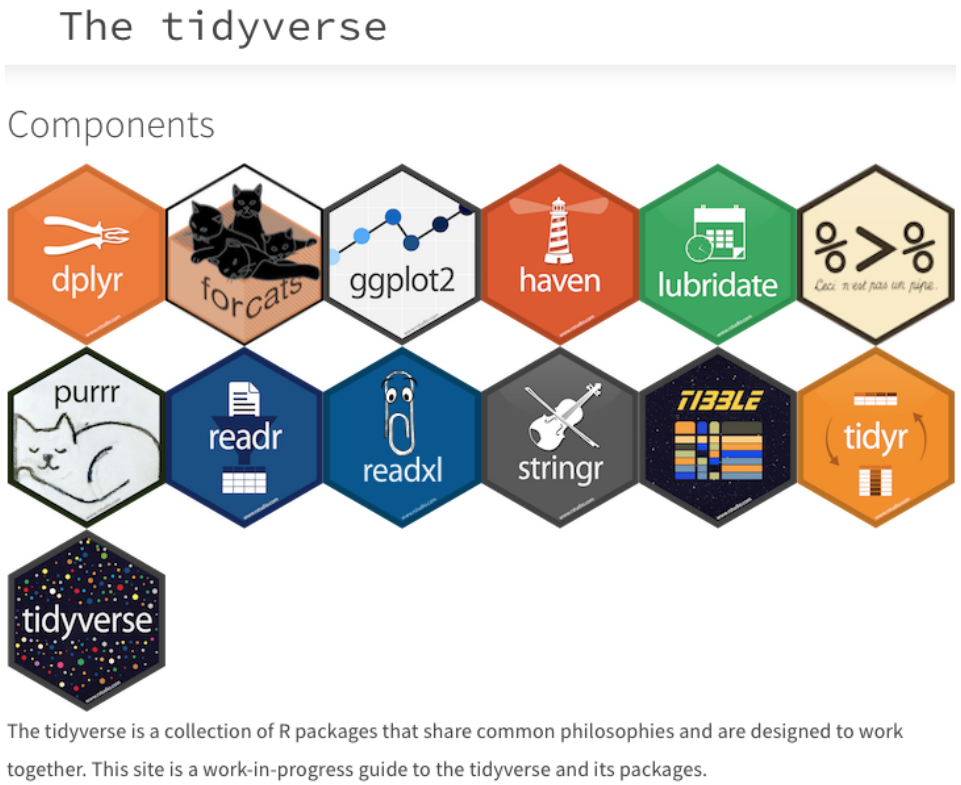
Type/Description	Encoding Methods	Example
Nominal Comparison A simple comparison of the categorical subdivisions of one or more measures in no particular order	• Bars only (horizontal or vertical)	
Time Series Multiple instances of one or more measures taken at equidistant points in time	• Lines to emphasize overall pattern • Bars to emphasize individual values • Points connected by lines to slightly emphasize individual values while still highlighting the overall pattern • Always place time on the horizontal axis	
Ranking Categorical subdivisions of a measure ordered by size (either descending or ascending)	• Bars only (horizontal or vertical) • To highlight high values, sort in descending order • To highlight low values, sort in ascending order	
Part-to-Whole Measures of individual categorical subdivisions as ratios to the whole	• Bars only (horizontal or vertical) • Use stacked bars only when you must display measures of the whole as well as the parts	

Type/Description	Encoding Methods	Example
Deviation Categorical subdivisions of a measure compared to a reference measure, expressed as the differences between them	• Lines to emphasize the overall pattern only when displaying deviation and time-series relationships together • Points connected by lines to slightly emphasize individual data points while also highlighting the overall pattern when displaying deviation and time-series relationships together • Bars to emphasize individual values, but limit to vertical bars when a time-series relationship is included • Always include a reference line to compare the measures of deviation against	
Frequency Distribution Counts of something per categorical subdivisions (intervals) of a quantitative range	• Vertical bars to emphasize individual values (called a *histogram*) • Lines to emphasize the overall pattern (called a *frequency polygon*)	
Correlation Comparisons of two paired sets of measures to determine if as one set goes up the other set goes either up or down in a corresponding manner, and if so, how strongly	• Points and a trend line in the form of a scatter plot • Bars may be used, arranged as a *paired bar graph* or a *correlation bar graph*, if scatter plots are unfamiliar • (Note: For descriptions of these graphs, see my book *Show Me the Numbers*.)	

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Python и *R* viz ecosystems

- **Python** 3.12 or higher (http://www.python.org)
  - Download (http://www.python.org/downloads/)
  - **Matplotlib** is a comprehensive library for creating static, animated, and interactive visualizations in Python (http://matplotlib.org)
    - + extensions
- **R** 4.5 or higher (http://www.r-project.org)
  - Download (http://cran.r-project.org/bin/windows/base/)
  - **ggplot2** is a system for declaratively creating graphics, based on The Grammar of Graphics (http://ggplot2.tidyverse.org)
    - You provide the data, tell ggplot2 how to map variables to aesthetics, what graphical primitives to use, and it takes care of the details

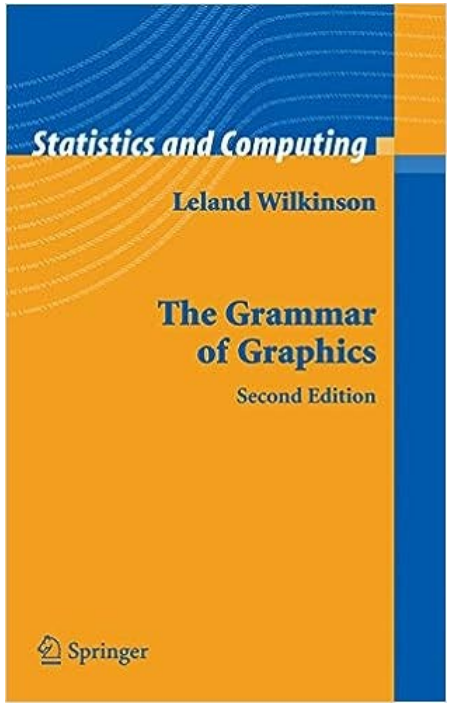


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The grammar of Graphics

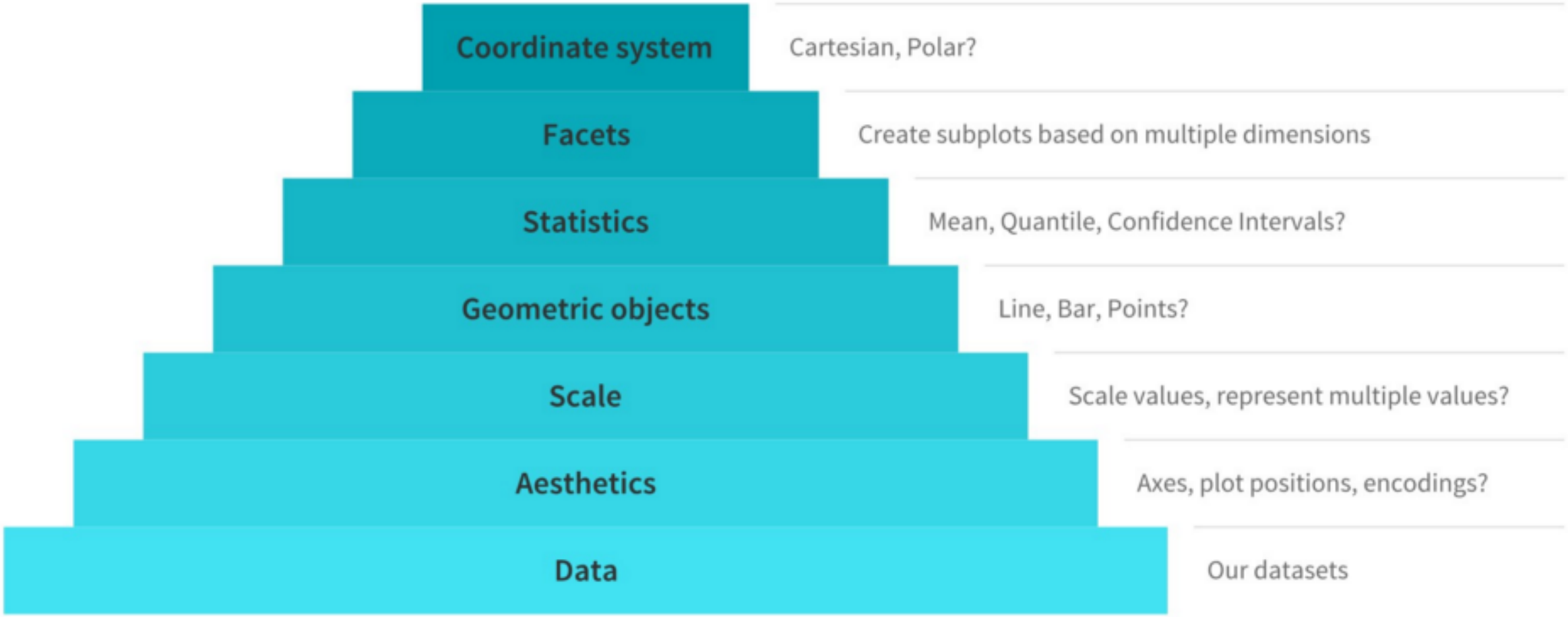
- A unique foundation for producing almost every quantitative graphic found in scientific journals, newspapers, statistical packages, and data visualization systems

- Wilkinson L. The Grammar of Graphics. 2nd ed. Springer, 2005. 709 p.



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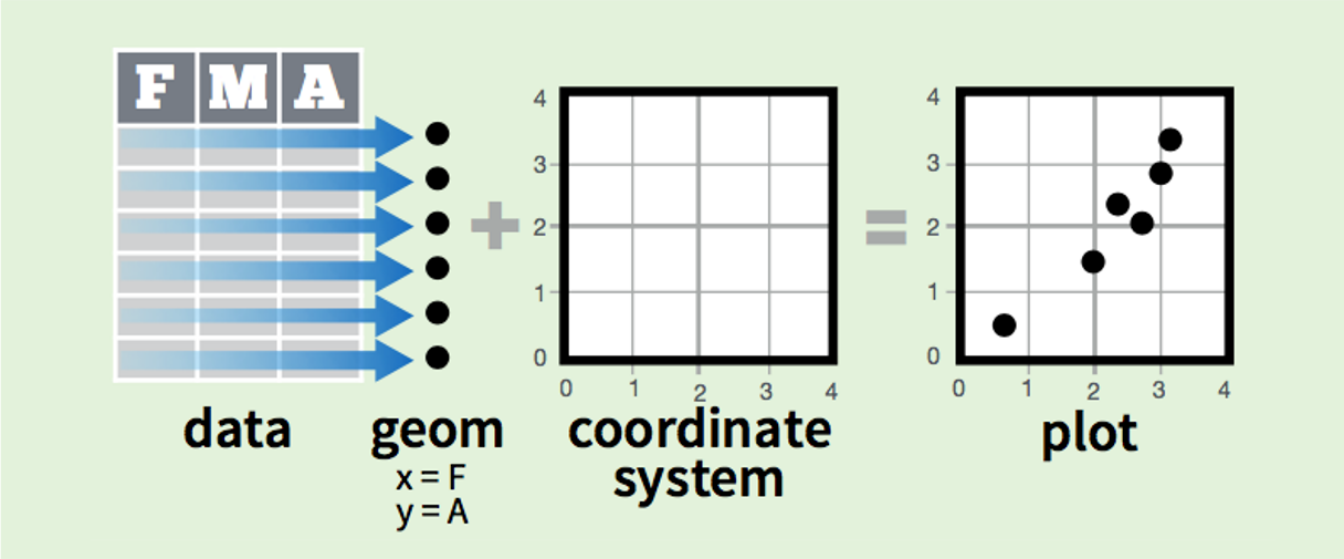
Major components of the Grammar of Graphics



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ggplot2 for R – most popular implementation

- **ggplot2**: Elegant Graphics for Data Analysis (http://ggplot2-book.org)
- ggplot2 and the grammar of graphics (http://r-unimelb.gitbook.io/rbook/putting-the-r-in-art/ggplot2-and-the-grammar-of-graphics)
- Data visualization with ggplot2: Cheatsheet (http://github.com/rstudio/cheatsheets/blob/main/data-visualization.pdf)

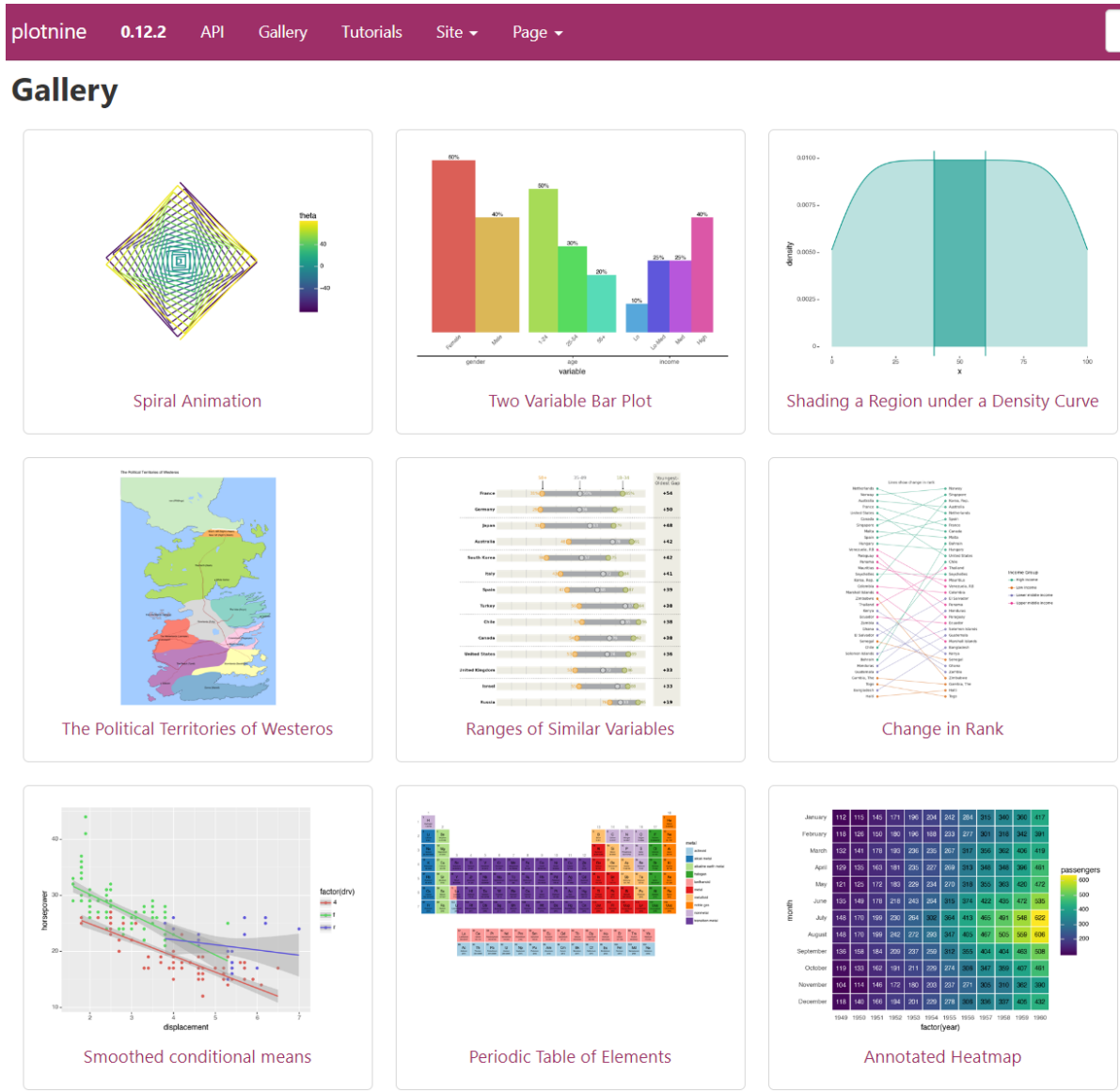


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plotnine for Python

- **plotnine** (http://plotnine.readthedocs.io)
  - An implementation of a grammar of graphics in Python based on ggplot2

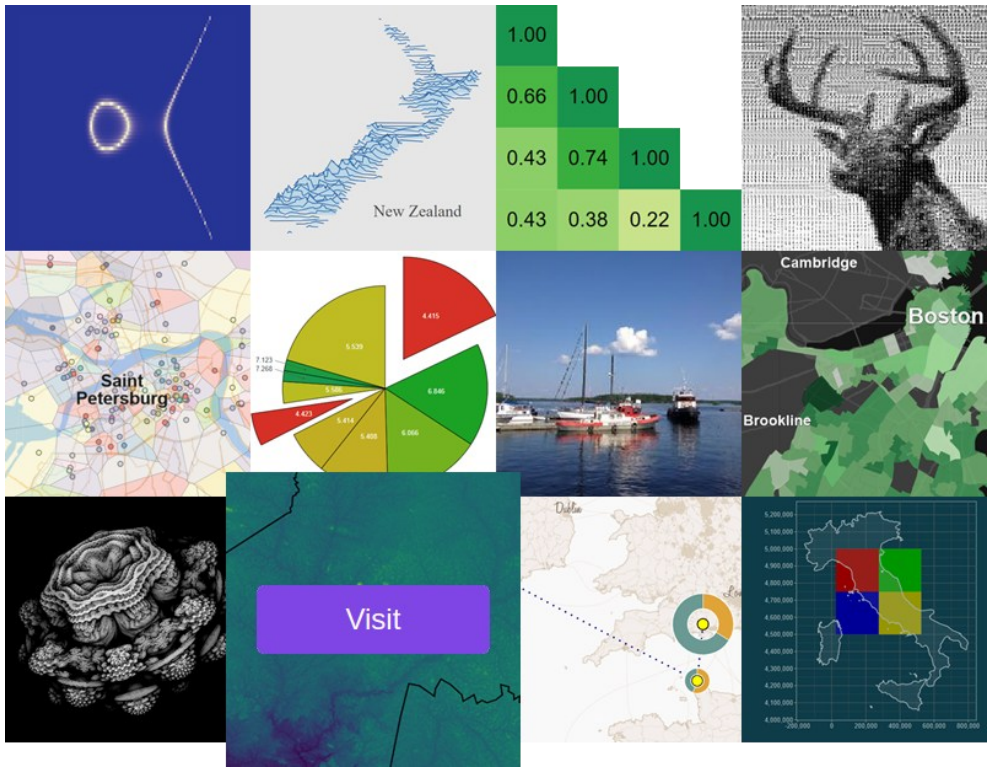


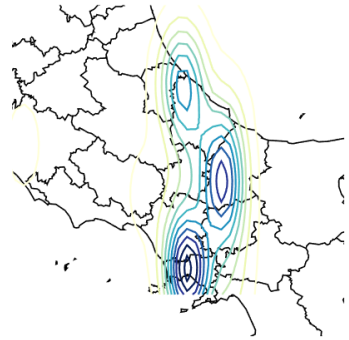


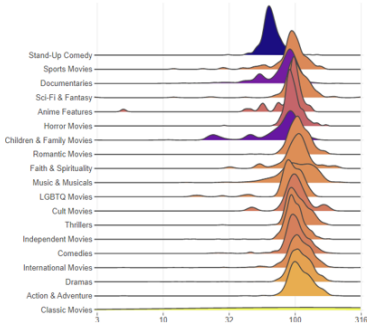
© 2026, Neznanov A.A.

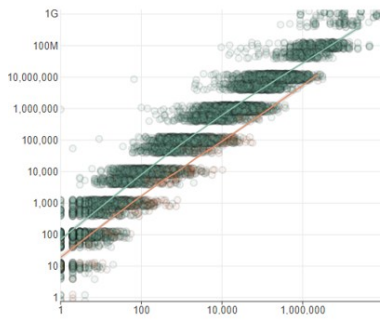
Modern declarative visualization in Python (and Kotlin)

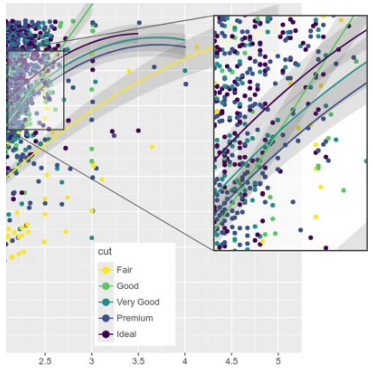
- **Lets-Plot** (http://lets-plot.org)
  - Multiplatform plotting library built on the principles of the Grammar of Graphics
  - Inspired by ggplot2!







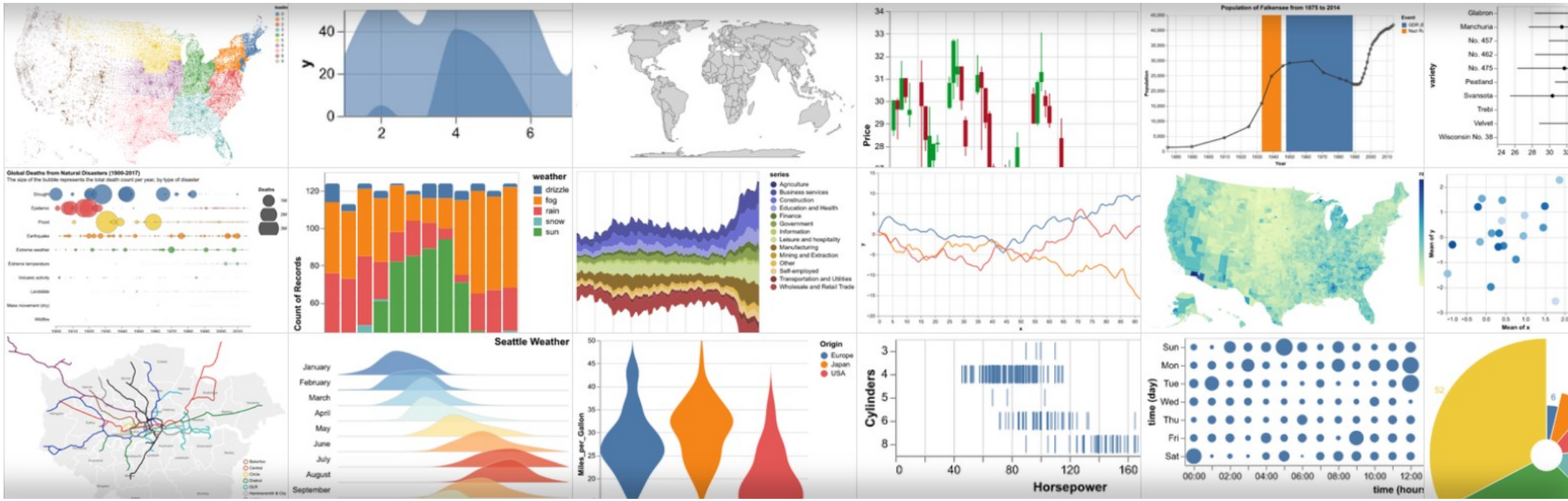




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Interactive Declarative Visualization in Python

- **Vega-Altair** (http://altair-viz.github.io)
  - Simple, friendly and consistent API, built on top of the powerful Vega-Lite grammar
  - Interactive **dynamic** charts!
    - Rich interactive UI with events and selectors



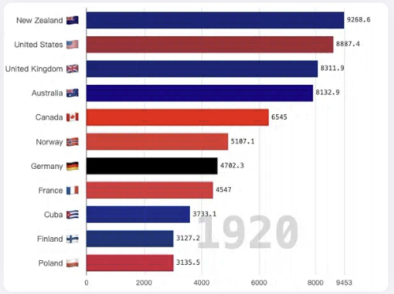
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JS (ECMA Script based languages)

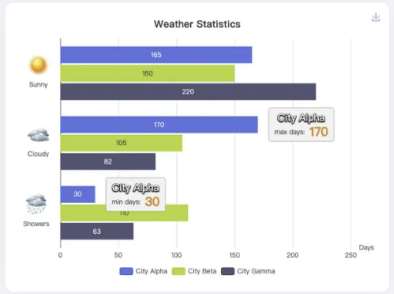
- **ECMAScript** 2017+
  - Standard ECMA-262 (http://www.ecma-international.org/publications-and-standards/standards/ecma-262/)
    - ECMAScript® 2023 (14th edition) language specification

- JavaScript/Jscript/JS = ECMAScript + I/O + framework (jQuery/React/Angular/…)
  - **Chart.js** (http://www.chartjs.org)
    - A set of frequently used chart types

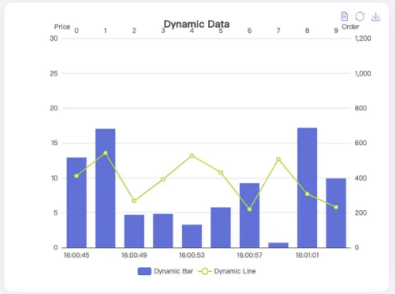
  - Apache **Echarts** (http://echarts.apache.org)
    - An Open Source JavaScript Visualization Library



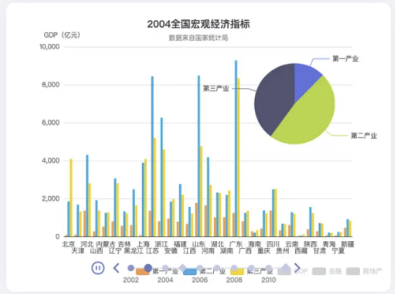
Bar Race



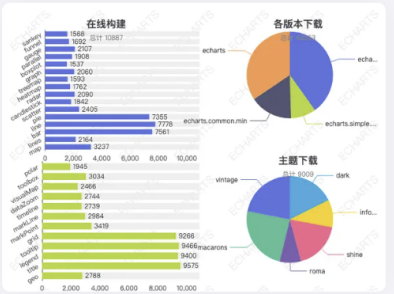
Weather Statistics



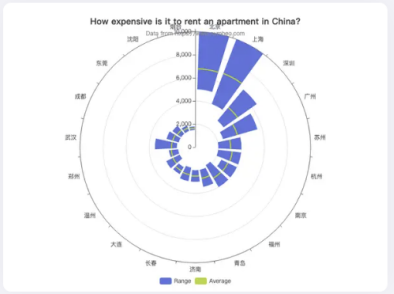
Dynamic Data



Finance Indices 2002



Watermark - ECharts Download

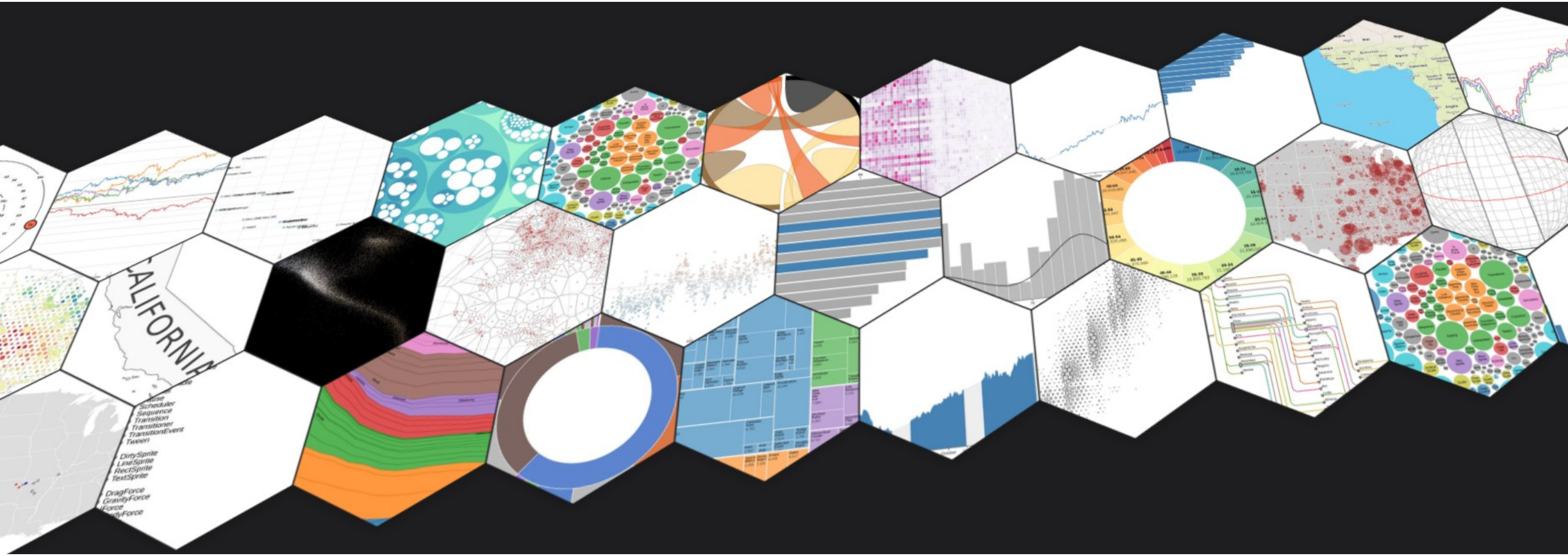


Bar Chart on Polar

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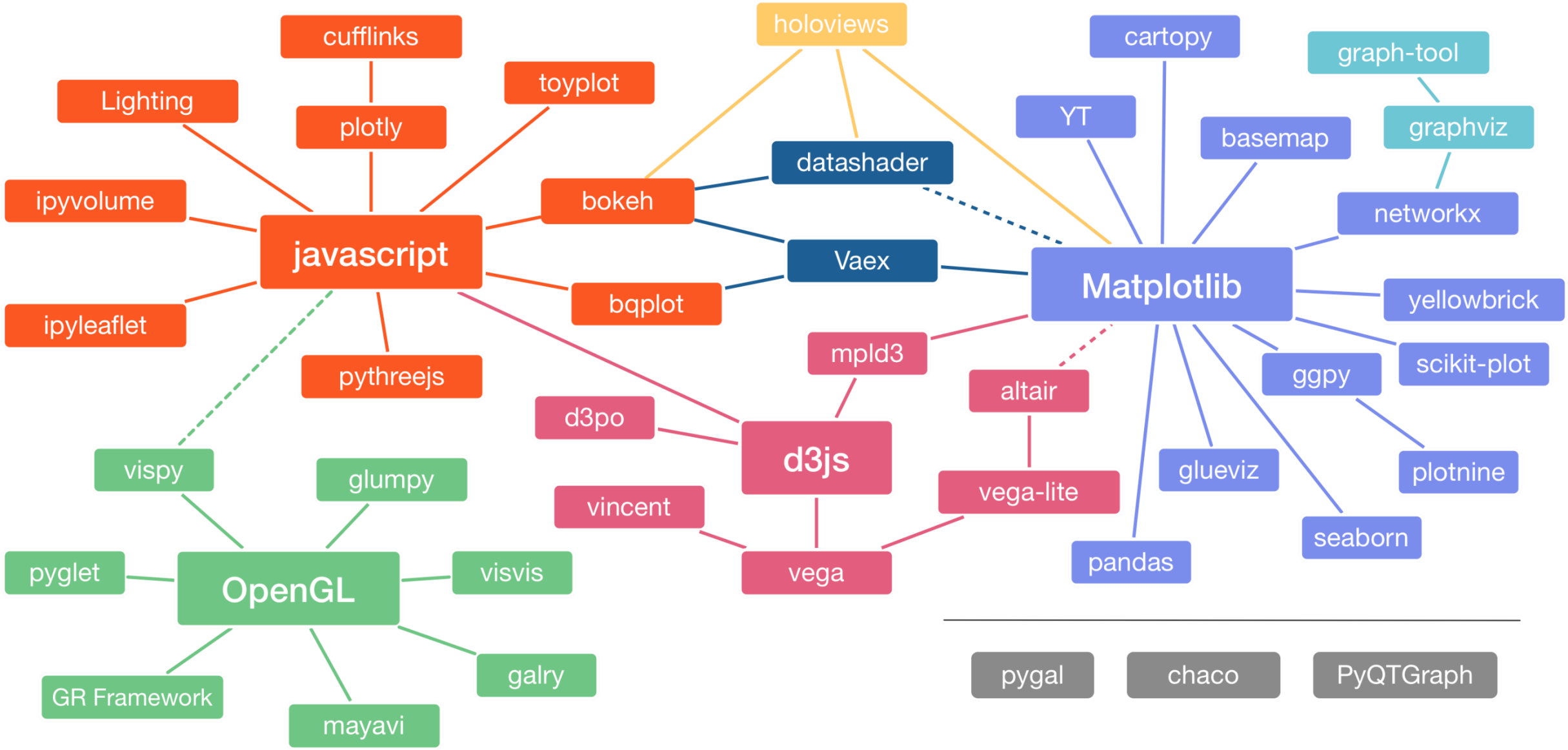
D3.js for JS – universal library for DataVis

- D3.js – Data-Driven Documents (http://d3js.org)
  - D3.js Gallery (http://christopheviau.com/d3list/gallery.html)



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“Legacy” Worlds of DataVis tools



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Two main problem of DataViz

Stupid ugly data presentation

Slow unoptimized vector rendering

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Effective data representation

- Apache **Arrow** – development platform for in-memory analytics (http://arrow.apache.org)
  - Reading and Writing the Apache **Parquet** Format (http://arrow.apache.org/docs/python/parquet.html)



- **Polars** – Lightning-fast DataFrame library for Rust and Python (http://www.pola.rs)
  - Lightning-fast DataFrame library for Rust and Python
  - Rust / Python
  - Lazy execution, end-to-end optimization
  - SQL-style declarative operations



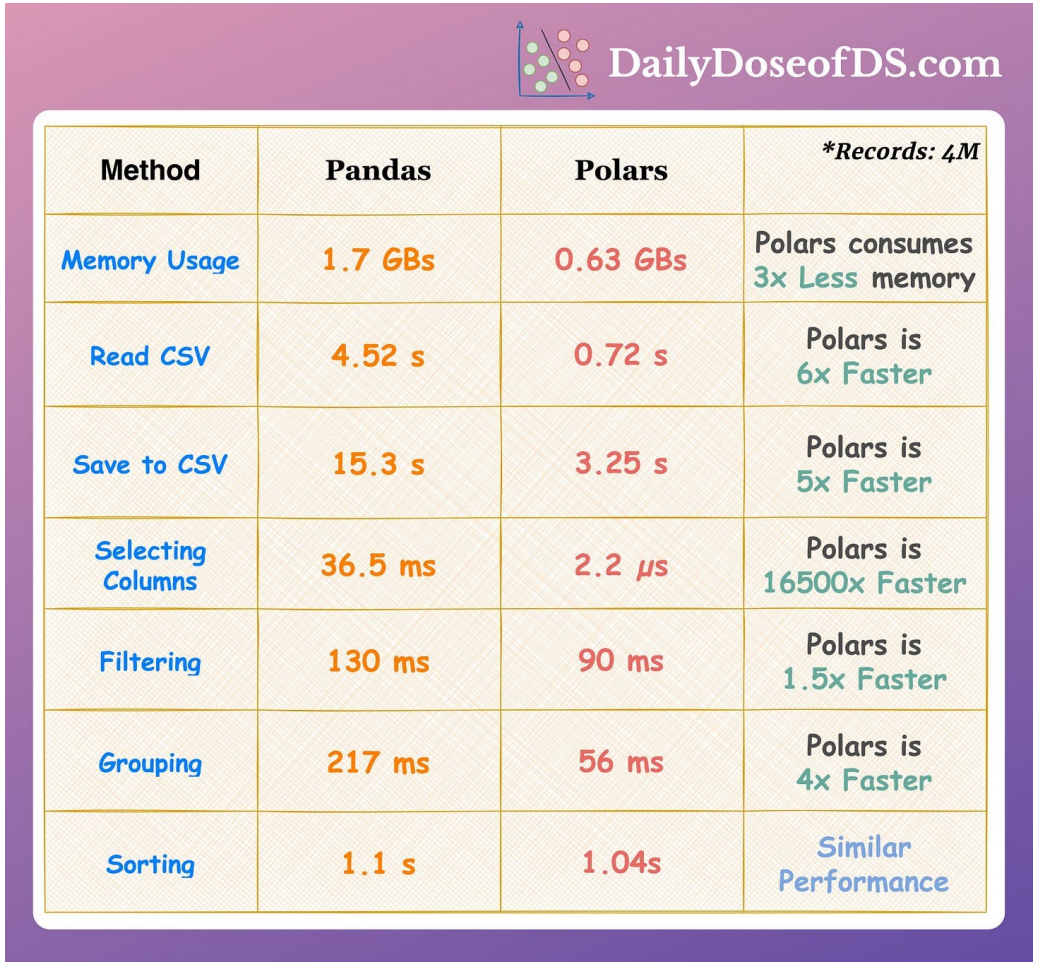
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Pandas 2.0 and legacy problems

- Chawla A. Pandas vs Polars — Run-time and Memory Comparison (http://www.blog.dailydoseofds.com/p/pandas-vs-polars-run-time-and-memory)

- 1.5.3 vs 0.17.1 ->

- Pandas 2.0
  - pd.options.mode.dtype_back = 'pyarrow' -> Oops…
    - Strange behavior (Booleans, string cache, …)
  - Gray zone of syntax
  - Legacy code problems

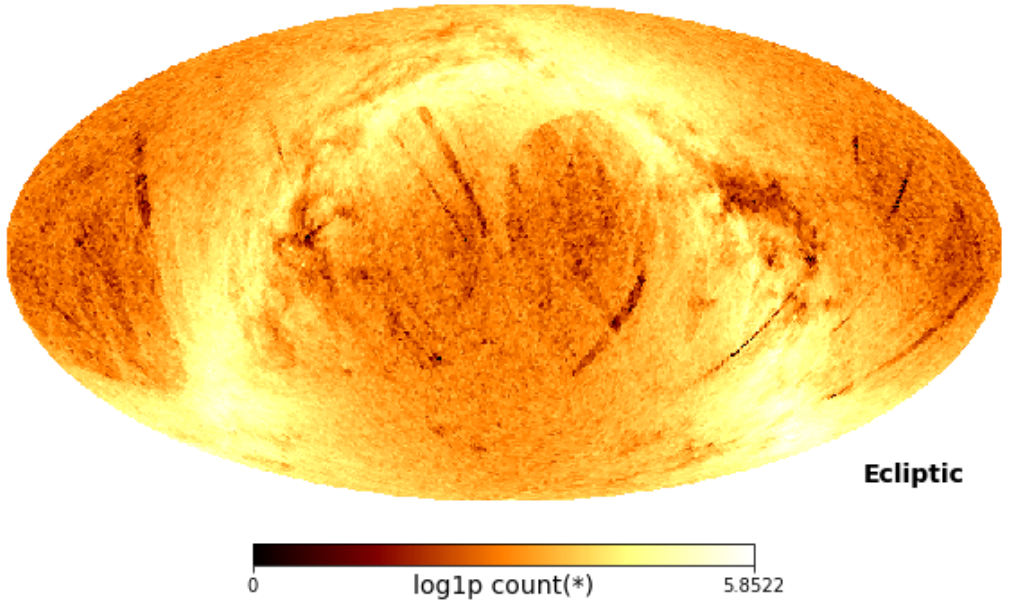


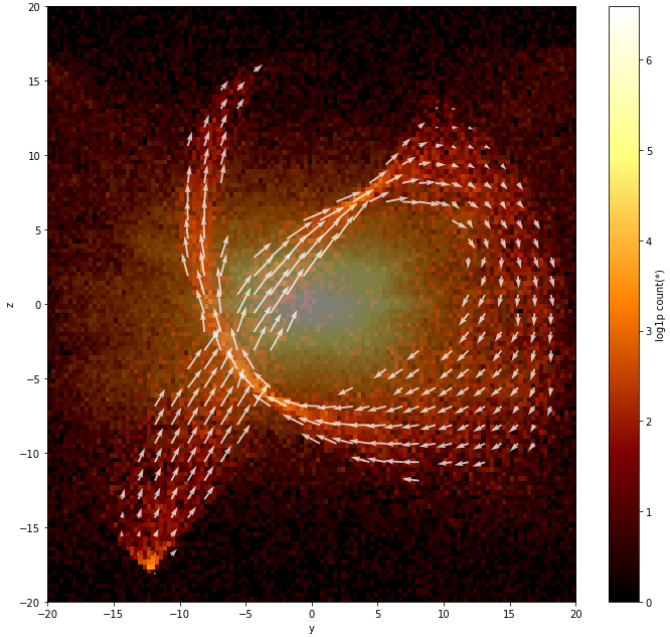
Method	Pandas	Polars	*Records: 4M
Memory Usage	1.7 GBs	0.63 GBs	Polars consumes 3x Less memory
Read CSV	4.52 s	0.72 s	Polars is 6x Faster
Save to CSV	15.3 s	3.25 s	Polars is 5x Faster
Selecting Columns	36.5 ms	2.2 μs	Polars is 16500x Faster
Filtering	130 ms	90 ms	Polars is 1.5x Faster
Grouping	217 ms	56 ms	Polars is 4x Faster
Sorting	1.1 s	1.04s	Similar Performance

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Effective data representation – platforms for mere mortals

- **VAEX** (http://vaex.io)
  - A python library for lazy Out-of-Core DataFrames (similar to Pandas), to visualize and explore big tabular datasets
  - It can calculate statistics such as mean, sum, count, standard deviation etc, on an N-dimensional grid up to a billion (10^9) objects/rows per second
  - Repo (http://github.com/vaexio/vaex)





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DuckDB for local OLAP

- **DuckDB** (http://duckdb.org)
  - An in-process SQL OLAP database management system

Why DuckDB?



Simple and portable

- In-process, serverless
- C++11, no dependencies, single-file build
- APIs for Python, R, Java, Julia, Swift, ...
- Runs on Windows, Linux, macOS, OpenBSD, ...

more →



Feature-rich

- Transactions, persistence
- Extensive SQL support
- Direct Parquet, CSV, and JSON querying
- Joins, aggregates, window functions

more →



Fast

- Optimized for analytics
- Vectorized and parallel engine
- Larger than memory processing
- Parallel Parquet, CSV, and NDJSON loaders

more →



Free and extensible

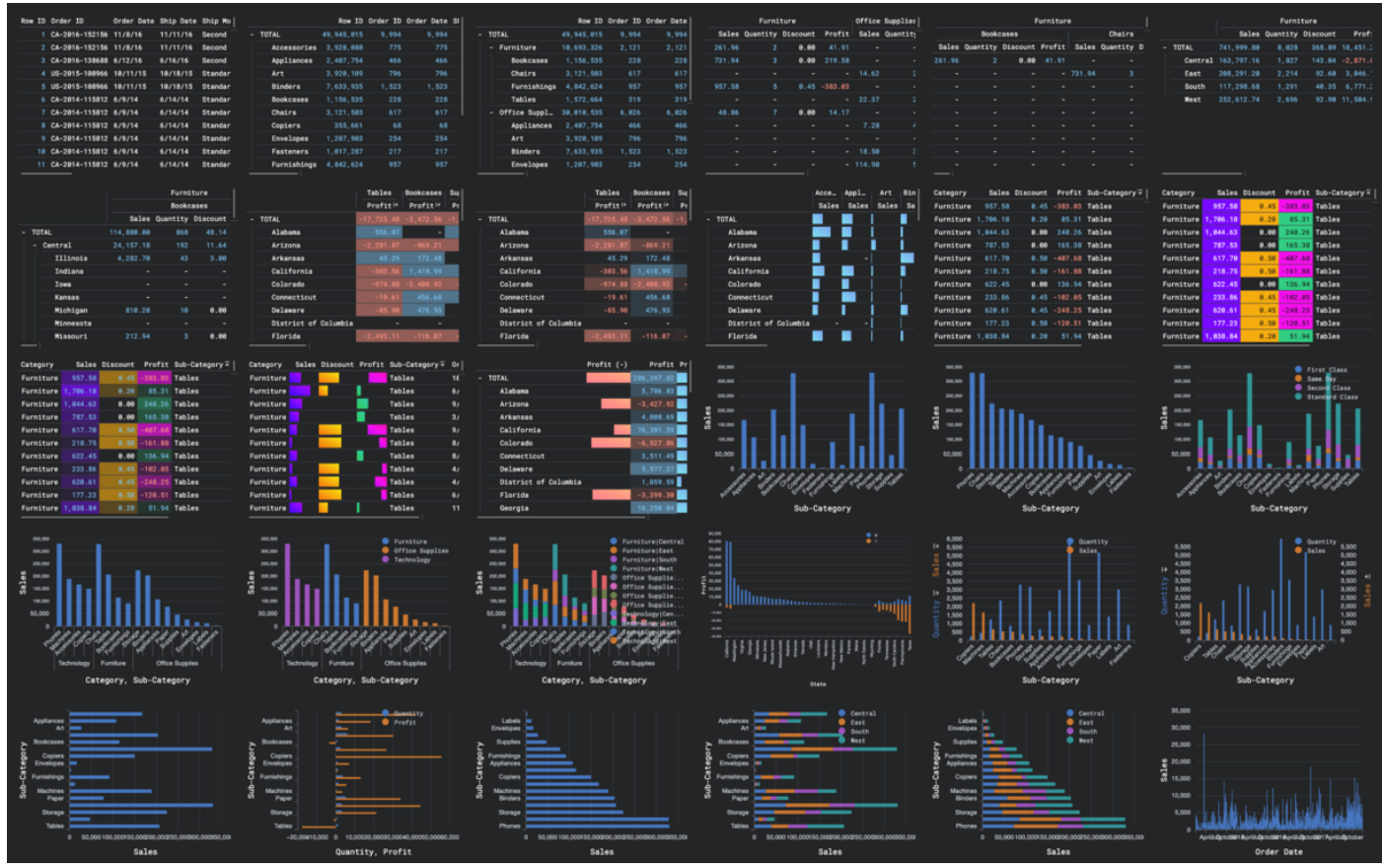
- Free & open-source
- Permissive MIT License
- Flexible extension mechanism

more →

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Perspective for JS and Python

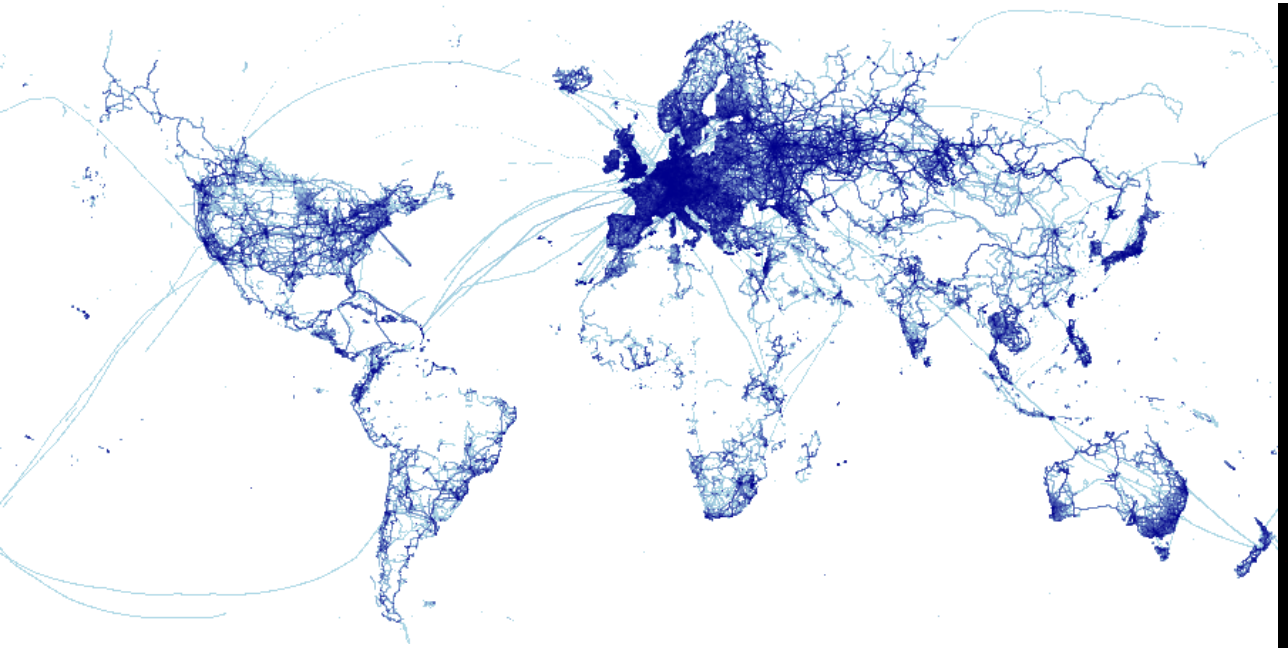
- **Perspective** (http://perspective.finos.org)
  - An interactive analytics and data visualization component, which is especially well-suited for large and/or streaming datasets

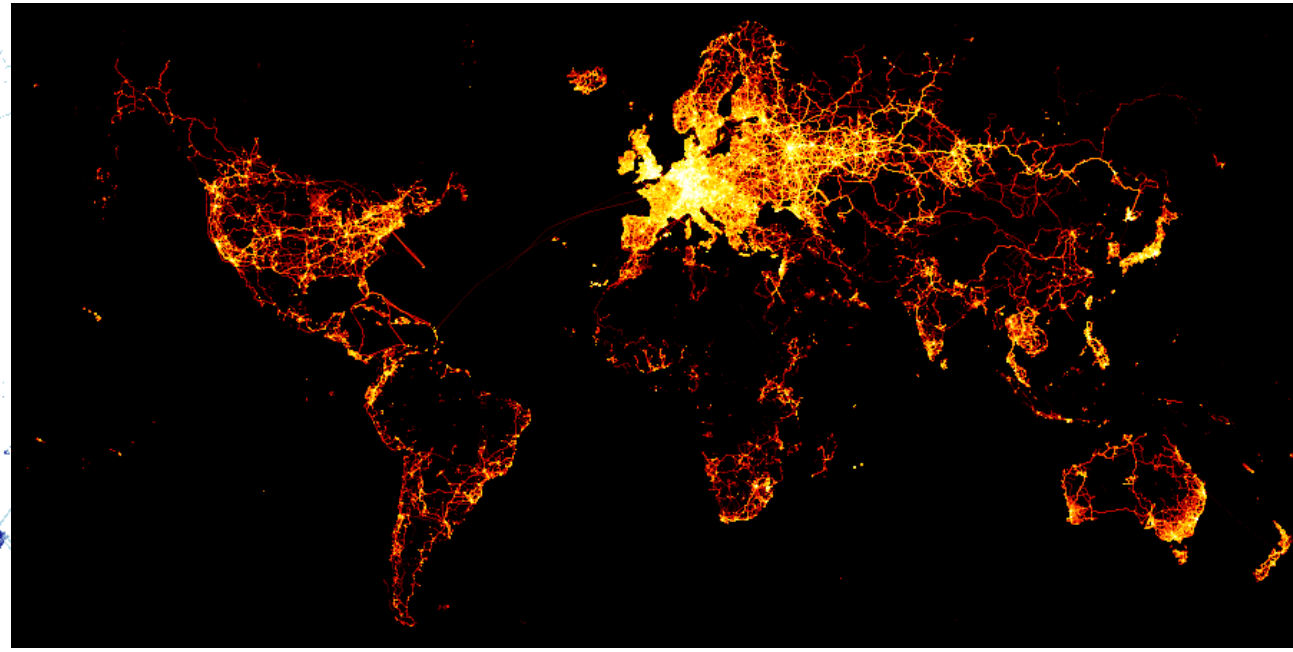


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Effective rasterization

- **Datashader** (http://datashader.org)
  - A graphics pipeline system for creating meaningful representations of large datasets quickly and flexibly
  - ✓ DataShader – Plotting Pitfalls (http://datashader.org/user_guide/Plotting_Pitfalls.html)
    - Overplotting, Oversaturation, Undersampling, Undersaturation, Underutilized range, Nonuniform colormapping, …

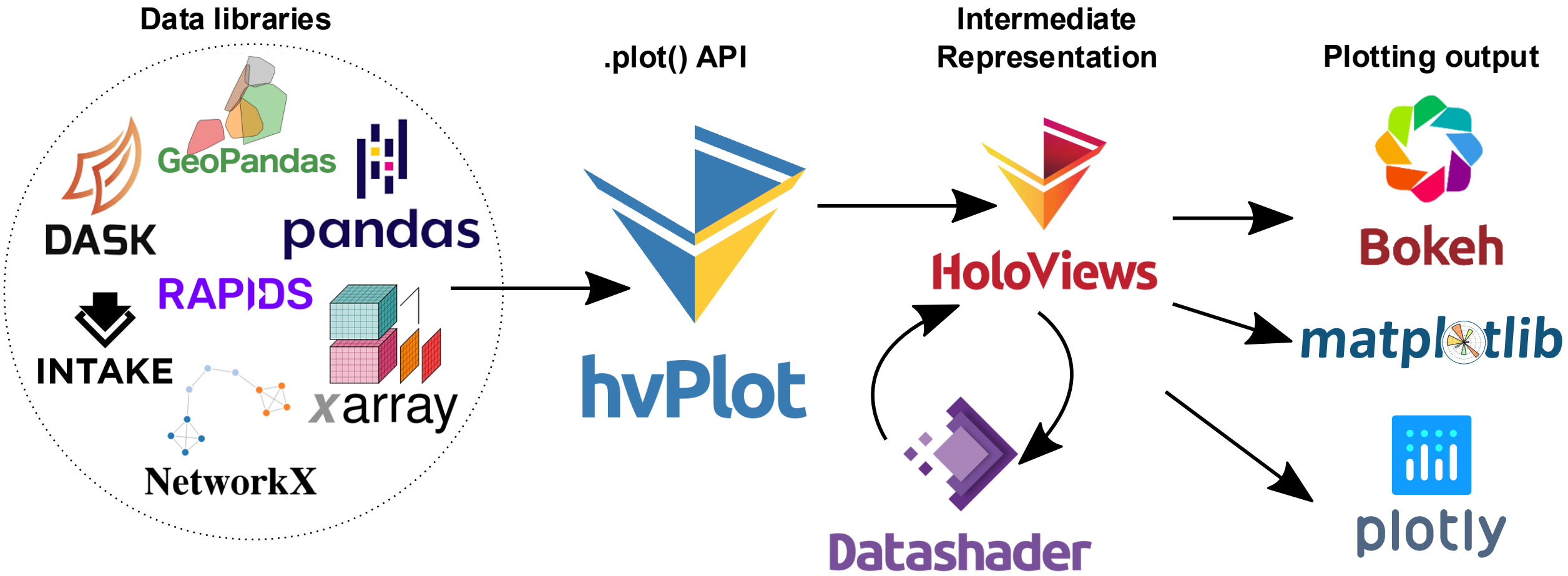




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hvPlot interactive

- **hvPlot** (http://hvplot.holoviz.org)
  - hvPlot Explorer – Example (http://panel-gallery.pyviz.demo.anaconda.com/hvplot_explore)



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NCVis for Python – Clusters

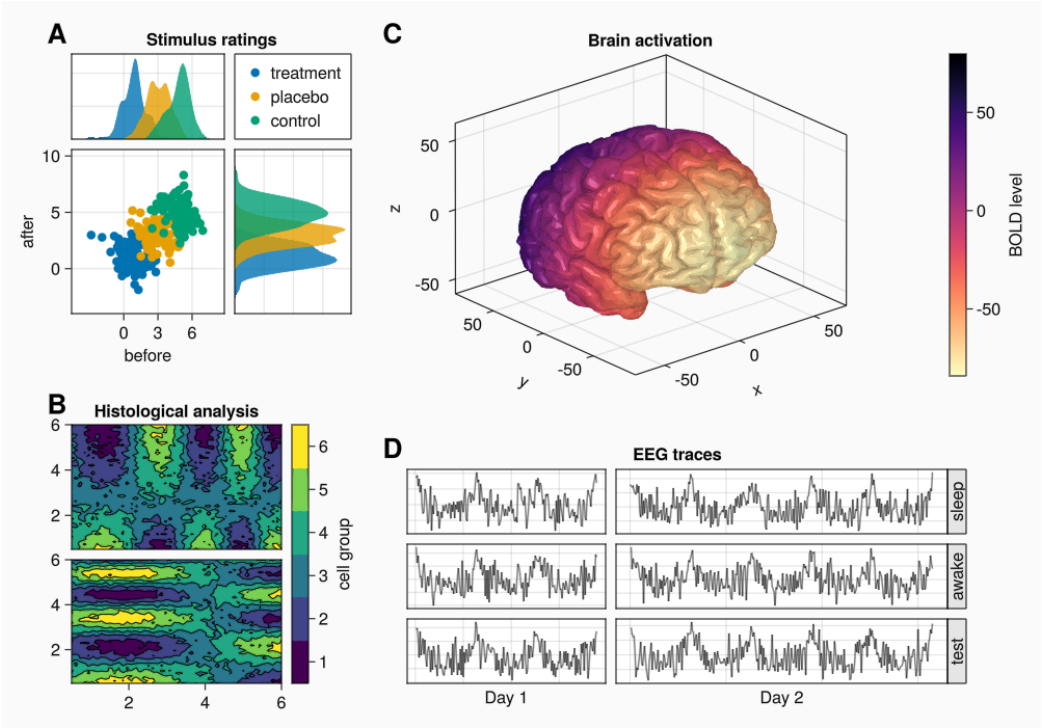
- NCVis (http://github.com/stat-ml/ncvis)
  - An efficient solution for data visualization and dimensionality reduction
  - Python / C++



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Makie for Julia

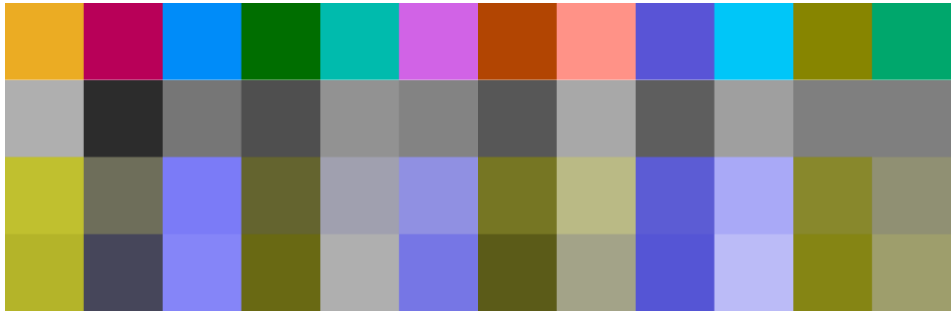
- **Makie** (http://makie.org)
  - A modern plotting library for Julia
  - Adequate backend + GPU via OpenGL



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Palette selection and optimization

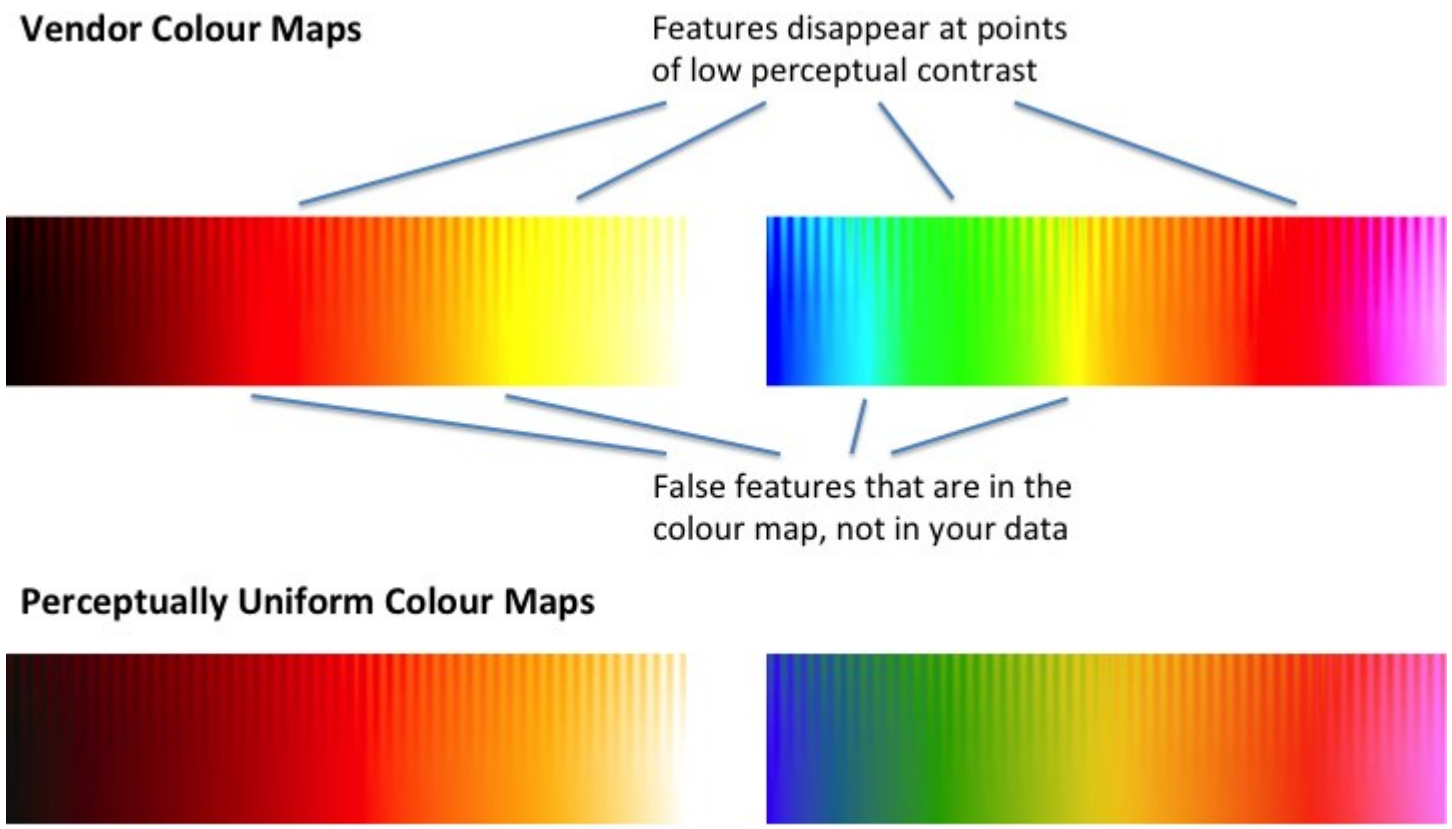
- A huge field for research and creativity!
  - [2013-09-09] *Mastering Multi-hued Color Scales with Chroma.js* (http://vis4.net/blog/posts/mastering-multi-hued-color-scales)
  - [2015-12-01] *Finding the Right Color Palettes for Data Visualizations* (http://blog.invisionapp.com/finding-the-right-color-palettes-for-data-visualizations)
- Some other tools:
  - ***Colorpicker for data*** *by tristen* (http://tristen.ca/hcl-picker)
  - ***Paletton.com*** *by Petr Staníček* (http://paletton.com)
  - ***COPASO****: Color Palette Software* (http://www.colourlovers.com/copaso/ColorPaletteSoftware)
  - ***Chroma.js Color Palette Helper*** *by Gregor Aisch* (http://www.vis4.net/palettes)
- ***Optimal qualitative colour palettes***
  by Anton Tsitsulin (http://tsitsul.in/blog/coloropt/)

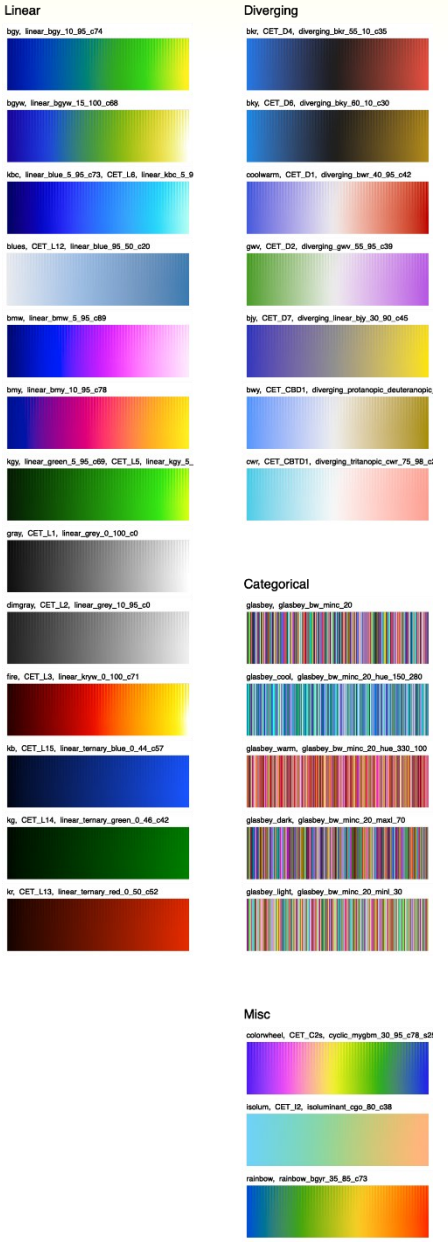


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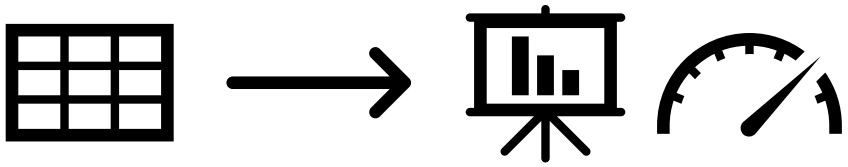
Ready palettes (color maps)

- **ColorCET** – Perceptually Uniform Colour Maps (http://colorcet.com)
  - Colorcet Python Repo (http://github.com/holoviz/colorcet)





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DASHBOARDING

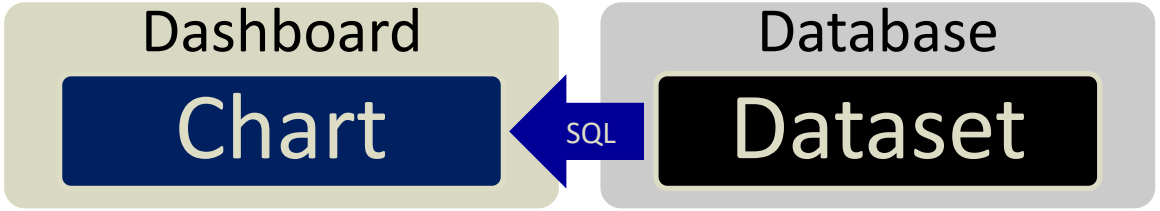
- ✓ Concepts
- ✓ Classification of dashboards
- ✓ BI and classical/modern methodology

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Dashboarding

- **Dashboard** –
  1. A program that displays a set of indicators and tools on the screen in the form of a control panel of a device
  2. This interactive panel itself (UI) on the computer screen

- Main terminology:

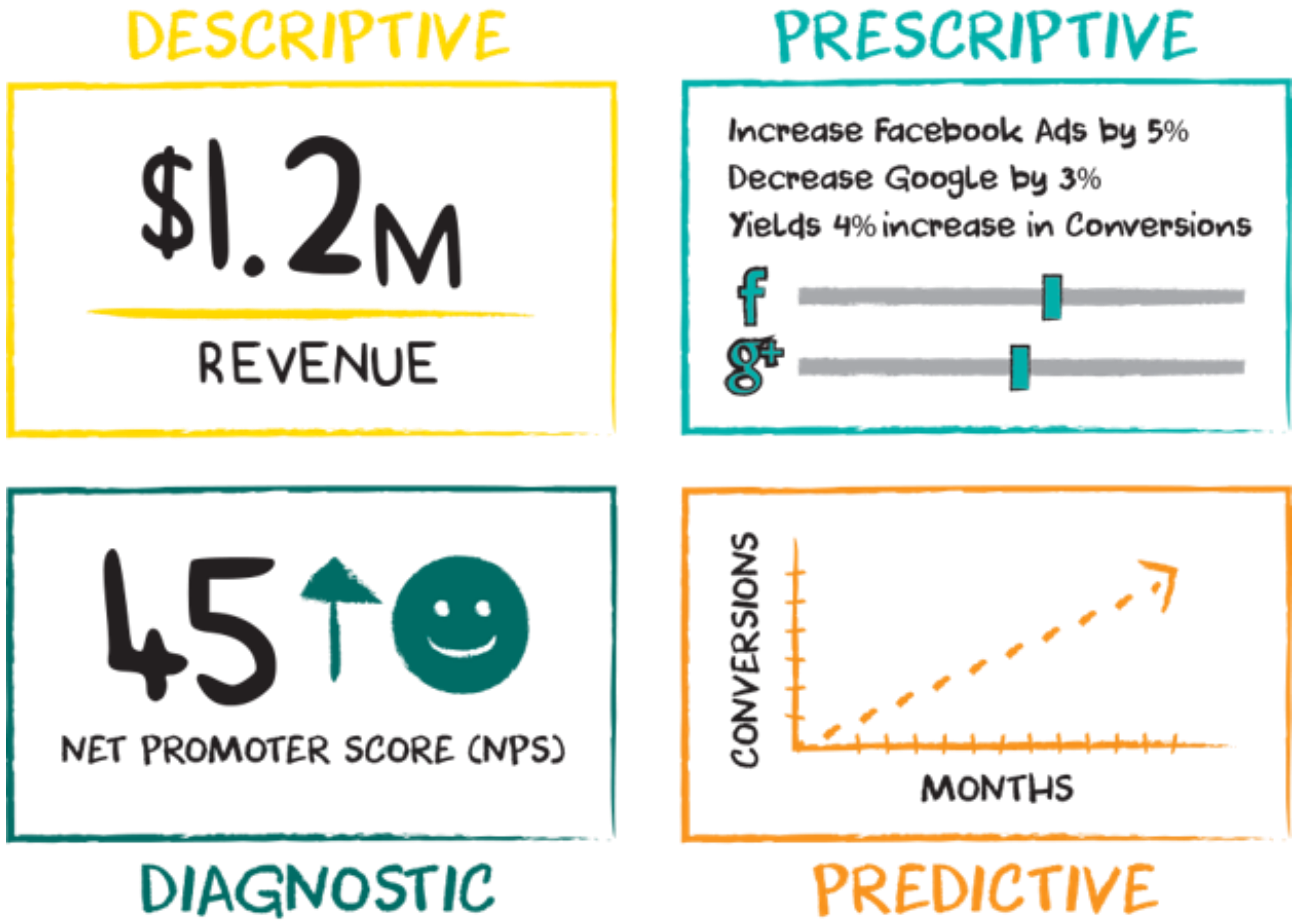


- Some tools:
  - *Microsoft Power BI – Interactive Data Visualization BI Tools* (http://powerbi.microsoft.com)
    - *Power BI Custom Visuals Community* (http://powerbi.microsoft.com/en-us/developers/custom-visualization/)
  - *Tableau Software* (http://www.tableau.com)
  - *Apache Superset* (http://superset.apache.org)
  - …

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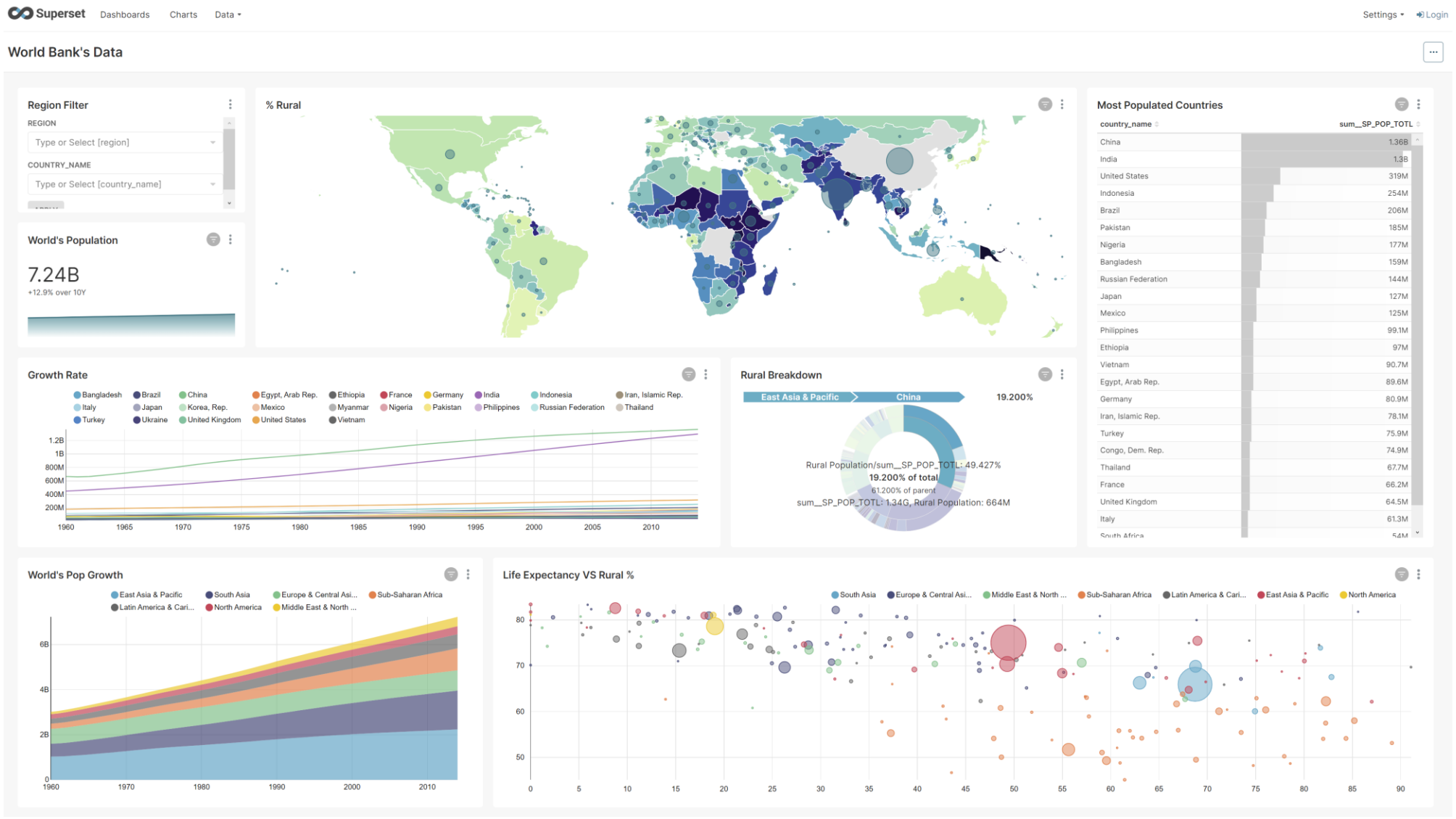
The Purpose of Dashboards

- Basic types of dashboards:
  - **Descriptive** (exploratory)
    - Description of current situation
  - **Diagnostic** (monitoring)
    - Changes tracking
  - **Predictive** (forecasting)
    - Changes prediction
  - **Prescriptive** (recommending)
    - Identifying the causes of changes and their relationships with some recommendations
  - **Combined**



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Official example dashboard of Apache Superset

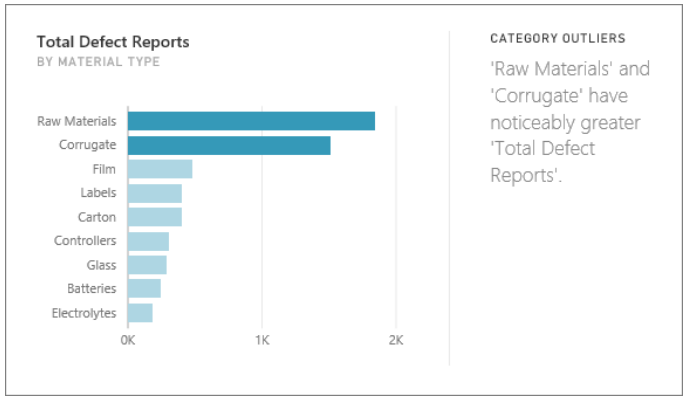


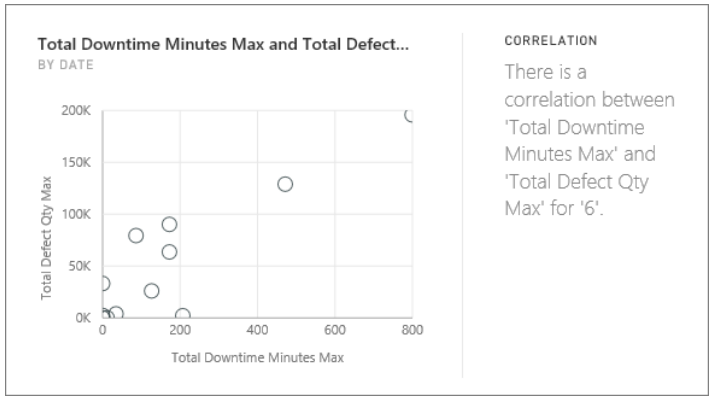
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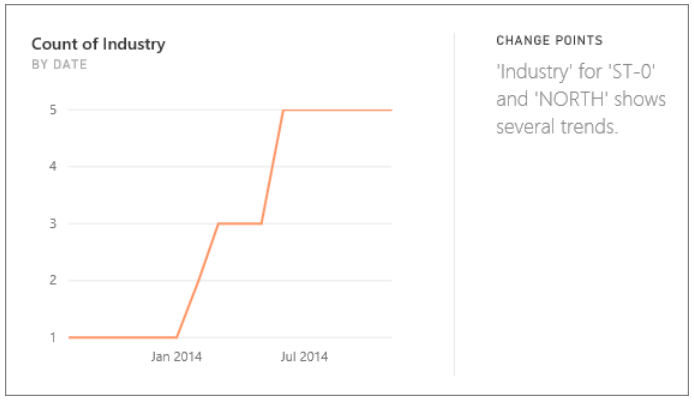
Infographics and dashboards

- Dashboards from an Analyst's and Designer's perspective
  - A Guide to Creating Dashboards People Love to Use (http://www.cpoc.org/assets/Data/guide_to_dashboard_design1.pdf)
  - Excel Dashboards – Tutorials, Templates, & Examples (http://chandoo.org/wp/excel-dashboards)

- How do I choose the visualization type, data presentation, color scheme, and other options?
  - AI can help, for example: Microsoft Power BI Insights –
    - Types of insights supported by Power BI (http://docs.microsoft.com/en-us/power-bi/service-insight-types)



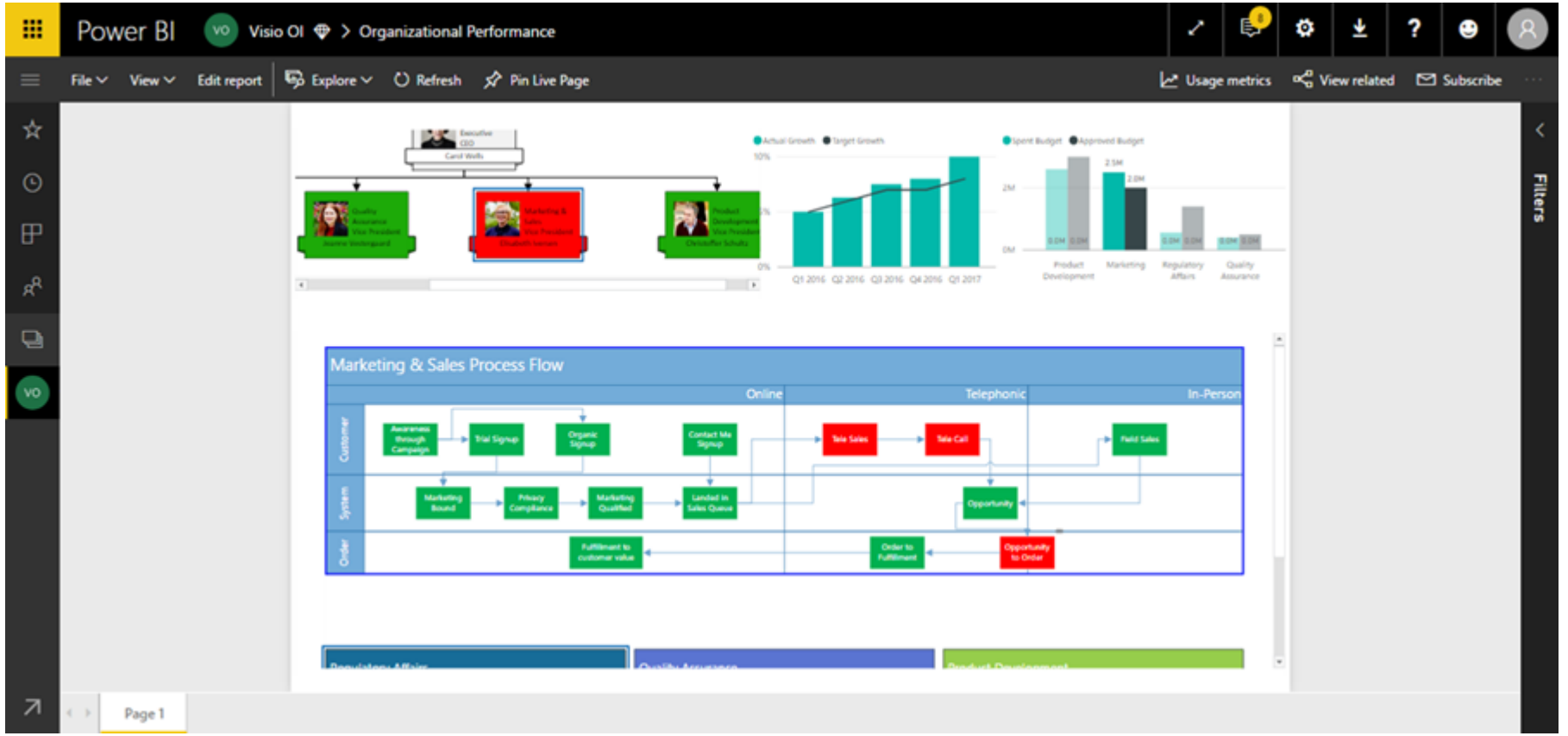




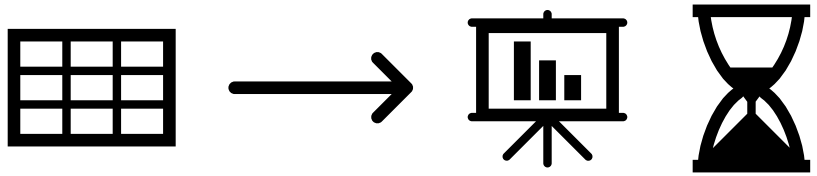
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Microsoft Visio

- *Microsoft Visio* (http://products.office.com/ru-ru/visio/flowchart-software)
  - *Microsoft Power BI integration*:



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TEMPORAL DATA VISUALIZATION

- ✓ Essentials
- ✓ Standartization
  - ✓ ISO 8601
- ✓ Classification
  - ✓ TimeViz Browser
- ✓ Tools

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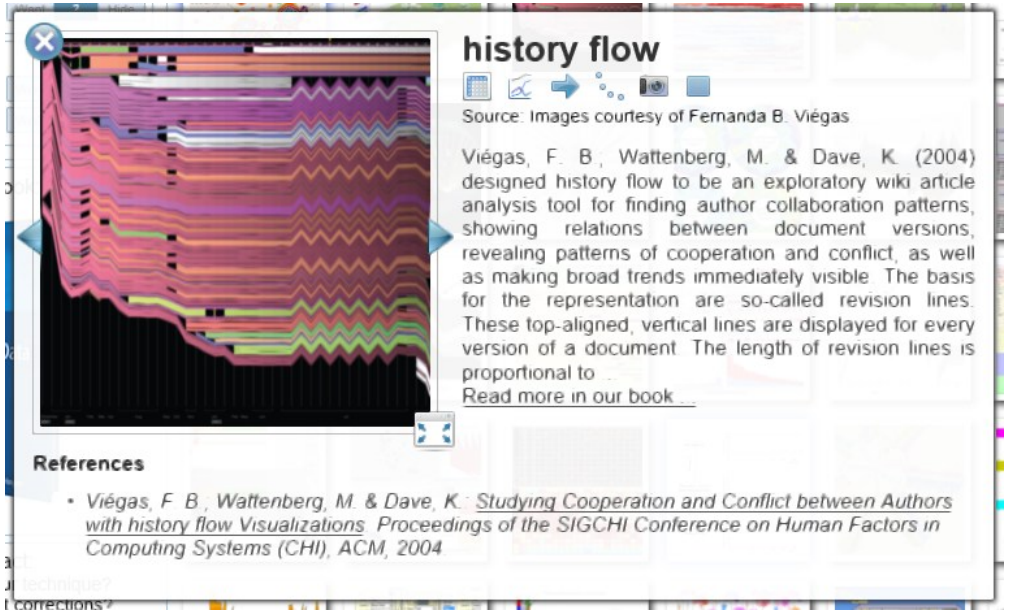
Temporal data (time series)

- Axes of time
  - Problem of different scales at different intervals
  - Linearity problem
- Timestamps
  - What to do with many-many-many events?
  - ISO 8601 (http://www.iso.org/iso-8601-date-and-time-format.html)!
- Time aggregation
  - With constant/adaptive granularity
    - Nanosecond/second/minute/hour/day/month/year/...
- Temporal data summarization
  - By number/characteristics of events
- Adaptability!
  - Data measurements
  - How to display more than one dimension?

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The TimeViz Browser – Temporal Data Visualization

- Visualization of Time-Oriented Data. 2003 (http://timeviz.net)
- TimeViz Browser 2.0 (http://browser.timeviz.net)
  - Classification + descriptions
    - Example of description:





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uPlot for JS – time series

- **µPlot** (http://github.com/leeoniya/uPlot)
  - A fast, memory-efficient Canvas 2D-based chart for plotting time series, lines, areas, ohlc & bars;
  - From a cold start it can create an interactive chart containing **150,000 data points in 90ms**, scaling linearly at ~31,000 pts/ms



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Conclusion

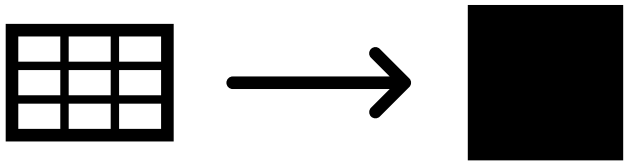
- **Object-attribute data** is a most common mindset and toolset in data representation

- Good traits
  - Comprehensive ontological description
  - Formal Concept Analysis as a foundational basis
  - Basic semantic interpretability
  - Full set of data models at all levels + data manipulation
  - Finally, a proper and effective implementation of dataframes
  - Declarative DataViz

- Bad traits
  - The problems of terminological diversity at all levels
  - Data manipulation is not grounded enough
  - The data quality, especially historical data, leaves much to be desired
  - No full adequately grounded object-attribute data manipulation pipeline in open-source world

No **Big Data** Vibe Here

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THE END

- ✓ Questions?



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